

Prompt Life v0.27.7 Final Implementation Report

This front matter summarizes the implementation. The full review packet, glossary path order, per-card question bank, and QA screenshots follow in the same PDF.

1. Summary of what changed

Fixed Journey checkpoint retry behavior, added stable per-attempt checkpoint state, generated a draft-only multi-question checkpoint bank for all 39 cards, generated glossary learning-path order, added checkpoint audits, and produced HTML/PDF review packets with screenshots.

2. Wrong-answer reveal bug

Fixed. Wrong answers no longer reveal or style the correct answer.

3. Stable feedback identity

Fixed. Feedback/highlighting follows stable choice IDs after shuffling, not visible A/B/C/D positions.

4. Correct answer hidden after wrong choices

Confirmed in browser QA at 390px and 320px. Previous wrong choices can remain lightly marked as tried.

5. Review attempt reset

Confirmed. Returning to completed Alignment opens a fresh visible checkpoint attempt with no selected answer and no correct reveal while durable completion remains.

6. Draft questions for all 39 cards

Yes. Every Journey learning card has 2-5 draft checkpoint questions.

7. Total draft questions

136

8. Total answer choices

544

9. Total wrong-answer distractors

408

10. Question count distribution

2 questions: 5 cards; 3 questions: 12 cards; 4 questions: 20 cards; 5 questions: 2 cards.

11. First six learning cards

What Is an LLM? - live 1, draft 2, objective: Explain What Is an LLM? as part of the day in the life of a prompt.; Where LLMs Fit - live 1, draft 2, objective: Explain Where LLMs Fit as part of the day in the life of a prompt.; Rationalists vs Empiricists - live 1, draft 3, objective: Explain Rationalists vs Empiricists as part of the day in the life of a prompt.; Training - live 1, draft 4, objective: Explain Training as part of the day in the life of a prompt.; Pretraining - live 1, draft 4, objective: Explain Pretraining as part of the day in the life of a prompt.; Overfitting and Generalization - live 1, draft 3, objective: Explain Overfitting and Generalization as part of the day in the life of a prompt.

12. Glossary learning-path order

Generated JSON and Markdown files with 134 terms, stage/card mapping, confidence, nearby terms, and distractor-use notes.

13. Glossary-near distractors

Used. Wrong choices record represented misconceptions, represented terms where applicable, source, and rationale.

14. Draft bank status

Draft-only. The live Journey remains on the existing single-question checkpoints.

15. Debug preview flag

None added. Wiring a preview mode is intentionally deferred until review.

16. Learner-facing copy changes

Visible app version changed to v0.27.7. Wrong-answer ARIA/class behavior now marks attempted wrong answers as tried. No Journey lesson copy or Play copy changed.

17. Checkpoint data/model/logic changes

Added attemptedWrongChoiceIds, completedQuestionIds, per-attempt shuffle seed, completion-gated advance logic, and tried-answer styling.

18. localStorage behavior

No new localStorage keys. Preserved promptlife:v1:lastLocation, lessonId, progress, reflections, gameInsights, exerciseProgress, promptRunProgress, traceComplete, learningTourComplete, playChallenges, glossaryDojo, plus legacy pl.* keys. Existing promptlife:v1:choiceOrderSeed is still read/preserved; per-attempt checkpoint seed is memory-only.

19. Verification commands

npm run typecheck: passed. npm run build: passed with existing large-chunk warning. npm run build:pages: passed with existing large-chunk warning. npm run audit:answers: passed. npm run audit:checkpoints: passed.

20. Mobile QA

320px and 390px checkpoint states had no horizontal overflow. Wrong-answer retry state kept the correct answer hidden. Buttons can scroll clear of the bottom nav.

21. Desktop/laptop QA

Desktop checkpoint state fit without horizontal overflow; Journey core detail blocks stayed one full-width column.

22. Files changed

README.md, docs/REVIEW_NOTES.md, package.json, package-lock.json, scripts/audit-answers.mjs, scripts/audit-checkpoints.mjs, scripts/generate-checkpoint-bank-v0277.mjs, src/main.tsx, src/styles/global.css, docs/journey/checkpoints/*v0-27-7*, docs/journey/prompt-life-v0-27-7-checkpoint-question-bank-review.html/pdf, docs/journey/screenshots/v0-27-7*.png.

23. Reports, PDFs, data, glossary files, screenshots

Created draft JSON/MD, glossary-order JSON/MD, HTML review packet, this PDF review packet, and v0.27.7 screenshots for 390px wrong/second-wrong/correct, 320px wrong, desktop checkpoint, desktop one-column, and review reset.

Screenshots Included Later In This PDF

docs/journey/screenshots/v0-27-7-checkpoint-before-390.png

docs/journey/screenshots/v0-27-7-checkpoint-wrong-no-reveal-390.png

docs/journey/screenshots/v0-27-7-checkpoint-second-wrong-390.png

docs/journey/screenshots/v0-27-7-checkpoint-correct-390.png

docs/journey/screenshots/v0-27-7-checkpoint-wrong-no-reveal-320.png

docs/journey/screenshots/v0-27-7-review-attempt-reset-390.png

docs/journey/screenshots/v0-27-7-checkpoint-desktop.png

docs/journey/screenshots/v0-27-7-core-one-column-desktop.png

Prompt Life v0.27.7

Journey Checkpoint Question Bank Draft, Distractor Strategy, and Review Packet

Generated from `docs/journey/checkpoints/checkpoint-question-bank-v0-27-7-draft.json` and `docs/journey/checkpoints/glossary-learning-path-order-v0-27-7.json`.

Metric	Value
Journey stages	8
Learning cards	39
Total proposed checkpoint questions	136
Total answer choices	544
Total wrong-answer distractors	408
Question count distribution	2 questions: 5 cards, 3 questions: 12 cards, 4 questions: 20 cards, 5 questions: 2 cards
Every question has 4 choices	Yes
Every question has exactly 1 correct answer	Yes
Every wrong choice has specific feedback	Yes
Every choice has a stable ID	Yes
Draft status	review-draft (not live by default)

This packet is a review artifact. The multi-question bank is draft-only in v0.27.7; the live app still uses the reviewed single-question checkpoints until the bank is approved and wired into learner mode.

Distractor Strategy

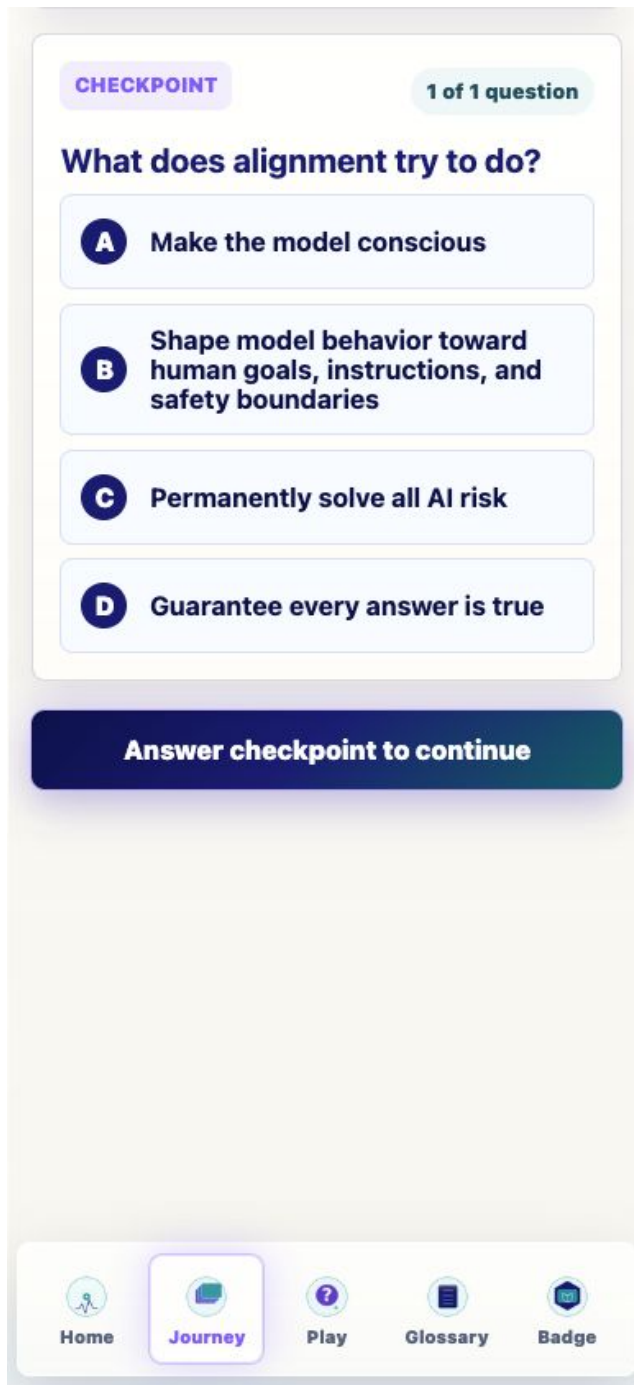
Misconception-first distractors using glossary proximity and Journey-path-near terms.

- * Wrong answers are authored as plausible learner misunderstandings, not random false statements.
- * Glossary-path-near terms supply many distractors: same card, same stage, nearby stage, or explicitly confusable metadata.
- * Each wrong choice records represented misconception, represented glossary term when applicable, distractor source, and rationale.
- * Stable IDs keep feedback attached to the underlying choice after answer order is shuffled.
- * The teaching goal is boundary literacy: mind vs model, training vs inference, context vs memory, probability vs truth, retrieval vs learning, diffusion vs autoregression, token vs word, embedding vs hidden state, logits vs probabilities, attention vs awareness, and temporary state vs durable weights.

Sources inspected/reused: `src/data/content.ts`, `src/data/glossaryDojoMeta.ts`, `src/features/glossary-dojo/engine.ts`, `scripts/audit-answers.mjs`

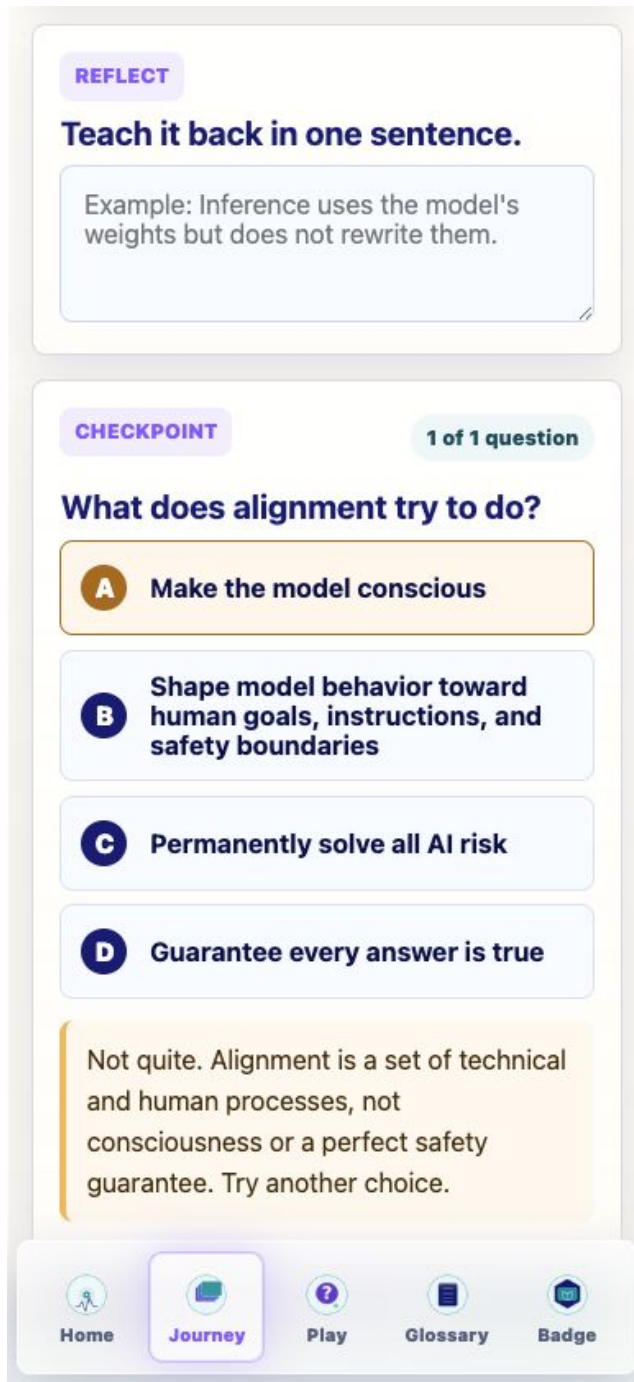
UI QA Screenshots

390px checkpoint before selection



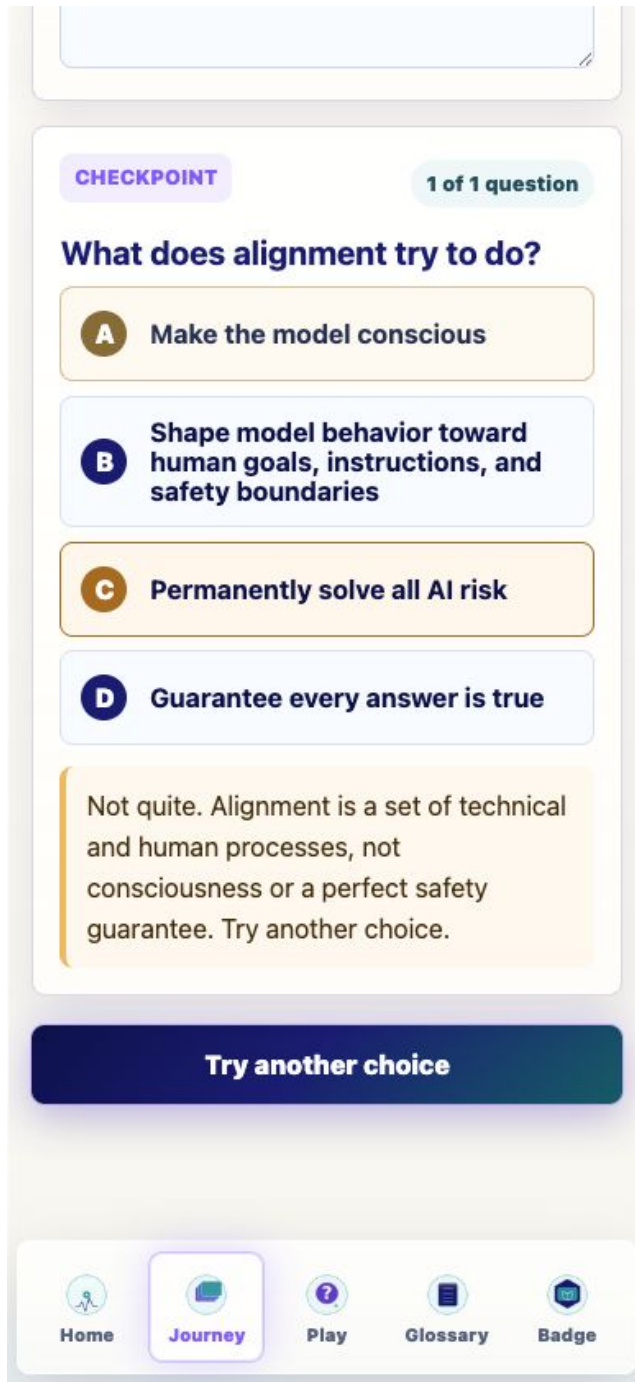
docs/journey/screenshots/v0-27-7-checkpoint-before-390.png

390px wrong answer: correct answer remains hidden



docs/journey/screenshots/v0-27-7-checkpoint-wrong-no-reveal-390.png

390px second wrong answer: feedback follows selected choice



docs/journey/screenshots/v0-27-7-checkpoint-second-wrong-390.png

390px correct answer: final button state

Teach it back in one sentence.

Example: Inference uses the model's weights but does not rewrite them.

CHECKPOINT

1 of 1 question

What does alignment try to do?

- A** Make the model conscious
- B** Shape model behavior toward human goals, instructions, and safety boundaries
- C** Permanently solve all AI risk
- D** Guarantee every answer is true

Insight unlocked. Alignment shapes behavior, but it is not magic morality.

Next learning card



Home



Journey



Play



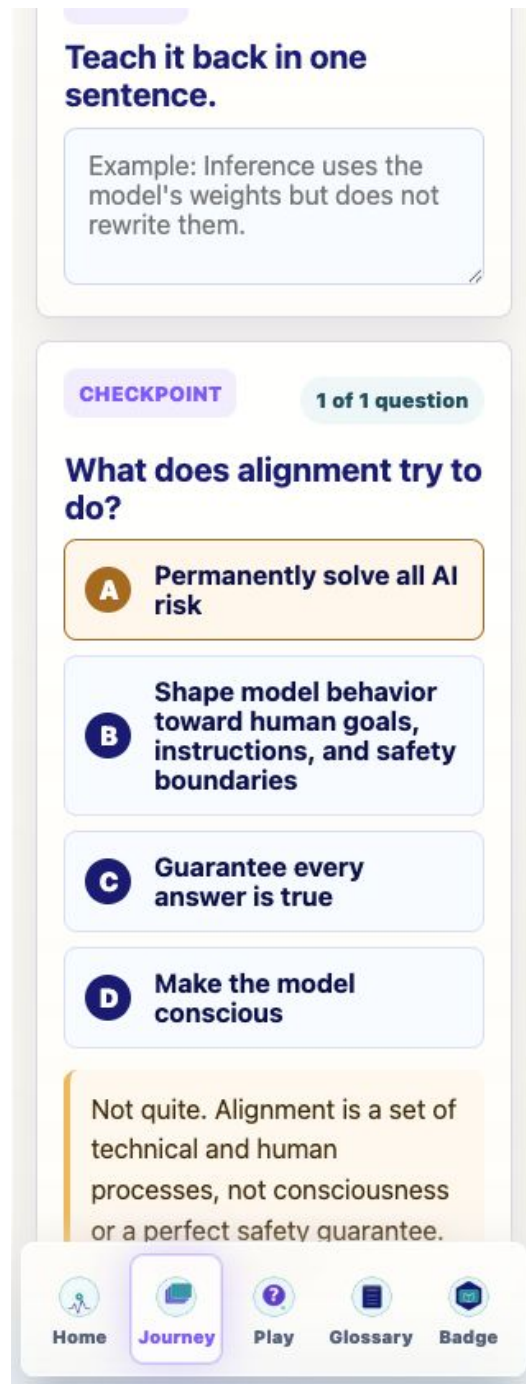
Glossary



Badge

docs/journey/screenshots/v0-27-7-checkpoint-correct-390.png

320px wrong answer: no horizontal overflow



docs/journey/screenshots/v0-27-7-checkpoint-wrong-no-reveal-320.png

390px review reset: no previous selection retained

Teach it back in one sentence.

Reflection is available in Learn mode.

CHECKPOINT

1 of 1 question

What does alignment try to do?

- A** Make the model conscious
- B** Guarantee every answer is true
- C** Permanently solve all AI risk
- D** Shape model behavior toward human goals, instructions, and safety boundaries

Return to Journey



Home



Journey



Play



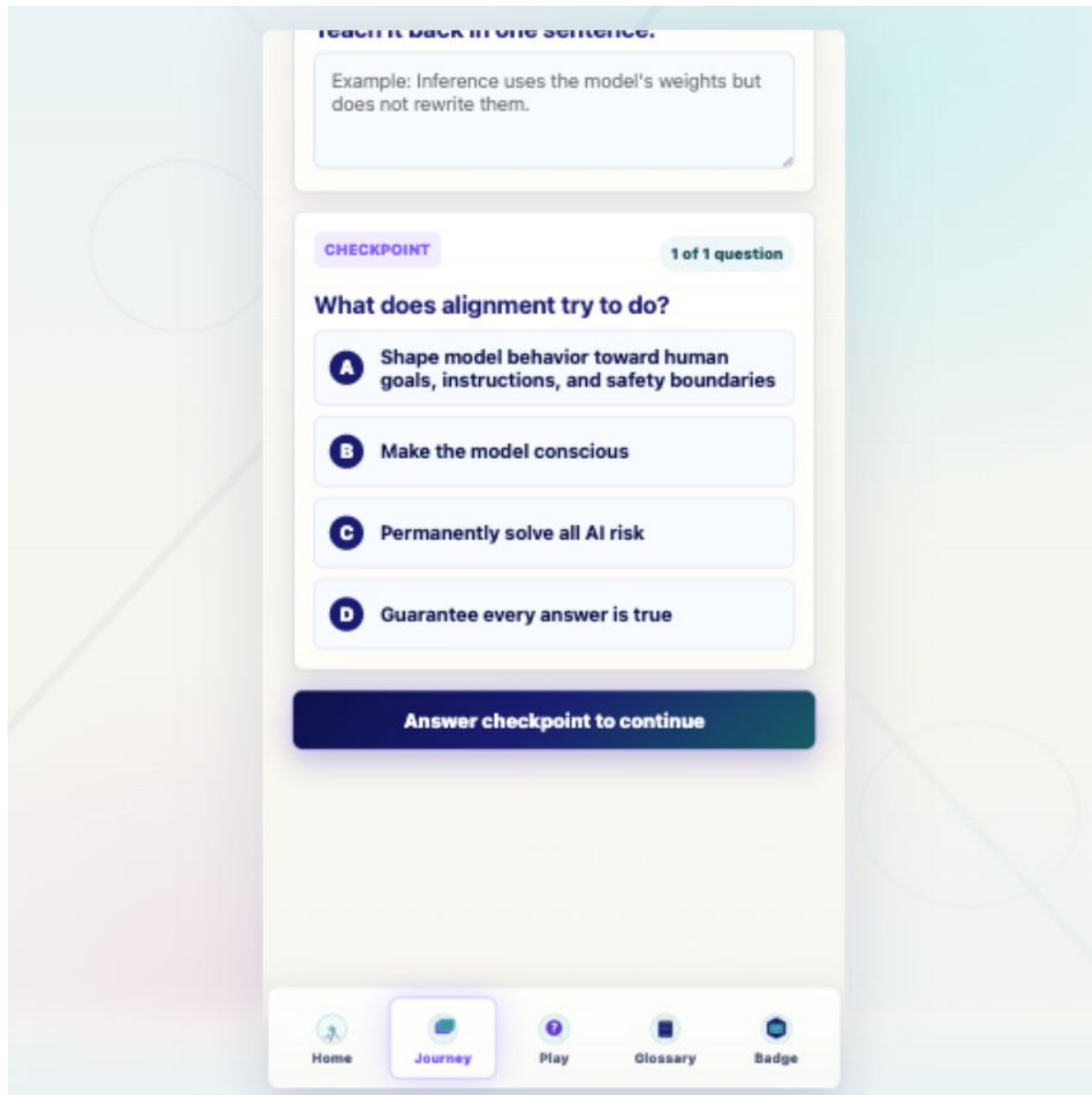
Glossary



Badge

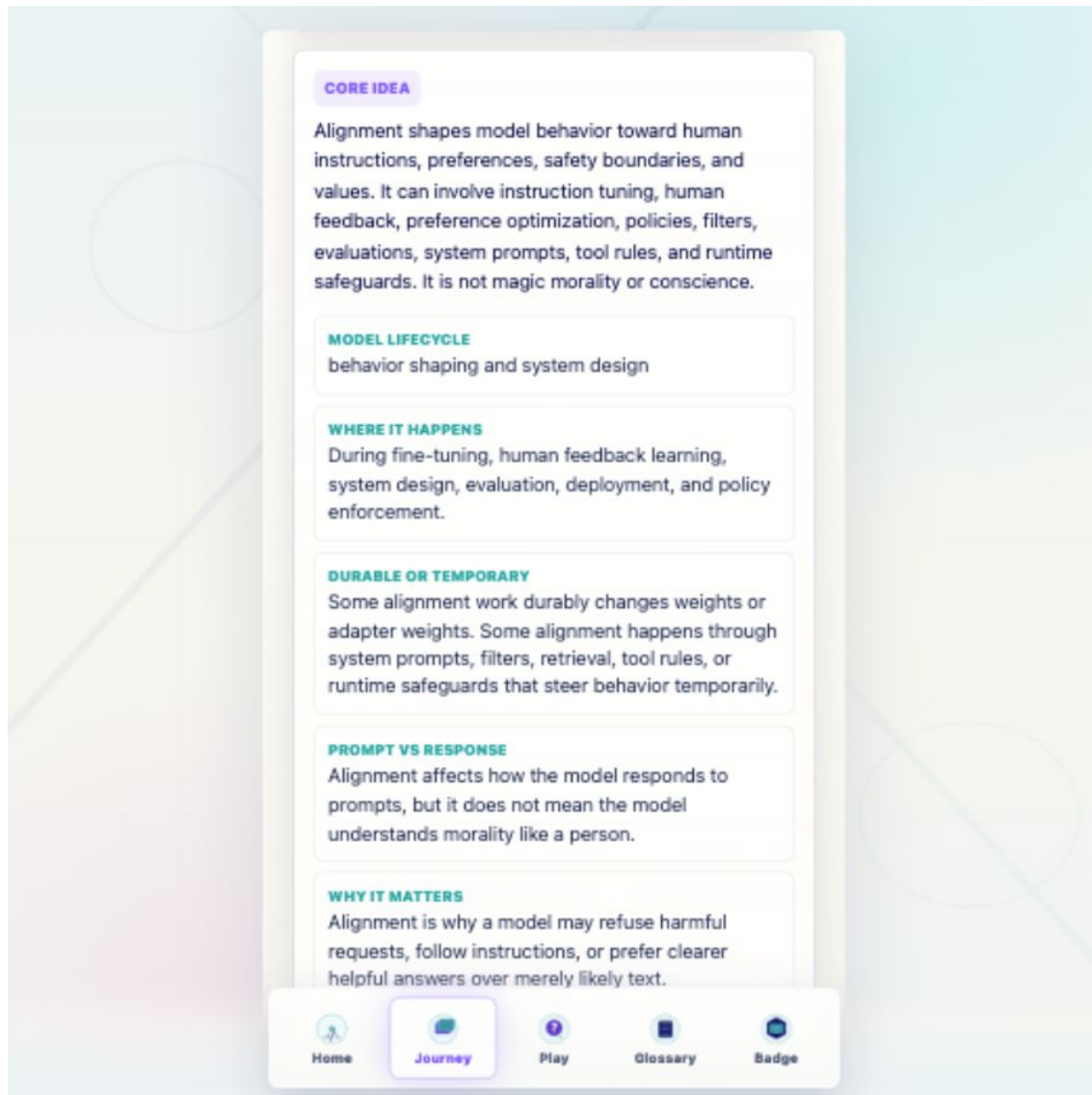
docs/journey/screenshots/v0-27-7-review-attempt-reset-390.png

Desktop checkpoint state



docs/journey/screenshots/v0-27-7-checkpoint-desktop.png

Desktop Journey detail blocks remain one column



docs/journey/screenshots/v0-27-7-core-one-column-desktop.png

Glossary Learning Path Order

Full glossary order with inferred/explicit mapping, nearby terms, and distractor-use notes. Inferred mappings are marked by the generator and should be reviewed by a human before using them as policy.

1. Alignment (*alignment*)

Stage/card: 1. Before Morning / Alignment - explicit learning-path hint

Definition: The effort to shape AI behavior toward human instructions, preferences, safety boundaries, and values.

Related: instruction-tuning, human-feedback-learning, rlhf, policy, guardrail

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Alignment can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

2. Evaluation (*evaluation*)

Stage/card: 1. Before Morning / Alignment - explicit learning-path hint

Definition: The process of testing model behavior against examples, tasks, rubrics, or safety checks.

Related: validation-data, generalization, alignment

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Evaluation can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

3. Guardrail (*guardrail*)

Stage/card: 1. Before Morning / Alignment - explicit learning-path hint

Definition: A system constraint intended to keep AI behavior within safer or more useful boundaries.

Related: alignment, policy, evaluation

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Guardrail can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

4. Human feedback learning (*human-feedback-learning*)

Stage/card: 1. Before Morning / Alignment - explicit learning-path hint

Definition: Training that uses human judgments, rankings, or preferences to shape future model behavior.

Related: rlhf, preference-optimization, alignment

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Human feedback learning can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

5. Instruction tuning (*instruction-tuning*)

Stage/card: 1. Before Morning / Alignment - explicit learning-path hint

Definition: Targeted training that helps a model follow instructions more usefully.

Related: fine-tuning, alignment, human-feedback-learning

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Instruction tuning can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

6. Policy (*policy*)

Stage/card: 1. Before Morning / Alignment - explicit learning-path hint

Definition: A rule, guideline, or system behavior constraint used to shape what an AI system should or should not do.

Related: alignment, guardrail, evaluation

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Policy can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

7. Preference optimization (preference-optimization)

Stage/card: 1. Before Morning / Alignment - explicit learning-path hint

Definition: Training that nudges a model toward preferred responses and away from less-preferred ones.

Related: human-feedback-learning, rlhf, alignment

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Preference optimization can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

8. RLHF (rlhf)

Stage/card: 1. Before Morning / Alignment - explicit learning-path hint

Definition: Reinforcement learning from human feedback, a family of methods that uses human preference signals to shape model behavior.

Related: human-feedback-learning, alignment, preference-optimization

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: RLHF can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

9. Adapter (adapter)

Stage/card: 1. Before Morning / Fine-Tuning - explicit learning-path hint

Definition: A smaller set of trained parameters added to or used with a base model to adapt behavior.

Related: fine-tuning, parameter, alignment

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Adapter can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

10. Fine-tuning (fine-tuning)

Stage/card: 1. Before Morning / Fine-Tuning - explicit learning-path hint

Definition: Additional targeted training after broad pretraining.

Related: pretraining, adapter, alignment, instruction-tuning

Confusable: prompt, rag, pretraining

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Fine-tuning is useful as a distractor when learners might confuse it with prompt, rag, pretraining.

11. Generalization (generalization)

Stage/card: 1. Before Morning / Overfitting and Generalization - explicit learning-path hint

Definition: The ability of learned patterns to work on new examples, not only examples seen during training.

Related: overfitting, validation-data, evaluation

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Generalization can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

12. Overfitting (overfitting)

Stage/card: 1. Before Morning / Overfitting and Generalization - explicit learning-path hint

Definition: A training failure mode where a model fits training examples too narrowly and performs worse on new examples.

Related: generalization, validation-data, training

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Overfitting can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

13. Validation data (validation-data)

Stage/card: 1. Before Morning / Overfitting and Generalization - explicit learning-path hint

Definition: Held-out examples used to check whether a model is learning patterns that work beyond training examples.

Related: overfitting, generalization, evaluation

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Validation data can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

14. Next token (next-token)

Stage/card: 1. Before Morning / Pretraining - explicit learning-path hint

Definition: The single token being scored and selected at the current generation step.

Related: generated-token, logits, sampling

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Next token can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

15. Pretraining (pretraining)

Stage/card: 1. Before Morning / Pretraining - explicit learning-path hint

Definition: The broad first stage of training that updates model weights by learning next-token patterns from large datasets.

Related: training, training-data, loss, weight-update

Confusable: fine-tuning, inference

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Pretraining is useful as a distractor when learners might confuse it with fine-tuning, inference.

16. Empiricism (empiricism)

Stage/card: 1. Before Morning / Rationalists vs Empiricists - explicit learning-path hint

Definition: An experience-first tradition that emphasizes learning from examples, observation, and evidence.

Related: rationalism, deep-learning, training

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Empiricism can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

17. Loss (loss)

Stage/card: 1. Before Morning / Rationalists vs Empiricists - explicit learning-path hint

Definition: A number that measures how far the model's prediction is from the target answer during training.

Related: training, weight, weight-update, pretraining, fine-tuning

Confusable: probability, confidence, evaluation

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Loss is useful as a distractor when learners might confuse it with probability, confidence, evaluation.

18. Rationalism (rationalism)

Stage/card: 1. Before Morning / Rationalists vs Empiricists - explicit learning-path hint

Definition: A reason-first tradition that emphasizes explicit principles, rules, and logic.

Related: symbolic-ai, empiricism, deep-learning

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Rationalism can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

19. Parameter (parameter)

Stage/card: 1. Before Morning / Training - explicit learning-path hint

Definition: A model value learned during training, often a weight or bias.

Related: weight, training, fine-tuning

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Parameter can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

20. Training (training)

Stage/card: 1. Before Morning / Training - explicit learning-path hint

Definition: The durable process of updating weights from prediction error or another training signal.

Related: loss, weight-update, pretraining, fine-tuning

Confusable: inference, fine-tuning, rag

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Training is useful as a distractor when learners might confuse it with inference, fine-tuning, rag.

21. Training data (training-data)

Stage/card: 1. Before Morning / Training - explicit learning-path hint

Definition: Examples used during training to shape model weights.

Related: pretraining, fine-tuning, validation-data

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Training data can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

22. Weight (weight)

Stage/card: 1. Before Morning / Training - explicit learning-path hint

Definition: A learned number inside the model that shapes how signals are transformed.

Related: parameter, weight-update, training

Confusable: hidden state, memory, parameter

Nearby terms: llm, prompt, response, token, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: Weight is useful as a distractor when learners might confuse it with hidden state, memory, parameter.

23. Weight update (weight-update)

Stage/card: 1. Before Morning / Training - explicit learning-path hint

Definition: A change to model weights made during training to reduce error or improve a preference objective.

Related: weight, training, loss

Confusable: prompt, context window, inference

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Weight update is useful as a distractor when learners might confuse it with prompt, context window, inference.

24. LLM (llm)

Stage/card: 1. Before Morning / What Is an LLM? - explicit learning-path hint

Definition: A large language model is a generative AI system focused on language and code, usually generating one response token at a time.

Related: ai, generative-ai, deep-learning, diffusion, multimodal

Confusable: ai, generative-ai, rag

Nearby terms: prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: LLM is useful as a distractor when learners might confuse it with ai, generative-ai, rag.

25. AI (ai)

Stage/card: 1. Before Morning / Where LLMs Fit - explicit learning-path hint

Definition: The broad field of building systems that perform tasks associated with intelligence.

Related: machine-learning, symbolic-ai, generative-ai, llm

Confusable: llm, machine-learning, generative-ai

Nearby terms: llm, prompt, response, token, weight, inference, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: AI is useful as a distractor when learners might confuse it with llm, machine-learning, generative-ai.

26. Classical machine learning (classical-machine-learning)

Stage/card: 1. Before Morning / Where LLMs Fit - explicit learning-path hint

Definition: Machine-learning methods that learn from data without usually using very large deep neural networks.

Related: machine-learning, deep-learning, ai

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: Classical machine learning can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

27. Deep learning (deep-learning)

Stage/card: 1. Before Morning / Where LLMs Fit - explicit learning-path hint

Definition: A kind of machine learning that uses many learned layers, often large neural networks, to transform data representations.

Related: machine-learning, neural network, generative-ai, parameter

Confusable: machine-learning, symbolic-ai

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, generative-ai, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: Deep learning is useful as a distractor when learners might confuse it with machine-learning, symbolic-ai.

28. Diffusion model (diffusion)

Stage/card: 1. Before Morning / Where LLMs Fit - explicit learning-path hint

Definition: A generative AI model family that often starts with noise and iteratively denoises patterns.

Related: generative-ai, llm, multimodal

Confusable: llm, autoregression, multimodal

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, multimodal, symbolic-ai, rule-based-ai

Distractor notes: Diffusion model is useful as a distractor when learners might confuse it with llm, autoregression, multimodal.

29. Foundation model (*foundation-model*)

Stage/card: 1. Before Morning / Where LLMs Fit - explicit learning-path hint

Definition: A broadly trained model that can be adapted or prompted for many downstream tasks.

Related: pretraining, fine-tuning, llm, multimodal

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Foundation model can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

30. Generative AI (*generative-ai*)

Stage/card: 1. Before Morning / Where LLMs Fit - explicit learning-path hint

Definition: AI systems that create new text, images, audio, code, video, or other media.

Related: llm, diffusion, multimodal, sampling

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: Generative AI can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

31. Machine learning (*machine-learning*)

Stage/card: 1. Before Morning / Where LLMs Fit - explicit learning-path hint

Definition: A major branch of AI where systems learn patterns from data rather than relying only on hand-written rules.

Related: ai, classical-machine-learning, deep-learning, training

Confusable: ai, deep-learning

Nearby terms: llm, prompt, response, token, weight, inference, ai, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: Machine learning is useful as a distractor when learners might confuse it with ai, deep-learning.

32. Multimodal AI (*multimodal*)

Stage/card: 1. Before Morning / Where LLMs Fit - explicit learning-path hint

Definition: AI that works with more than one media type, such as text, images, audio, video, or code.

Related: generative-ai, diffusion, embedding, vector

Confusable: llm, diffusion

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, symbolic-ai, rule-based-ai

Distractor notes: Multimodal AI is useful as a distractor when learners might confuse it with llm, diffusion.

33. Rule-based AI (*rule-based-ai*)

Stage/card: 1. Before Morning / Where LLMs Fit - explicit learning-path hint

Definition: AI built around explicit rules, conditions, logic, or procedures written or configured by people.

Related: symbolic-ai, policy, guardrail, deep-learning

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Rule-based AI can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

34. Symbolic AI (*symbolic-ai*)

Stage/card: 1. Before Morning / Where LLMs Fit - explicit learning-path hint

Definition: An AI tradition based on explicit symbols, logic, and hand-written rules.

Related: rule-based-ai, rationalism, deep-learning, policy

Confusable: rule-based-ai, deep-learning

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, rule-based-ai

Distractor notes: Symbolic AI is useful as a distractor when learners might confuse it with rule-based-ai, deep-learning.

35. Embedding (embedding)

Stage/card: 2. Morning Commute / Embeddings - explicit learning-path hint

Definition: A learned starting vector retrieved for a token ID.

Related: embedding-table, token-id, vector, hidden state

Confusable: hidden state, definition, memory

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Embedding is useful as a distractor when learners might confuse it with hidden state, definition, memory.

36. Embedding table (embedding-table)

Stage/card: 2. Morning Commute / Embeddings - explicit learning-path hint

Definition: The learned table that maps token IDs to embedding vectors.

Related: embedding, token-id, training

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Embedding table can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

37. Forward pass (forward-pass)

Stage/card: 2. Morning Commute / Inference - explicit learning-path hint

Definition: One run through the model that turns the current context into next-token scores.

Related: inference, hidden state, logits

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Forward pass can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

38. Inference (inference)

Stage/card: 2. Morning Commute / Inference - explicit learning-path hint

Definition: Normal model use: fixed learned weights process the current context to produce scores for the next response token.

Related: forward-pass, hidden state, logits, sampling

Confusable: training, pretraining, fine-tuning

Nearby terms: llm, prompt, response, token, weight, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: Inference is useful as a distractor when learners might confuse it with training, pretraining, fine-tuning.

39. Completion (completion)

Stage/card: 2. Morning Commute / Prompt vs Response - explicit learning-path hint

Definition: The generated text that follows a prompt, assembled from response tokens.

Related: response, generated-token, autoregression

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Completion can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

40. Decoding step (decoding-step)

Stage/card: 2. Morning Commute / Prompt vs Response - explicit learning-path hint

Definition: The inference step that turns next-token scores into one chosen next token.

Related: logits, softmax, sampling

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Decoding step can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

41. Generated token (*generated-token*)

Stage/card: 2. Morning Commute / Prompt vs Response - explicit learning-path hint

Definition: One newly chosen response token appended to the context during generation.

Related: response-tokens, sampling, autoregression

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Generated token can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

42. Input context (*input-context*)

Stage/card: 2. Morning Commute / Prompt vs Response - explicit learning-path hint

Definition: Everything currently visible to the model: prompt tokens, conversation history, retrieved documents, and already-generated response tokens.

Related: context window, prompt-tokens, response-tokens, rag

Confusable: context window, prompt, memory, system-prompt

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Input context is useful as a distractor when learners might confuse it with context window, prompt, memory.

43. Model output (*model-output*)

Stage/card: 2. Morning Commute / Prompt vs Response - explicit learning-path hint

Definition: The visible response assembled from generated tokens.

Related: None listed

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Model output can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

44. Prompt (*prompt*)

Stage/card: 2. Morning Commute / Prompt vs Response - explicit learning-path hint

Definition: The input given to the model, including user text, system instructions, conversation history, retrieved material, or other context.

Related: prompt-tokens, input-context, response

Confusable: response, system-prompt, context window

Nearby terms: llm, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: Prompt is useful as a distractor when learners might confuse it with response, system-prompt, context window.

45. Prompt token (*prompt-tokens*)

Stage/card: 2. Morning Commute / Prompt vs Response - explicit learning-path hint

Definition: A token that is given to the model as part of the current context before the next response token is chosen.

Related: prompt, input-context, response-tokens

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Prompt token can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

46. Response (response)

Stage/card: 2. Morning Commute / Prompt vs Response - explicit learning-path hint

Definition: The sequence of tokens the model generates after processing the current context.

Related: response-tokens, generated-token, completion

Confusable: prompt, completion

Nearby terms: llm, prompt, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: Response is useful as a distractor when learners might confuse it with prompt, completion.

47. Response token (response-tokens)

Stage/card: 2. Morning Commute / Prompt vs Response - explicit learning-path hint

Definition: A token generated by the model during response generation.

Related: response, generated-token, autoregression

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Response token can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

48. Response-so-far (response-so-far)

Stage/card: 2. Morning Commute / Prompt vs Response - explicit learning-path hint

Definition: The generated response tokens currently visible in context before the next token is chosen.

Related: response, response-tokens, generated-token, autoregression, context window

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Response-so-far can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

49. Activation (activation)

Stage/card: 2. Morning Commute / Tensors - explicit learning-path hint

Definition: A temporary numerical value or pattern produced inside the model during a forward pass.

Related: hidden state, inference, tensor

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Activation can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

50. Tensor (tensor)

Stage/card: 2. Morning Commute / Tensors - explicit learning-path hint

Definition: An organized block of numbers, often holding many token vectors at once.

Related: vector, activation, weight-tensor

Confusable: vector, weight-tensor

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Tensor is useful as a distractor when learners might confuse it with vector, weight-tensor.

51. Weight tensor (weight-tensor)

Stage/card: 2. Morning Commute / Tensors - explicit learning-path hint

Definition: A shaped block of learned model parameters.

Related: weight, parameter, tensor

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Weight tensor can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

52. Token ID (token-id)

Stage/card: 2. Morning Commute / Token IDs - explicit learning-path hint

Definition: The number assigned to a token so the model can look up its learned starting vector.

Related: token, tokenizer, embedding-table

Confusable: token, embedding

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Token ID is useful as a distractor when learners might confuse it with token, embedding.

53. Token (token)

Stage/card: 2. Morning Commute / Tokenization - explicit learning-path hint

Definition: A model-readable chunk of text, such as a word, word piece, punctuation mark, space, or other tokenizer chunk.

Related: tokenization, tokenizer, token-id

Confusable: token-id, word

Nearby terms: llm, prompt, response, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai, rule-based-ai

Distractor notes: Token is useful as a distractor when learners might confuse it with token-id, word.

54. Tokenization (tokenization)

Stage/card: 2. Morning Commute / Tokenization - explicit learning-path hint

Definition: The step that splits text into model-readable tokens that can be represented as token IDs.

Related: tokenizer, token, token-id

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Tokenization can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

55. Tokenizer (tokenizer)

Stage/card: 2. Morning Commute / Tokenization - explicit learning-path hint

Definition: The rules and vocabulary that split text into tokens and map those tokens to IDs.

Related: tokenization, token, token-id

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Tokenizer can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

56. Distributed representation (distributed-representation)

Stage/card: 2. Morning Commute / Vectors - explicit learning-path hint

Definition: A representation where information is spread across many vector dimensions rather than stored in one neat slot.

Related: vector, feature, hidden state

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Distributed representation can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

57. Feature (feature)

Stage/card: 2. Morning Commute / Vectors - explicit learning-path hint

Definition: A learned numerical pattern that helps represent something useful for prediction.

Related: vector, distributed-representation, MLP

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Feature can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

58. Vector (vector)

Stage/card: 2. Morning Commute / Vectors - explicit learning-path hint

Definition: A list of numbers that represents features of something, such as a token.

Related: feature, distributed-representation, tensor

Confusable: tensor, embedding

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Vector is useful as a distractor when learners might confuse it with tensor, embedding.

59. Attention (attention)

Stage/card: 3. Workday / Attention - explicit learning-path hint

Definition: A mechanism that assigns relevance weights between token positions in the current context.

Related: relevance-weight, hidden state, MLP

Confusable: human attention, MLP, hidden state

Nearby terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion

Distractor notes: Attention is useful as a distractor when learners might confuse it with human attention, MLP, hidden state.

60. Relevance weight (relevance-weight)

Stage/card: 3. Workday / Attention - explicit learning-path hint

Definition: A temporary numerical strength showing how much one token position should use information from another position.

Related: attention, input-context

Confusable: None listed

Nearby terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion

Distractor notes: Relevance weight can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

61. Hidden state (hidden state)

Stage/card: 3. Workday / Hidden States - explicit learning-path hint

Definition: A temporary context-shaped internal vector for a token at a particular model layer.

Related: embedding, activation, temporary-activation, forward-pass

Confusable: embedding, memory, activation

Nearby terms: inference, forward-pass, input-context, weight, activation, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion, tokenization

Distractor notes: Hidden state is useful as a distractor when learners might confuse it with embedding, memory, activation.

62. Temporary activation (temporary-activation)

Stage/card: 3. Workday / Hidden States - explicit learning-path hint

Definition: A temporary internal numerical value or pattern created during the current forward pass.

Related: activation, hidden state, attention

Confusable: None listed

Nearby terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion

Distractor notes: Temporary activation can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

63. Layer (layer)

Stage/card: 3. Workday / Layers - explicit learning-path hint

Definition: A repeated transformer block that refines hidden states.

Related: transformer-layer, attention, MLP, hidden state

Confusable: None listed

Nearby terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion

Distractor notes: Layer can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

64. Normalization (normalization)

Stage/card: 3. Workday / Layers - explicit learning-path hint

Definition: A numerical adjustment that keeps internal values in a more stable range as they move through layers.

Related: transformer-layer, activation

Confusable: None listed

Nearby terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion

Distractor notes: Normalization can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

65. Residual path (residual-path)

Stage/card: 3. Workday / Layers - explicit learning-path hint

Definition: A connection that carries earlier information forward around a transformation inside a layer.

Related: transformer-layer, hidden state

Confusable: None listed

Nearby terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion

Distractor notes: Residual path can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

66. Transformer layer (transformer-layer)

Stage/card: 3. Workday / Layers - explicit learning-path hint

Definition: A repeated block in a transformer that applies attention, MLP transformations, residual paths, and normalization to hidden states.

Related: layer, attention, MLP

Confusable: None listed

Nearby terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion

Distractor notes: Transformer layer can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

67. Feed-forward network (feed-forward-network)

Stage/card: 3. Workday / MLP - explicit learning-path hint

Definition: A network where values move through learned transformations in one direction for the current computation.

Related: MLP, multi-layer-perceptron

Confusable: None listed

Nearby terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion

Distractor notes: Feed-forward network can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

68. MLP (MLP)

Stage/card: 3. Workday / MLP - explicit learning-path hint

Definition: A feed-forward network inside a transformer layer that reshapes each token position's feature vector.

Related: multi-layer-perceptron, feed-forward-network, feature, hidden state

Confusable: attention, neural network

Nearby terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion

Distractor notes: MLP is useful as a distractor when learners might confuse it with attention, neural network.

69. Multi-layer perceptron (multi-layer-perceptron)

Stage/card: 3. Workday / MLP - explicit learning-path hint

Definition: The expanded name for MLP: a small stack of learned transformations applied to a vector.

Related: MLP, feed-forward-network

Confusable: None listed

Nearby terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling, prompt, response, prompt-tokens, response-tokens, generated-token, completion

Distractor notes: Multi-layer perceptron can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

70. Logits (logits)

Stage/card: 4. Decision Room / Logits - explicit learning-path hint

Definition: Raw scores for possible next tokens before probabilities.

Related: hidden state, softmax, probability

Confusable: probability, softmax

Nearby terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer, MLP, multi-layer-perceptron, feed-forward-network, feature, transformer-layer, residual-path, normalization

Distractor notes: Logits is useful as a distractor when learners might confuse it with probability, softmax.

71. Sampling (sampling)

Stage/card: 4. Decision Room / Sampling - explicit learning-path hint

Definition: The decoding step that chooses one generated token from next-token probabilities.

Related: probability, generated-token, temperature

Confusable: softmax, temperature

Nearby terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer, MLP, multi-layer-perceptron, feed-forward-network, feature, transformer-layer, residual-path, normalization

Distractor notes: Sampling is useful as a distractor when learners might confuse it with softmax, temperature.

72. Temperature (temperature)

Stage/card: 4. Decision Room / Sampling - explicit learning-path hint

Definition: A decoding setting that changes how sharp or spread out next-token probabilities feel during sampling.

Related: sampling, probability, softmax

Confusable: None listed

Nearby terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer, MLP, multi-layer-perceptron, feed-forward-network, feature, transformer-layer, residual-path, normalization

Distractor notes: Temperature can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

73. Top-k (top-k)

Stage/card: 4. Decision Room / Sampling - explicit learning-path hint

Definition: An optional sampling setting that keeps only the k most likely next-token candidates.

Related: sampling, temperature, top-p

Confusable: None listed

Nearby terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer, MLP, multi-layer-perceptron, feed-forward-network, feature, transformer-layer, residual-path, normalization

Distractor notes: Top-k can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

74. Top-p (top-p)

Stage/card: 4. Decision Room / Sampling - explicit learning-path hint

Definition: An optional sampling setting that keeps the smallest set of likely tokens whose probabilities add up to a chosen threshold.

Related: sampling, temperature, top-k

Confusable: None listed

Nearby terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer, MLP, multi-layer-perceptron, feed-forward-network, feature, transformer-layer, residual-path, normalization

Distractor notes: Top-p can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

75. Probability (probability)

Stage/card: 4. Decision Room / Softmax - explicit learning-path hint

Definition: A model-assigned chance for a candidate next token, not a truth guarantee.

Related: softmax, sampling, hallucination

Confusable: None listed

Nearby terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer, MLP, multi-layer-perceptron, feed-forward-network, feature, transformer-layer, residual-path, normalization

Distractor notes: Probability can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

76. Softmax (softmax)

Stage/card: 4. Decision Room / Softmax - explicit learning-path hint

Definition: A function that turns next-token raw scores into probabilities that sum to one.

Related: logits, probability, sampling

Confusable: logits, sampling

Nearby terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer, MLP, multi-layer-perceptron, feed-forward-network, feature, transformer-layer, residual-path, normalization

Distractor notes: Softmax is useful as a distractor when learners might confuse it with logits, sampling.

77. Autoregression (autoregression)

Stage/card: 5. The Day Repeats / Autoregression - explicit learning-path hint

Definition: Generating a response by choosing one token, appending it, and repeating.

Related: None listed

Confusable: completion, diffusion

Nearby terms: logits, hidden state, next-token, generated-token, softmax, probability, decoding-step, sampling, response-tokens, temperature, top-k, top-p, completion, response-so-far

Distractor notes: Autoregression is useful as a distractor when learners might confuse it with completion, diffusion.

78. Context window (context window)

Stage/card: 5. The Day Repeats / Context Window - explicit learning-path hint

Definition: The limited amount of tokens or media the model can consider at once.

Related: None listed

Confusable: memory, training-data, rag

Nearby terms: logits, hidden state, next-token, generated-token, softmax, probability, decoding-step, sampling, response-tokens, temperature, top-k, top-p, autoregression, completion

Distractor notes: Context window is useful as a distractor when learners might confuse it with memory, training-data, rag.

79. Memory (memory)

Stage/card: 5. The Day Repeats / Context Window - explicit learning-path hint

Definition: A durable store of information available later.

Related: None listed

Confusable: None listed

Nearby terms: logits, hidden state, next-token, generated-token, softmax, probability, decoding-step, sampling, response-tokens, temperature, top-k, top-p, autoregression, completion

Distractor notes: Memory can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

80. Citation (citation)

Stage/card: 5. The Day Repeats / Grounding - explicit learning-path hint

Definition: A pointer to a source, document, passage, data item, or tool result that can support checking a claim.

Related: grounding, rag, retrieval, hallucination

Confusable: None listed

Nearby terms: logits, hidden state, next-token, generated-token, softmax, probability, decoding-step, sampling, response-tokens, temperature, top-k, top-p, autoregression, completion

Distractor notes: Citation can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

81. Grounding (grounding)

Stage/card: 5. The Day Repeats / Grounding - explicit learning-path hint

Definition: Connecting a model response to available evidence, such as retrieved documents, cited sources, data, or tool results.

Related: None listed

Confusable: citation, truth, rag

Nearby terms: logits, hidden state, next-token, generated-token, softmax, probability, decoding-step, sampling, response-tokens, temperature, top-k, top-p, autoregression, completion

Distractor notes: Grounding is useful as a distractor when learners might confuse it with citation, truth, rag.

82. Hallucination (hallucination)

Stage/card: 5. The Day Repeats / Hallucinations - explicit learning-path hint

Definition: A confident model output that is unsupported, incorrect, or fabricated.

Related: None listed

Confusable: lying, uncertainty

Nearby terms: logits, hidden state, next-token, generated-token, softmax, probability, decoding-step, sampling, response-tokens, temperature, top-k, top-p, autoregression, completion

Distractor notes: Hallucination is useful as a distractor when learners might confuse it with lying, uncertainty.

83. RAG (rag)

Stage/card: 5. The Day Repeats / RAG and Retrieval - explicit learning-path hint

Definition: Retrieval-augmented generation: retrieving outside information and placing it into the model context before generation.

Related: None listed

Confusable: retrieval, grounding, fine-tuning, training

Nearby terms: logits, hidden state, next-token, generated-token, softmax, probability, decoding-step, sampling, response-tokens, temperature, top-k, top-p, autoregression, completion

Distractor notes: RAG is useful as a distractor when learners might confuse it with retrieval, grounding, fine-tuning.

84. Retrieval (retrieval)

Stage/card: 5. The Day Repeats / RAG and Retrieval - explicit learning-path hint

Definition: Finding external material and placing it into the current context for the model to use.

Related: None listed

Confusable: training, memory

Nearby terms: logits, hidden state, next-token, generated-token, softmax, probability, decoding-step, sampling, response-tokens, temperature, top-k, top-p, autoregression, completion

Distractor notes: Retrieval is useful as a distractor when learners might confuse it with training, memory.

85. Denoising (denoising)

Stage/card: 6. Twilight: The Wider Landscape / Diffusion vs Autoregression - explicit learning-path hint

Definition: A generation process that repeatedly refines a noisy pattern toward a clearer output.

Related: diffusion, generative-ai, representation

Confusable: None listed

Nearby terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference, context window, retrieval, prompt-tokens, memory, rag, grounding

Distractor notes: Denoising can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

86. In-context learning (in-context learning)

Stage/card: 6. Twilight: The Wider Landscape / How AI Learns - explicit learning-path hint

Definition: Behavior change caused by examples or instructions inside the prompt during inference.

Related: None listed

Confusable: None listed

Nearby terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference, context window, retrieval, prompt-tokens, memory, rag, grounding

Distractor notes: In-context learning can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

87. Representation (representation)

Stage/card: 6. Twilight: The Wider Landscape / Multimodal AI - explicit learning-path hint

Definition: A model-readable form of information, often numerical, that lets a system process text, images, audio, code, or other inputs.

Related: embedding, vector, multimodal

Confusable: None listed

Nearby terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference, context window, retrieval, prompt-tokens, memory, rag, grounding

Distractor notes: Representation can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

88. Compute (compute)

Stage/card: 6. Twilight: The Wider Landscape / The Perfect Storm - explicit learning-path hint

Definition: The processing power used to train, run, or evaluate AI systems.

Related: data-center, energy-use, efficient-inference

Confusable: None listed

Nearby terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference, context window, retrieval, prompt-tokens, memory, rag, grounding

Distractor notes: Compute can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

89. Data center (data-center)

Stage/card: 6. Twilight: The Wider Landscape / The Perfect Storm - explicit learning-path hint

Definition: A facility that houses servers, networking, power, and cooling for digital services, including AI workloads.

Related: compute, energy-use, water-use, carbon-emissions

Confusable: None listed

Nearby terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference, context window, retrieval, prompt-tokens, memory, rag, grounding

Distractor notes: Data center can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

90. Economic incentives (economic-incentives)

Stage/card: 6. Twilight: The Wider Landscape / The Perfect Storm - explicit learning-path hint

Definition: Money, competition, demand, and platform pressures that encourage organizations to build, deploy, and scale AI systems.

Related: perfect-storm-term, compute, human-feedback-labor

Confusable: None listed

Nearby terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference, context window, retrieval, prompt-tokens, memory, rag, grounding

Distractor notes: Economic incentives can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

91. Human feedback labor (human-feedback-labor)

Stage/card: 6. Twilight: The Wider Landscape / The Perfect Storm - explicit learning-path hint

Definition: Human work used to label data, rank outputs, evaluate behavior, or make model responses more useful.

Related: human-feedback-learning, alignment, training-data

Confusable: None listed

Nearby terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference, context window, retrieval, prompt-tokens, memory, rag, grounding

Distractor notes: Human feedback labor can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

92. Perfect storm (perfect-storm-term)

Stage/card: 6. Twilight: The Wider Landscape / The Perfect Storm - explicit learning-path hint

Definition: A convergence of data, compute, storage, deep-learning methods, human labor, and incentives that made modern LLMs possible.

Related: training-data, compute, data-center, deep-learning

Confusable: None listed

Nearby terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference, context window, retrieval, prompt-tokens, memory, rag, grounding

Distractor notes: Perfect storm can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

93. Storage (storage)

Stage/card: 6. Twilight: The Wider Landscape / The Perfect Storm - explicit learning-path hint

Definition: Digital capacity used to hold datasets, model checkpoints, logs, indexes, and other infrastructure needed around AI systems.

Related: training-data, model-checkpoint, data-center

Confusable: None listed

Nearby terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference, context window, retrieval, prompt-tokens, memory, rag, grounding

Distractor notes: Storage can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

94. Collective intelligence (collective-intelligence-term)

Stage/card: 7. Midnight Ledger / Collective Intelligence, Extracted - explicit learning-path hint

Definition: The shared human knowledge, language, art, code, research, documentation, and culture that leave patterns models can learn from.

Related: training-data, data-provenance, copyright, compensation

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Collective intelligence can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

95. Compensation (compensation)

Stage/card: 7. Midnight Ledger / Collective Intelligence, Extracted - explicit learning-path hint

Definition: Payment, credit, licensing, or other return for the use of someone's labor, data, or creative work.

Related: copyright, consent, data-provenance

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Compensation can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

96. Consent (consent)

Stage/card: 7. Midnight Ledger / Collective Intelligence, Extracted - explicit learning-path hint

Definition: Permission from people or creators for a specific use of their data or work.

Related: data-provenance, copyright, compensation

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Consent can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

97. Copyright (copyright)

Stage/card: 7. Midnight Ledger / Collective Intelligence, Extracted - explicit learning-path hint

Definition: Legal rights that protect certain creative works and shape how they may be copied, licensed, transformed, or used.

Related: data-provenance, consent, compensation

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Copyright can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

98. Data provenance (data-provenance)

Stage/card: 7. Midnight Ledger / Collective Intelligence, Extracted - explicit learning-path hint

Definition: Information about where data came from, how it was collected, and what rights or limits apply to its use.

Related: training-data, consent, copyright, governance

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Data provenance can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

99. Bias (bias)

Stage/card: 7. Midnight Ledger / Costs We Must Count - explicit learning-path hint

Definition: A pattern of unfair, skewed, or systematically harmful behavior in data, model outputs, or deployment.

Related: governance, human-review, responsible-ai

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Bias can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

100. Carbon emissions (carbon-emissions)

Stage/card: 7. Midnight Ledger / Costs We Must Count - explicit learning-path hint

Definition: Climate-warming emissions associated with the energy and supply chains used by AI systems.

Related: energy-use, data-center, responsible-ai

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Carbon emissions can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

101. Deskilling (deskilling)

Stage/card: 7. Midnight Ledger / Costs We Must Count - explicit learning-path hint

Definition: Loss or weakening of human skills when tools take over too much of the practice, judgment, or feedback loop.

Related: labor-disruption, human-review, human-centered-ai-term

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Deskilling can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

102. E-waste (e-waste)

Stage/card: 7. Midnight Ledger / Costs We Must Count - explicit learning-path hint

Definition: Discarded electronic equipment and hardware that can create material, pollution, and disposal burdens.

Related: environmental-footprint, data-center

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: E-waste can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

103. Energy use (energy-use)

Stage/card: 7. Midnight Ledger / Costs We Must Count - explicit learning-path hint

Definition: The electricity required to train, run, cool, and support an AI system.

Related: compute, data-center, carbon-emissions

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Energy use can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

104. Environmental footprint (environmental-footprint)

Stage/card: 7. Midnight Ledger / Costs We Must Count - explicit learning-path hint

Definition: The environmental effects associated with an AI system, including energy, water, carbon, land, hardware, and waste impacts.

Related: data-center, energy-use, water-use, carbon-emissions

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Environmental footprint can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

105. Information pollution (information-pollution)

Stage/card: 7. Midnight Ledger / Costs We Must Count - explicit learning-path hint

Definition: Low-quality, misleading, synthetic, or unsupported content that makes an information environment harder to trust.

Related: hallucination, source-review, governance

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Information pollution can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

106. Labor disruption (labor-disruption)

Stage/card: 7. Midnight Ledger / Costs We Must Count - explicit learning-path hint

Definition: Changes to jobs, tasks, wages, working conditions, or power caused by AI deployment.

Related: deskilling, human-centered-ai-term, governance

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Labor disruption can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

107. Water use (water-use)

Stage/card: 7. Midnight Ledger / Costs We Must Count - explicit learning-path hint

Definition: Water that may be used directly or indirectly for cooling and electricity generation around AI infrastructure.

Related: data-center, energy-use, governance

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Water use can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

108. Concentration of power (concentration-of-power)

Stage/card: 7. Midnight Ledger / Risk vs Myth - explicit learning-path hint

Definition: A situation where data, compute, capital, or platform control is held by a small number of actors.

Related: governance, data-center, responsible-ai

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Concentration of power can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

109. Overreliance (overreliance)

Stage/card: 7. Midnight Ledger / Risk vs Myth - explicit learning-path hint

Definition: Depending on AI output too much without enough human review, evidence checking, or local judgment.

Related: human-review, uncertainty, accountability

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Overreliance can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

110. Privacy (privacy)

Stage/card: 7. Midnight Ledger / Risk vs Myth - explicit learning-path hint

Definition: The practice of protecting sensitive data from exposure in prompts, logs, outputs, and integrations.

Related: None listed

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Privacy can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

111. Prompt Injection (prompt-injection)

Stage/card: 7. Midnight Ledger / Risk vs Myth - explicit learning-path hint

Definition: An attack or failure mode where malicious or conflicting text tries to override intended instructions.

Related: None listed

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Prompt Injection can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

112. Risk literacy (risk-literacy)

Stage/card: 7. Midnight Ledger / Risk vs Myth - explicit learning-path hint

Definition: The practice of separating mechanism-based AI risks from myths or magical stories about what models are.

Related: privacy, prompt-injection, governance

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Risk literacy can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

113. Tool use (tool-use)

Stage/card: 7. Midnight Ledger / Risk vs Myth - explicit learning-path hint

Definition: A system setup where an AI model can call external tools, APIs, files, databases, or actions.

Related: prompt-injection, privacy, governance

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Tool use can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

114. Vendor lock-in (vendor-lock-in)

Stage/card: 7. Midnight Ledger / Risk vs Myth - explicit learning-path hint

Definition: Dependence on one provider, platform, format, or contract that makes switching difficult.

Related: governance, responsible-ai, accountability

Confusable: None listed

Nearby terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference, diffusion, autoregression, generative-ai, sampling

Distractor notes: Vendor lock-in can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

115. Accessibility (accessibility)

Stage/card: 8. New Dawn / Benefits Worth Taking Seriously - explicit learning-path hint

Definition: Designing tools and materials so more people can perceive, understand, navigate, and participate.

Related: translation, summarization, human-review

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Accessibility can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

116. Benefits (benefits)

Stage/card: 8. New Dawn / Benefits Worth Taking Seriously - explicit learning-path hint

Definition: Useful help a model or AI system can provide when the task, evidence, limits, and human review fit the situation.

Related: accessibility, translation, summarization, human-review

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Benefits can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

117. Summarization (summarization)

Stage/card: 8. New Dawn / Benefits Worth Taking Seriously - explicit learning-path hint

Definition: Condensing longer material into a shorter version that preserves key points for a purpose.

Related: grounding, rag, human-review

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Summarization can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

118. Translation (translation)

Stage/card: 8. New Dawn / Benefits Worth Taking Seriously - explicit learning-path hint

Definition: Rendering meaning from one language or mode into another.

Related: multimodal, accessibility, human-review

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Translation can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

119. Efficient inference (efficient-inference)

Stage/card: 8. New Dawn / Better AI Is a Choice - explicit learning-path hint

Definition: Running a model in ways that use less time, hardware, energy, or money for the task.

Related: compute, model-distillation, responsible-ai

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Efficient inference can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

120. Governance (governance)

Stage/card: 8. New Dawn / Better AI Is a Choice - explicit learning-path hint

Definition: The policies, processes, oversight, evaluation, and accountability structures that guide AI use.

Related: policy, human-review, responsible-ai

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Governance can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

121. Model distillation (model-distillation)

Stage/card: 8. New Dawn / Better AI Is a Choice - explicit learning-path hint

Definition: A method where a smaller model is trained to imitate useful behavior from a larger model.

Related: efficient-inference, training, responsible-ai

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Model distillation can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

122. Responsible AI (responsible-ai)

Stage/card: 8. New Dawn / Better AI Is a Choice - explicit learning-path hint

Definition: AI designed, deployed, and evaluated with attention to benefits, risks, rights, evidence, accountability, and affected communities.

Related: governance, human-centered-ai-term, human-review

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Responsible AI can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

123. Effective prompting (effective-prompting)

Stage/card: 8. New Dawn / Effective Prompting from Model Literacy - explicit learning-path hint

Definition: Building a useful current context with a clear task, constraints, examples, evidence needs, uncertainty request, and output format.

Related: prompt, context window, human-review, uncertainty

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Effective prompting can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

124. Human review (human-review)

Stage/card: 8. New Dawn / Effective Prompting from Model Literacy - explicit learning-path hint

Definition: A person checking, interpreting, editing, approving, or rejecting AI output before it is used.

Related: human-centered-ai-term, governance, uncertainty

Confusable: evaluation, responsible-ai

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Human review is useful as a distractor when learners might confuse it with evaluation, responsible-ai.

125. Source review (source-review)

Stage/card: 8. New Dawn / Effective Prompting from Model Literacy - explicit learning-path hint

Definition: Checking claims against credible evidence before treating them as reliable teaching content.

Related: grounding, citation, governance

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Source review can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

126. Uncertainty (uncertainty)

Stage/card: 8. New Dawn / Effective Prompting from Model Literacy - explicit learning-path hint

Definition: A signal that the answer may be incomplete, unsupported, ambiguous, or in need of verification.

Related: grounding, hallucination, human-review

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Uncertainty can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

127. Accountability (accountability)

Stage/card: 8. New Dawn / Human-Centered AI - explicit learning-path hint

Definition: Clear responsibility for decisions, harms, corrections, and oversight around an AI system.

Related: human-review, governance, responsible-ai

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste, labor-disruption

Distractor notes: Accountability can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

128. Common good (common-good)

Stage/card: 8. New Dawn / Human-Centered AI - explicit learning-path hint

Definition: The shared conditions that help people and communities flourish.

Related: human-centered-ai-term, governance, responsible-ai

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Common good can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

129. Dignity (dignity)

Stage/card: 8. New Dawn / Human-Centered AI - explicit learning-path hint

Definition: The worth of persons that should not be reduced to speed, profit, prediction, or automation.

Related: human-centered-ai-term, common-good, governance

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Dignity can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

130. Human-centered AI (human-centered-ai-term)

Stage/card: 8. New Dawn / Human-Centered AI - explicit learning-path hint

Definition: An approach that judges AI by how well it supports human dignity, learning, responsibility, creativity, relationships, and the common good.

Related: dignity, common-good, human-review, governance

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Human-centered AI can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

131. Model literacy (model-literacy)

Stage/card: 8. New Dawn / Model Literate Synthesis - explicit learning-path hint

Definition: Understanding enough about how models work and how they affect people to use and govern them wisely.

Related: training, inference, context window, responsible-ai

Confusable: None listed

Nearby terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability, environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste

Distractor notes: Model literacy can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

132. Model Checkpoint (model-checkpoint)

Stage/card: . Related / Foundational / - inferred or unmapped

Definition: A saved version of model parameters at a point in training or fine-tuning.

Related: None listed

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Model Checkpoint can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

133. Neural network (neural network)

Stage/card: . Related / Foundational / - inferred or unmapped

Definition: A layered model made of learned numerical transformations.

Related: None listed

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: Neural network can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

134. System prompt (system-prompt)

Stage/card: . Related / Foundational / - inferred or unmapped

Definition: A high-priority instruction supplied by the AI system to steer behavior during the current run.

Related: alignment, guardrail, in-context learning

Confusable: None listed

Nearby terms: llm, prompt, response, token, weight, inference, ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, diffusion, multimodal, symbolic-ai

Distractor notes: System prompt can be used as a same-stage or nearby-stage distractor when the question asks for a more specific mechanism.

Learning-Card Review Sections

1. What Is an LLM?

Stage: 1. Before Morning | **Card ID:** what-is-llm | **Card in stage:** 1

Primary learning objective: Explain What Is an LLM? as part of the day in the life of a prompt.

Secondary objectives: A large language model is a learned system that turns current context into probabilities for the next token., This card describes the whole model lifecycle, but especially normal inference: current context to next-token probabilities to generated response token., The weights were changed during training or fine-tuning., Prompt tokens are given to the model., Next, we place LLMs inside the wider AI family: AI, machine learning, generative AI, and LLMs.

Key distinctions: The weights were changed during training or fine-tuning., Prompt tokens are given to the model., An LLM is a conscious reader or a database.

Misconceptions to correct: An LLM is a conscious reader or a database.

Related glossary terms: llm, prompt, response, token, weight, inference

Nearby glossary distractor candidates: alignment, adapter, instruction-tuning, fine-tuning, human-feedback-learning, rlhf, preference-optimization, policy, guardrail, evaluation

Why this matters: This definition lowers the mystery. The model can be powerful without being a mind, database, or hand-written rulebook.

Existing Live Checkpoint

Question count: 1

Question: Which description is most accurate?

Correct: A learned prediction system

- A learned prediction system (correct)
- A conscious reader Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook.
- A hand-written rulebook Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook.
- A search engine that knows every file Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook.

Proposed Draft Question Bank

Proposed count: 2. **Rationale:** Light or narrow concept card; two checks should confirm the core distinction without overloading the learner.

Question 1: Which description best fits What Is an LLM in this learning card?

ID: draft-what-is-llm-q1 | **Type:** definition | **Correct choice ID:** choice-draft-what-is-llm-q1-correct

Misconception tags: None listed

Glossary terms: llm, prompt, response, token, weight, inference

choice-draft-what-is-llm-q1-correct [CORRECT]: A learned prediction system

Feedback: An LLM uses learned weights to turn context into next-token probabilities.

choice-draft-what-is-llm-q1-a-conscious-reader [DISTRACTOR]: A conscious reader

Feedback: Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook.

Misconception: A conscious reader | **Represented term:** llm | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-what-is-llm-q1-a-hand-written-rulebook [DISTRACTOR]: A hand-written rulebook

Feedback: Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook.

Misconception: A hand-written rulebook | **Represented term:** symbolic-ai | **Source:** same-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-what-is-llm-q1-a-search-engine-that-knows-every-file [DISTRACTOR]: A search engine that knows every file

Feedback: Not quite. The model is powerful, but it is not conscious, not a database, and not a hand-coded rulebook.

Misconception: A search engine that knows every file | **Represented term:** rag | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does What Is an LLM? most directly teach?

ID: draft-what-is-llm-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-what-is-llm-q2-correct

Misconception tags: None listed

Glossary terms: llm, prompt, response, token, weight, inference

choice-draft-what-is-llm-q2-correct [CORRECT]: The weights were changed during training or fine-tuning.

Feedback: Good distinction. The weights were changed during training or fine-tuning.

choice-draft-what-is-llm-q2-rag-1 [DISTRACTOR]: It means the model permanently learned the retrieved document.

Feedback: This choice points toward RAG. RAG matters nearby, but this question is asking about What Is an LLM?.

Misconception: Not training, fine-tuning, permanent memory, or a guarantee that the answer is true. | Represented term: rag | Source: explicitly confusable glossary metadata

Rationale: Tempting because RAG is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific What Is an LLM? mechanism or distinction.

choice-draft-what-is-llm-q2-deep-learning-2 [DISTRACTOR]: It means a hand-written rulebook is doing the whole task.

Feedback: This choice points toward Deep learning. Deep learning matters nearby, but this question is asking about What Is an LLM?.

Misconception: Not magic, not all machine learning, and not a hand-coded rulebook. | Represented term: deep-learning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Deep learning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific What Is an LLM? mechanism or distinction.

choice-draft-what-is-llm-q2-autoregression-3 [DISTRACTOR]: It denoises a whole image from noise.

Feedback: This choice points toward Autoregression. Autoregression matters nearby, but this question is asking about What Is an LLM?.

Misconception: Autoregression is being treated as the same idea as What Is an LLM?. | Represented term: autoregression | Source: explicitly confusable glossary metadata

Rationale: Tempting because Autoregression is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific What Is an LLM? mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

2. Where LLMs Fit

Stage: 1. Before Morning | **Card ID:** where-llms-fit | **Card in stage:** 2

Primary learning objective: Explain Where LLMs Fit as part of the day in the life of a prompt.

Secondary objectives: A large language model is one kind of generative AI system, built with deep-learning methods inside the broader field of artificial intelligence., This is a map of AI categories, not a step inside one model run., This card describes families of systems., LLMs generate response tokens; other generative systems may generate pixels, audio, video frames, or other outputs., The first card explains what an LLM is.

Key distinctions: This card describes families of systems., LLMs generate response tokens; other generative systems may generate pixels, audio, video frames, or other outputs., All AI is an LLM, or all generative AI works like ChatGPT.

Misconceptions to correct: All AI is an LLM, or all generative AI works like ChatGPT.

Related glossary terms: ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, llm, diffusion, multimodal, symbolic-ai, rule-based-ai, foundation-model

Nearby glossary distractor candidates: rationalism, empiricism, loss

Why this matters: Learners should not treat "AI," "machine learning," "generative AI," and "LLM" as interchangeable words.

Existing Live Checkpoint

Question count: 1

Question: Which statement is most accurate?

Correct: An LLM is one kind of generative AI inside the broader AI family.

- An LLM is one kind of generative AI inside the broader AI family. (correct)
- All AI systems are LLMs. Not quite. Many AI systems are not large language models.
- Diffusion models and LLMs work the same way. Not quite. Diffusion usually denoises patterns, while LLM text generation is autoregressive.
- Machine learning and AI mean exactly the same thing. Not quite. Machine learning is a major branch within the broader AI field.

Proposed Draft Question Bank

Proposed count: 2. **Rationale:** Light or narrow concept card; two checks should confirm the core distinction without overloading the learner.

Question 1: Which statement best explains Where LLMs Fit in this learning card?

ID: draft-where-llms-fit-q1 | **Type:** definition | **Correct choice ID:** choice-draft-where-llms-fit-q1-correct

Misconception tags: None listed

Glossary terms: ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, llm, diffusion, multimodal, symbolic-ai, rule-based-ai, foundation-model

choice-draft-where-llms-fit-q1-correct [CORRECT]: An LLM is one kind of generative AI inside the broader AI family.

Feedback: Insight unlocked. LLMs are one important branch, not the whole AI forest.

choice-draft-where-llms-fit-q1-all-ai-systems-are-llms [DISTRACTOR]: All AI systems are LLMs.

Feedback: Not quite. Many AI systems are not large language models.

Misconception: All AI systems are LLMs. | Represented term: ai | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-where-llms-fit-q1-diffusion-models-and-llms-work-the-same-way [DISTRACTOR]: Diffusion models and LLMs work the same way.

Feedback: Not quite. Diffusion usually denoises patterns, while LLM text generation is autoregressive.

Misconception: Diffusion models and LLMs work the same way. | Represented term: diffusion | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-where-llms-fit-q1-machine-learning-and-ai-mean-exactly-the-same-thing [DISTRACTOR]: Machine learning and AI mean exactly the same thing.

Feedback: Not quite. Machine learning is a major branch within the broader AI field.

Misconception: Machine learning and AI mean exactly the same thing. | Represented term: machine-learning | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Where LLMs Fit most directly teach?

ID: draft-where-llms-fit-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-where-llms-fit-q2-correct

Misconception tags: None listed

Glossary terms: ai, machine-learning, classical-machine-learning, deep-learning, generative-ai, llm, diffusion, multimodal, symbolic-ai, rule-based-ai, foundation-model

choice-draft-where-llms-fit-q2-correct [CORRECT]: This card describes families of systems.

Feedback: Good distinction. This card describes families of systems.

choice-draft-where-llms-fit-q2-parameter-1 [DISTRACTOR]: It is mainly about Parameter, not the mechanism named in Where LLMs Fit.

Feedback: This choice points toward Parameter. Parameter matters nearby, but this question is asking about Where LLMs Fit.

Misconception: Often used interchangeably with weights in casual explanations. | Represented term: parameter | Source: explicitly confusable glossary metadata

Rationale: Tempting because Parameter is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Where LLMs Fit mechanism or distinction.

choice-draft-where-llms-fit-q2-sampling-2 [DISTRACTOR]: It writes the whole answer at once.

Feedback: This choice points toward Sampling. Sampling matters nearby, but this question is asking about Where LLMs Fit.

Misconception: Not writing the whole response, permanently teaching the model, or proving truth. | Represented term: sampling | Source: explicitly confusable glossary metadata

Rationale: Tempting because Sampling is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Where LLMs Fit mechanism or distinction.

choice-draft-where-llms-fit-q2-rag-3 [DISTRACTOR]: It means the model permanently learned the retrieved document.

Feedback: This choice points toward RAG. RAG matters nearby, but this question is asking about Where LLMs Fit.

Misconception: Not training, fine-tuning, permanent memory, or a guarantee that the answer is true. | Represented term: rag | Source: explicitly confusable glossary metadata

Rationale: Tempting because RAG is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Where LLMs Fit mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

3. Rationalists vs Empiricists

Stage: 1. Before Morning | **Card ID:** history | **Card in stage:** 3

Primary learning objective: Explain Rationalists vs Empiricists as part of the day in the life of a prompt.

Secondary objectives: Symbolic AI tries to reason with explicit rules and symbols; deep learning learns useful patterns from data by adjusting weights., This is historical and architectural background., Deep learning changes weights during training., A prompt is not matched against one giant rulebook., If LLMs are learned systems, then the next question is: how are they learned?

Key distinctions: Deep learning changes weights during training., A prompt is not matched against one giant rulebook., LLMs are hand-coded rulebooks, or symbolic AI is obsolete.

Misconceptions to correct: LLMs are hand-coded rulebooks, or symbolic AI is obsolete.

Related glossary terms: rationalism, empiricism, symbolic-ai, deep-learning, training, loss, weight

Nearby glossary distractor candidates: foundation-model, rule-based-ai, multimodal, diffusion, generative-ai, training-data, classical-machine-learning, machine-learning, parameter, ai, weight-update

Why this matters: This helps learners stop expecting LLMs to behave like ordinary hand-written programs.

Existing Live Checkpoint

Question count: 1

Question: Which phrase best describes the LLM tradition?

Correct: Learned patterns from data

- Learned patterns from data (correct)
- Only fixed if-then rules Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights.
- A spreadsheet macro Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights.
- A conscious reasoning engine Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: Which phrase best describes how LLMs use learned patterns rather than only hand-written rules?

ID: draft-history-q1 | **Type:** definition | **Correct choice ID:** choice-draft-history-q1-correct

Misconception tags: None listed

Glossary terms: rationalism, empiricism, symbolic-ai, deep-learning, training, loss, weight

choice-draft-history-q1-correct [CORRECT]: Learned patterns from data

Feedback: LLMs are learned prediction systems shaped by data.

choice-draft-history-q1-only-fixed-if-then-rules [DISTRACTOR]: Only fixed if-then rules

Feedback: Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights.

Misconception: Only fixed if-then rules | **Represented term:** symbolic-ai | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-history-q1-a-spreadsheet-macro [DISTRACTOR]: A spreadsheet macro

Feedback: Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights.

Misconception: A spreadsheet macro | **Represented term:** symbolic-ai | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-history-q1-a-conscious-reasoning-engine [DISTRACTOR]: A conscious reasoning engine

Feedback: Not quite. Rules still matter in software, but LLM fluency mostly comes from learned weights.

Misconception: A conscious reasoning engine | **Represented term:** llm | **Source:** same-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Rationalists vs Empiricists most directly teach?

ID: draft-history-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-history-q2-correct

Misconception tags: None listed

Glossary terms: rationalism, empiricism, symbolic-ai, deep-learning, training, loss, weight

choice-draft-history-q2-correct [CORRECT]: Deep learning changes weights during training.

Feedback: Good distinction. Deep learning changes weights during training.

choice-draft-history-q2-machine-learning-1 [DISTRACTOR]: It means all AI and machine learning are exactly the same thing.

Feedback: This choice points toward Machine learning. Machine learning matters nearby, but this question is asking about Rationalists vs Empiricists.

Misconception: Not exactly the same thing as all AI. | Represented term: machine-learning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Machine learning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Rationalists vs Empiricists mechanism or distinction.

choice-draft-history-q2-neural-network-2 [DISTRACTOR]: It is mainly about Neural network, not the mechanism named in Rationalists vs Empiricists.

Feedback: This choice points toward Neural network. Neural network matters nearby, but this question is asking about Rationalists vs Empiricists.

Misconception: Not a biological brain. | Represented term: neural network | Source: explicitly confusable glossary metadata

Rationale: Tempting because Neural network is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Rationalists vs Empiricists mechanism or distinction.

choice-draft-history-q2-generative-ai-3 [DISTRACTOR]: It is mainly about Generative AI, not the mechanism named in Rationalists vs Empiricists.

Feedback: This choice points toward Generative AI. Generative AI matters nearby, but this question is asking about Rationalists vs Empiricists.

Misconception: Not automatically an LLM and not automatically truthful. | Represented term: generative-ai | Source: explicitly confusable glossary metadata

Rationale: Tempting because Generative AI is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Rationalists vs Empiricists mechanism or distinction.

Question 3: Which common confusion does Rationalists vs Empiricists help learners avoid?

ID: draft-history-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-history-q3-correct

Misconception tags: None listed

Glossary terms: rationalism, empiricism, symbolic-ai, deep-learning, training, loss, weight

choice-draft-history-q3-correct [CORRECT]: It corrects the misconception: LLMs are hand-coded rulebooks, or symbolic AI is obsolete.

Feedback: Good distinction. Rationalists vs Empiricists is clearer when that misconception is separated from the mechanism.

choice-draft-history-q3-neural-network-1 [DISTRACTOR]: It is mainly about Neural network, not the mechanism named in Rationalists vs Empiricists.

Feedback: This choice points toward Neural network. Neural network matters nearby, but this question is asking about Rationalists vs Empiricists.

Misconception: Not a biological brain. | Represented term: neural network | Source: explicitly confusable glossary metadata

Rationale: Tempting because Neural network is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Rationalists vs Empiricists mechanism or distinction.

choice-draft-history-q3-generative-ai-2 [DISTRACTOR]: It is mainly about Generative AI, not the mechanism named in Rationalists vs Empiricists.

Feedback: This choice points toward Generative AI. Generative AI matters nearby, but this question is asking about Rationalists vs Empiricists.

Misconception: Not automatically an LLM and not automatically truthful. | Represented term: generative-ai | Source: explicitly confusable glossary metadata

Rationale: Tempting because Generative AI is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Rationalists vs Empiricists mechanism or distinction.

choice-draft-history-q3-parameter-3 [DISTRACTOR]: It is mainly about Parameter, not the mechanism named in Rationalists vs Empiricists.

Feedback: This choice points toward Parameter. Parameter matters nearby, but this question is asking about Rationalists vs Empiricists.

Misconception: Often used interchangeably with weights in casual explanations. | Represented term: parameter | Source: explicitly confusable glossary metadata

Rationale: Tempting because Parameter is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Rationalists vs Empiricists mechanism or distinction.

Questions to Review Checklist

☐ Are the learning objectives accurate?

☐ Does each question restate the topic clearly?

☐ Is one answer clearly best?

- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

4. Training

Stage: 1. Before Morning | **Card ID:** training | **Card in stage:** 4

Primary learning objective: Explain Training as part of the day in the life of a prompt.

Secondary objectives: Training is the process that adjusts model weights by comparing predictions with target answers, such as actual next tokens., Before ordinary use, in a separate optimization process., Training durably changes weights., Training examples may look like text prompts and completions, but the purpose is different: the system is updating weights, not merely producing a live response for a user., Pretraining is the broad first version of this training loop.

Key distinctions: Training durably changes weights., Training examples may look like text prompts and completions, but the purpose is different: the system is updating weights, not merely producing a live response for a user., When I chat with the model, it automatically trains itself.

Misconceptions to correct: When I chat with the model, it automatically trains itself.

Related glossary terms: training, loss, weight, weight-update, parameter, training-data, inference, forward-pass

Nearby glossary distractor candidates: next-token, pretraining, overfitting, generalization, validation-data

Why this matters: This is the dividing line between learning and using. Many AI myths come from confusing training with inference.

Existing Live Checkpoint

Question count: 1

Question: What makes training different from inference?

Correct: Training durably updates weights

- Training durably updates weights (correct)
- Training only reads the current context window Not quite. Training is a weight-changing process, not ordinary response generation.
- Training is the same as sampling Not quite. Training is a weight-changing process, not ordinary response generation.
- Training writes the final answer Not quite. Training is a weight-changing process, not ordinary response generation.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: What makes training different from inference?

ID: draft-training-q1 | **Type:** definition | **Correct choice ID:** choice-draft-training-q1-correct

Misconception tags: None listed

Glossary terms: training, loss, weight, weight-update, parameter, training-data, inference, forward-pass

choice-draft-training-q1-correct [CORRECT]: Training durably updates weights

Feedback: Training changes the model; inference uses the model.

choice-draft-training-q1-training-only-reads-the-current-context-window [DISTRACTOR]: Training only reads the current context window

Feedback: Not quite. Training is a weight-changing process, not ordinary response generation.

Misconception: Training only reads the current context window | Represented term: context window | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-training-q1-training-is-the-same-as-sampling [DISTRACTOR]: Training is the same as sampling

Feedback: Not quite. Training is a weight-changing process, not ordinary response generation.

Misconception: Training is the same as sampling | Represented term: training | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-training-q1-training-writes-the-final-answer [DISTRACTOR]: Training writes the final answer

Feedback: Not quite. Training is a weight-changing process, not ordinary response generation.

Misconception: Training writes the final answer | Represented term: training | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Training most directly teach?

ID: draft-training-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-training-q2-correct

Misconception tags: None listed

Glossary terms: training, loss, weight, weight-update, parameter, training-data, inference, forward-pass

choice-draft-training-q2-correct [CORRECT]: Training durably changes weights.

Feedback: Good distinction. Training durably changes weights.

choice-draft-training-q2-pretraining-1 [DISTRACTOR]: It happens again from scratch during every user prompt.

Feedback: This choice points toward Pretraining. Pretraining matters nearby, but this question is asking about Training.

Misconception: Often confused with fine-tuning or perfect source recall. | Represented term: pretraining | Source: explicitly confusable glossary metadata

Rationale: Tempting because Pretraining is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Training mechanism or distinction.

choice-draft-training-q2-probability-2 [DISTRACTOR]: It is a truth score for the claim.

Feedback: This choice points toward Probability. Probability matters nearby, but this question is asking about Training.

Misconception: Not truth, proof, confidence, or grounding by itself. | Represented term: probability | Source: explicitly confusable glossary metadata

Rationale: Tempting because Probability is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Training mechanism or distinction.

choice-draft-training-q2-evaluation-3 [DISTRACTOR]: It is mainly about Evaluation, not the mechanism named in Training.

Feedback: This choice points toward Evaluation. Evaluation matters nearby, but this question is asking about Training.

Misconception: Not the same as training unless results feed a training process. | Represented term: evaluation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Evaluation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Training mechanism or distinction.

Question 3: Which common confusion does Training help learners avoid?

ID: draft-training-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-training-q3-correct

Misconception tags: None listed

Glossary terms: training, loss, weight, weight-update, parameter, training-data, inference, forward-pass

choice-draft-training-q3-correct [CORRECT]: It corrects the misconception: When I chat with the model, it automatically trains itself.

Feedback: Good distinction. Training is clearer when that misconception is separated from the mechanism.

choice-draft-training-q3-probability-1 [DISTRACTOR]: It is a truth score for the claim.

Feedback: This choice points toward Probability. Probability matters nearby, but this question is asking about Training.

Misconception: Not truth, proof, confidence, or grounding by itself. | Represented term: probability | Source: explicitly confusable glossary metadata

Rationale: Tempting because Probability is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Training mechanism or distinction.

choice-draft-training-q3-evaluation-2 [DISTRACTOR]: It is mainly about Evaluation, not the mechanism named in Training.

Feedback: This choice points toward Evaluation. Evaluation matters nearby, but this question is asking about Training.

Misconception: Not the same as training unless results feed a training process. | Represented term: evaluation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Evaluation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Training mechanism or distinction.

choice-draft-training-q3-hidden-state-3 [DISTRACTOR]: It is the token's fixed starting vector before context matters.

Feedback: This choice points toward Hidden state. Hidden state matters nearby, but this question is asking about Training.

Misconception: Often confused with embeddings, weights, saved memory, or visible response text. | Represented term: hidden state | Source: explicitly confusable glossary metadata

Rationale: Tempting because Hidden state is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Training mechanism or distinction.

Question 4: How does Training connect to the prompt's Journey?

ID: draft-training-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-training-q4-correct

Misconception tags: None listed

Glossary terms: training, loss, weight, weight-update, parameter, training-data, inference, forward-pass

choice-draft-training-q4-correct [CORRECT]: Pretraining is the broad first version of this training loop.

Feedback: Good connection. Pretraining is the broad first version of this training loop.

choice-draft-training-q4-evaluation-1 [DISTRACTOR]: It is mainly about Evaluation, not the mechanism named in Training.

Feedback: This choice points toward Evaluation. Evaluation matters nearby, but this question is asking about Training.

Misconception: Not the same as training unless results feed a training process. | Represented term: evaluation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Evaluation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Training mechanism or distinction.

choice-draft-training-q4-hidden-state-2 [DISTRACTOR]: It is the token's fixed starting vector before context matters.

Feedback: This choice points toward Hidden state. Hidden state matters nearby, but this question is asking about Training.

Misconception: Often confused with embeddings, weights, saved memory, or visible response text. | Represented term: hidden state | Source: explicitly confusable glossary metadata

Rationale: Tempting because Hidden state is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Training mechanism or distinction.

choice-draft-training-q4-memory-3 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Training.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Training mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

5. Pretraining

Stage: 1. Before Morning | **Card ID:** pretraining | **Card in stage:** 5

Primary learning objective: Explain Pretraining as part of the day in the life of a prompt.

Secondary objectives: Pretraining is the broad first training phase where a model learns general patterns by repeatedly predicting tokens across large datasets., Before deployment, during the broad foundation-building stage., Pretraining durably changes weights., Pretraining creates the weights later used to process prompts and generate responses., Pretraining builds broad capability.

Key distinctions: Pretraining durably changes weights., Pretraining creates the weights later used to process prompts and generate responses., Pretraining means the model can perfectly recall everything it saw.

Misconceptions to correct: Pretraining means the model can perfectly recall everything it saw.

Related glossary terms: pretraining, training, training-data, weight, loss, next-token, inference

Nearby glossary distractor candidates: weight-update, parameter, overfitting, generalization, validation-data

Why this matters: Pretraining is why next-token prediction can become surprisingly capable. It installs broad pattern knowledge into weights.

Existing Live Checkpoint

Question count: 1

Question: During pretraining, what changes?

Correct: Model weights

- Model weights (correct)
- Only the current chat Not quite. Pretraining is a durable training process, not a temporary chat event.
- Only the user interface Not quite. Pretraining is a durable training process, not a temporary chat event.
- Nothing changes Not quite. Pretraining is a durable training process, not a temporary chat event.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: During pretraining, what changes?

ID: draft-pretraining-q1 | **Type:** definition | **Correct choice ID:** choice-draft-pretraining-q1-correct

Misconception tags: None listed

Glossary terms: pretraining, training, training-data, weight, loss, next-token, inference

choice-draft-pretraining-q1-correct [CORRECT]: Model weights

Feedback: Pretraining changes weights before normal use.

choice-draft-pretraining-q1-only-the-current-chat [DISTRACTOR]: Only the current chat

Feedback: Not quite. Pretraining is a durable training process, not a temporary chat event.

Misconception: Only the current chat | **Represented term:** inference | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-pretraining-q1-only-the-user-interface [DISTRACTOR]: Only the user interface

Feedback: Not quite. Pretraining is a durable training process, not a temporary chat event.

Misconception: Only the user interface | **Represented term:** training | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-pretraining-q1-nothing-changes [DISTRACTOR]: Nothing changes

Feedback: Not quite. Pretraining is a durable training process, not a temporary chat event.

Misconception: Nothing changes | **Represented term:** training-data | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Pretraining most directly teach?

ID: draft-pretraining-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-pretraining-q2-correct

Misconception tags: None listed

Glossary terms: pretraining, training, training-data, weight, loss, next-token, inference

choice-draft-pretraining-q2-correct [CORRECT]: Pretraining durably changes weights.

Feedback: Good distinction. Pretraining durably changes weights.

choice-draft-pretraining-q2-rag-1 [DISTRACTOR]: It means the model permanently learned the retrieved document.

Feedback: This choice points toward RAG. RAG matters nearby, but this question is asking about Pretraining.

Misconception: Not training, fine-tuning, permanent memory, or a guarantee that the answer is true. | Represented term: rag | Source: explicitly confusable glossary metadata

Rationale: Tempting because RAG is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Pretraining mechanism or distinction.

choice-draft-pretraining-q2-validation-data-2 [DISTRACTOR]: It is mainly about Validation data, not the mechanism named in Pretraining.

Feedback: This choice points toward Validation data. Validation data matters nearby, but this question is asking about Pretraining.

Misconception: Not training data if it is kept out of weight updates. | Represented term: validation-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Validation data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Pretraining mechanism or distinction.

choice-draft-pretraining-q2-hidden-state-3 [DISTRACTOR]: It is the token's fixed starting vector before context matters.

Feedback: This choice points toward Hidden state. Hidden state matters nearby, but this question is asking about Pretraining.

Misconception: Often confused with embeddings, weights, saved memory, or visible response text. | Represented term: hidden state | Source: explicitly confusable glossary metadata

Rationale: Tempting because Hidden state is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Pretraining mechanism or distinction.

Question 3: Which common confusion does Pretraining help learners avoid?

ID: draft-pretraining-q3 | Type: boundary distinction | Correct choice ID: choice-draft-pretraining-q3-correct

Misconception tags: None listed

Glossary terms: pretraining, training, training-data, weight, loss, next-token, inference

choice-draft-pretraining-q3-correct [CORRECT]: It corrects the misconception: Pretraining means the model can perfectly recall everything it saw.

Feedback: Good distinction. Pretraining is clearer when that misconception is separated from the mechanism.

choice-draft-pretraining-q3-validation-data-1 [DISTRACTOR]: It is mainly about Validation data, not the mechanism named in Pretraining.

Feedback: This choice points toward Validation data. Validation data matters nearby, but this question is asking about Pretraining.

Misconception: Not training data if it is kept out of weight updates. | Represented term: validation-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Validation data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Pretraining mechanism or distinction.

choice-draft-pretraining-q3-hidden-state-2 [DISTRACTOR]: It is the token's fixed starting vector before context matters.

Feedback: This choice points toward Hidden state. Hidden state matters nearby, but this question is asking about Pretraining.

Misconception: Often confused with embeddings, weights, saved memory, or visible response text. | Represented term: hidden state | Source: explicitly confusable glossary metadata

Rationale: Tempting because Hidden state is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Pretraining mechanism or distinction.

choice-draft-pretraining-q3-memory-3 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Pretraining.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Pretraining mechanism or distinction.

Question 4: How does Pretraining connect to the prompt's Journey?

ID: draft-pretraining-q4 | Type: concept relationship | Correct choice ID: choice-draft-pretraining-q4-correct

Misconception tags: None listed

Glossary terms: pretraining, training, training-data, weight, loss, next-token, inference

choice-draft-pretraining-q4-correct [CORRECT]: Pretraining builds broad capability.

Feedback: Good connection. Pretraining builds broad capability.

choice-draft-pretraining-q4-hidden-state-1 [DISTRACTOR]: It is the token's fixed starting vector before context matters.

Feedback: This choice points toward Hidden state. Hidden state matters nearby, but this question is asking about Pretraining.

Misconception: Often confused with embeddings, weights, saved memory, or visible response text. | Represented term: hidden state | Source: explicitly confusable glossary metadata

Rationale: Tempting because Hidden state is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Pretraining mechanism or distinction.

choice-draft-pretraining-q4-memory-2 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Pretraining.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Pretraining mechanism or distinction.

choice-draft-pretraining-q4-parameter-3 [DISTRACTOR]: It is mainly about Parameter, not the mechanism named in Pretraining.

Feedback: This choice points toward Parameter. Parameter matters nearby, but this question is asking about Pretraining.

Misconception: Often used interchangeably with weights in casual explanations. | Represented term: parameter | Source: explicitly confusable glossary metadata

Rationale: Tempting because Parameter is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Pretraining mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

6. Overfitting and Generalization

Stage: 1. Before Morning | **Card ID:** overfitting-generalization | **Card in stage:** 6

Primary learning objective: Explain Overfitting and Generalization as part of the day in the life of a prompt.

Secondary objectives: Overfitting happens when a model learns training examples too narrowly and performs poorly on new examples; generalization means learned patterns work beyond the examples seen during training., During training, evaluation, and model selection., Overfitting is about what gets installed into weights during training., A model that generalizes well can handle new prompts., Pretraining should build broad patterns.

Key distinctions: Overfitting is about what gets installed into weights during training., A model that generalizes well can handle new prompts., If a model performs well on training data, it has learned well.

Misconceptions to correct: If a model performs well on training data, it has learned well.

Related glossary terms: overfitting, generalization, validation-data, training-data, evaluation, training

Nearby glossary distractor candidates: next-token, pretraining

Why this matters: This helps learners see why more memorization is not the same as better intelligence.

Existing Live Checkpoint

Question count: 1

Question: Which is a sign of overfitting?

Correct: Great performance on training examples but poor performance on new examples

- Great performance on training examples but poor performance on new examples (correct)
- A model using a context window Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval.
- A model retrieving documents with RAG Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval.
- A model generating one token at a time Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: Which is a sign of overfitting in Overfitting and Generalization?

ID: draft-overfitting-generalization-q1 | **Type:** definition | **Correct choice ID:** choice-draft-overfitting-generalization-q1-correct

Misconception tags: None listed

Glossary terms: overfitting, generalization, validation-data, training-data, evaluation, training

choice-draft-overfitting-generalization-q1-correct [CORRECT]: Great performance on training examples but poor performance on new examples

Feedback: Generalization matters more than memorizing the answer key.

choice-draft-overfitting-generalization-q1-a-model-using-a-context-window [DISTRACTOR]: A model using a context window

Feedback: Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval.

Misconception: A model using a context window | Represented term: context window | Source: learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-overfitting-generalization-q1-a-model-retrieving-documents-with-rag [DISTRACTOR]: A model retrieving documents with RAG

Feedback: Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval.

Misconception: A model retrieving documents with RAG | Represented term: rag | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-overfitting-generalization-q1-a-model-generating-one-token-at-a-time [DISTRACTOR]: A model generating one token at a time

Feedback: Not quite. Overfitting is a training/evaluation failure mode, not context or retrieval.

Misconception: A model generating one token at a time | Represented term: token | Source: same-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Overfitting and Generalization most directly teach?

ID: draft-overfitting-generalization-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-overfitting-generalization-q2-correct

Misconception tags: None listed

Glossary terms: overfitting, generalization, validation-data, training-data, evaluation, training

choice-draft-overfitting-generalization-q2-correct [CORRECT]: Overfitting is about what gets installed into weights during training.

Feedback: Good distinction. Overfitting is about what gets installed into weights during training.

choice-draft-overfitting-generalization-q2-alignment-1 [DISTRACTOR]: It is mainly about Alignment, not the mechanism named in Overfitting and Generalization.

Feedback: This choice points toward Alignment. Alignment matters nearby, but this question is asking about Overfitting and Generalization.

Misconception: Aligned behavior does not guarantee truth or mean the model understands morality. | **Represented term:** alignment | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Alignment is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Overfitting and Generalization mechanism or distinction.

choice-draft-overfitting-generalization-q2-inference-2 [DISTRACTOR]: It durably trains the model on the current prompt.

Feedback: This choice points toward Inference. Inference matters nearby, but this question is asking about Overfitting and Generalization.

Misconception: Inference can use context during one run, but it does not normally train the model. | **Represented term:** inference | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Inference is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Overfitting and Generalization mechanism or distinction.

choice-draft-overfitting-generalization-q2-rag-3 [DISTRACTOR]: It means the model permanently learned the retrieved document.

Feedback: This choice points toward RAG. RAG matters nearby, but this question is asking about Overfitting and Generalization.

Misconception: Not training, fine-tuning, permanent memory, or a guarantee that the answer is true. | **Represented term:** rag | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because RAG is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Overfitting and Generalization mechanism or distinction.

Question 3: Which common confusion does Overfitting and Generalization help learners avoid?

ID: draft-overfitting-generalization-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-overfitting-generalization-q3-correct

Misconception tags: None listed

Glossary terms: overfitting, generalization, validation-data, training-data, evaluation, training

choice-draft-overfitting-generalization-q3-correct [CORRECT]: It corrects the misconception: If a model performs well on training data, it has learned well.

Feedback: Good distinction. Overfitting and Generalization is clearer when that misconception is separated from the mechanism.

choice-draft-overfitting-generalization-q3-inference-1 [DISTRACTOR]: It durably trains the model on the current prompt.

Feedback: This choice points toward Inference. Inference matters nearby, but this question is asking about Overfitting and Generalization.

Misconception: Inference can use context during one run, but it does not normally train the model. | **Represented term:** inference | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Inference is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Overfitting and Generalization mechanism or distinction.

choice-draft-overfitting-generalization-q3-rag-2 [DISTRACTOR]: It means the model permanently learned the retrieved document.

Feedback: This choice points toward RAG. RAG matters nearby, but this question is asking about Overfitting and Generalization.

Misconception: Not training, fine-tuning, permanent memory, or a guarantee that the answer is true. | **Represented term:** rag | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because RAG is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Overfitting and Generalization mechanism or distinction.

choice-draft-overfitting-generalization-q3-loss-3 [DISTRACTOR]: It is mainly about Loss, not the mechanism named in Overfitting and Generalization.

Feedback: This choice points toward Loss. Loss matters nearby, but this question is asking about Overfitting and Generalization.

Misconception: Truth, confidence, punishment, or a human sense of failure. | **Represented term:** loss | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Loss is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Overfitting and Generalization mechanism or distinction.

Questions to Review Checklist

[] Are the learning objectives accurate?

- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

7. Fine-Tuning

Stage: 1. Before Morning | **Card ID:** fine-tuning | **Card in stage:** 7

Primary learning objective: Explain Fine-Tuning as part of the day in the life of a prompt.

Secondary objectives: Fine-tuning is additional training that adapts a pretrained model toward a task, domain, style, or preference., After pretraining and before or between deployment cycles., Fine-tuning can durably change weights or add durable adapter behavior., Fine-tuning shapes future response patterns., Fine-tuning adapts a model; alignment is one major reason to fine-tune or otherwise shape behavior.

Key distinctions: Fine-tuning can durably change weights or add durable adapter behavior., Fine-tuning shapes future response patterns., Prompting once is the same as fine-tuning.

Misconceptions to correct: Prompting once is the same as fine-tuning.

Related glossary terms: fine-tuning, adapter, pretraining, training, prompt, rag, sampling, alignment

Nearby glossary distractor candidates: token, tokenizer, tokenization, model-output, decoding-step, input-context, completion, generated-token, response-so-far, response-tokens, prompt-tokens, response

Why this matters: This explains how one base model can become many differently behaving assistants.

Existing Live Checkpoint

Question count: 1

Question: Which action is closest to fine-tuning?

Correct: Additional training that shapes future responses

- Additional training that shapes future responses (correct)
- Typing one better prompt Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training.
- Retrieving one PDF into context Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training.
- Sampling the next token Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: Which action is closest to fine-tuning a model?

ID: draft-fine-tuning-q1 | **Type:** definition | **Correct choice ID:** choice-draft-fine-tuning-q1-correct

Misconception tags: None listed

Glossary terms: fine-tuning, adapter, pretraining, training, prompt, rag, sampling, alignment

choice-draft-fine-tuning-q1-correct [CORRECT]: Additional training that shapes future responses

Feedback: Fine-tuning changes future behavior more durably than a single prompt.

choice-draft-fine-tuning-q1-typing-one-better-prompt [DISTRACTOR]: Typing one better prompt

Feedback: Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training.

Misconception: Typing one better prompt | **Represented term:** rag | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-fine-tuning-q1-retrieving-one-pdf-into-context [DISTRACTOR]: Retrieving one PDF into context

Feedback: Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training.

Misconception: Retrieving one PDF into context | **Represented term:** context window | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-fine-tuning-q1-sampling-the-next-token [DISTRACTOR]: Sampling the next token

Feedback: Not quite. Prompting and RAG steer the current context; fine-tuning changes model behavior through training.

Misconception: Sampling the next token | **Represented term:** token | **Source:** same-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Fine-Tuning most directly teach?

ID: draft-fine-tuning-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-fine-tuning-q2-correct

Misconception tags: None listed

Glossary terms: fine-tuning, adapter, pretraining, training, prompt, rag, sampling, alignment

choice-draft-fine-tuning-q2-correct [CORRECT]: Fine-tuning can durably change weights or add durable adapter behavior.

Feedback: Good distinction. Fine-tuning can durably change weights or add durable adapter behavior.

choice-draft-fine-tuning-q2-inference-1 [DISTRACTOR]: It durably trains the model on the current prompt.

Feedback: This choice points toward Inference. Inference matters nearby, but this question is asking about Fine-Tuning.

Misconception: Inference can use context during one run, but it does not normally train the model. | Represented term: inference | Source: explicitly confusable glossary metadata

Rationale: Tempting because Inference is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Fine-Tuning mechanism or distinction.

choice-draft-fine-tuning-q2-training-data-2 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Fine-Tuning.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Fine-Tuning.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Fine-Tuning mechanism or distinction.

choice-draft-fine-tuning-q2-next-token-3 [DISTRACTOR]: It is mainly about Next token, not the mechanism named in Fine-Tuning.

Feedback: This choice points toward Next token. Next token matters nearby, but this question is asking about Fine-Tuning.

Misconception: Not the whole answer, a permanent memory, or a guarantee of truth. | Represented term: next-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Next token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Fine-Tuning mechanism or distinction.

Question 3: Which common confusion does Fine-Tuning help learners avoid?

ID: draft-fine-tuning-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-fine-tuning-q3-correct

Misconception tags: None listed

Glossary terms: fine-tuning, adapter, pretraining, training, prompt, rag, sampling, alignment

choice-draft-fine-tuning-q3-correct [CORRECT]: It corrects the misconception: Prompting once is the same as fine-tuning.

Feedback: Good distinction. Fine-Tuning is clearer when that misconception is separated from the mechanism.

choice-draft-fine-tuning-q3-training-data-1 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Fine-Tuning.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Fine-Tuning.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Fine-Tuning mechanism or distinction.

choice-draft-fine-tuning-q3-next-token-2 [DISTRACTOR]: It is mainly about Next token, not the mechanism named in Fine-Tuning.

Feedback: This choice points toward Next token. Next token matters nearby, but this question is asking about Fine-Tuning.

Misconception: Not the whole answer, a permanent memory, or a guarantee of truth. | Represented term: next-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Next token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Fine-Tuning mechanism or distinction.

choice-draft-fine-tuning-q3-loss-3 [DISTRACTOR]: It is mainly about Loss, not the mechanism named in Fine-Tuning.

Feedback: This choice points toward Loss. Loss matters nearby, but this question is asking about Fine-Tuning.

Misconception: Truth, confidence, punishment, or a human sense of failure. | Represented term: loss | Source: explicitly confusable glossary metadata

Rationale: Tempting because Loss is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Fine-Tuning mechanism or distinction.

Question 4: How does Fine-Tuning connect to the prompt's Journey?

ID: draft-fine-tuning-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-fine-tuning-q4-correct

Misconception tags: None listed

Glossary terms: fine-tuning, adapter, pretraining, training, prompt, rag, sampling, alignment

choice-draft-fine-tuning-q4-correct [CORRECT]: Fine-tuning adapts a model; alignment is one major reason to fine-tune or otherwise shape behavior.

Feedback: Good connection. Fine-tuning adapts a model; alignment is one major reason to fine-tune or otherwise shape behavior.

choice-draft-fine-tuning-q4-next-token-1 [DISTRACTOR]: It is mainly about Next token, not the mechanism named in Fine-Tuning.

Feedback: This choice points toward Next token. Next token matters nearby, but this question is asking about Fine-Tuning.

Misconception: Not the whole answer, a permanent memory, or a guarantee of truth. | Represented term: next-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Next token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Fine-Tuning mechanism or distinction.

choice-draft-fine-tuning-q4-loss-2 [DISTRACTOR]: It is mainly about Loss, not the mechanism named in Fine-Tuning.

Feedback: This choice points toward Loss. Loss matters nearby, but this question is asking about Fine-Tuning.

Misconception: Truth, confidence, punishment, or a human sense of failure. | Represented term: loss | Source: explicitly confusable glossary metadata

Rationale: Tempting because Loss is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Fine-Tuning mechanism or distinction.

choice-draft-fine-tuning-q4-weight-update-3 [DISTRACTOR]: It is mainly about Weight update, not the mechanism named in Fine-Tuning.

Feedback: This choice points toward Weight update. Weight update matters nearby, but this question is asking about Fine-Tuning.

Misconception: Not the same as adding a prompt to context. | Represented term: weight-update | Source: explicitly confusable glossary metadata

Rationale: Tempting because Weight update is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Fine-Tuning mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

8. Alignment

Stage: 1. Before Morning | **Card ID:** alignment | **Card in stage:** 8

Primary learning objective: Explain Alignment as part of the day in the life of a prompt.

Secondary objectives: Alignment is the effort to shape AI behavior toward human instructions, preferences, safety boundaries, and values., During fine-tuning, human feedback learning, system design, evaluation, deployment, and policy enforcement., Some alignment work durably changes weights or adapter weights., Alignment affects how the model responds to prompts, but it does not mean the model understands morality like a person., Fine-tuning can shape model behavior.

Key distinctions: Some alignment work durably changes weights or adapter weights., Alignment affects how the model responds to prompts, but it does not mean the model understands morality like a person., Aligned means morally good or fully trustworthy.

Misconceptions to correct: Aligned means morally good or fully trustworthy.

Related glossary terms: alignment, instruction-tuning, human-feedback-learning, rlhf, preference-optimization, policy, guardrail, evaluation, fine-tuning, system-prompt

Nearby glossary distractor candidates: adapter, inference, forward-pass

Why this matters: Alignment is why a model may refuse harmful requests, follow instructions, or prefer clearer helpful answers over merely likely text.

Existing Live Checkpoint

Question count: 1

Question: What does alignment try to do?

Correct: Shape model behavior toward human goals, instructions, and safety boundaries

- Shape model behavior toward human goals, instructions, and safety boundaries (correct)
- Make the model conscious Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee.
- Guarantee every answer is true Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee.
- Permanently solve all AI risk Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: What does alignment try to do?

ID: draft-alignment-q1 | **Type:** definition | **Correct choice ID:** choice-draft-alignment-q1-correct

Misconception tags: None listed

Glossary terms: alignment, instruction-tuning, human-feedback-learning, rlhf, preference-optimization, policy, guardrail, evaluation, fine-tuning, system-prompt

choice-draft-alignment-q1-correct [CORRECT]: Shape model behavior toward human goals, instructions, and safety boundaries

Feedback: Alignment shapes behavior, but it is not magic morality.

choice-draft-alignment-q1-make-the-model-conscious [DISTRACTOR]: Make the model conscious

Feedback: Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee.

Misconception: Make the model conscious | **Represented term:** llm | **Source:** same-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-alignment-q1-guarantee-every-answer-is-true [DISTRACTOR]: Guarantee every answer is true

Feedback: Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee.

Misconception: Guarantee every answer is true | **Represented term:** probability | **Source:** learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-alignment-q1-permanently-solve-all-ai-risk [DISTRACTOR]: Permanently solve all AI risk

Feedback: Not quite. Alignment is a set of technical and human processes, not consciousness or a perfect safety guarantee.

Misconception: Permanently solve all AI risk | **Represented term:** ai | **Source:** same-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Alignment most directly teach?

ID: draft-alignment-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-alignment-q2-correct

Misconception tags: None listed

Glossary terms: alignment, instruction-tuning, human-feedback-learning, rlhf, preference-optimization, policy, guardrail, evaluation, fine-tuning, system-prompt

choice-draft-alignment-q2-correct [CORRECT]: Some alignment work durably changes weights or adapter weights.

Feedback: Good distinction. Some alignment work durably changes weights or adapter weights.

choice-draft-alignment-q2-prompt-1 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about Alignment.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Alignment mechanism or distinction.

choice-draft-alignment-q2-rag-2 [DISTRACTOR]: It means the model permanently learned the retrieved document.

Feedback: This choice points toward RAG. RAG matters nearby, but this question is asking about Alignment.

Misconception: Not training, fine-tuning, permanent memory, or a guarantee that the answer is true. | Represented term: rag | Source: explicitly confusable glossary metadata

Rationale: Tempting because RAG is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Alignment mechanism or distinction.

choice-draft-alignment-q2-pretraining-3 [DISTRACTOR]: It happens again from scratch during every user prompt.

Feedback: This choice points toward Pretraining. Pretraining matters nearby, but this question is asking about Alignment.

Misconception: Often confused with fine-tuning or perfect source recall. | Represented term: pretraining | Source: explicitly confusable glossary metadata

Rationale: Tempting because Pretraining is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Alignment mechanism or distinction.

Question 3: Which common confusion does Alignment help learners avoid?

ID: draft-alignment-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-alignment-q3-correct

Misconception tags: None listed

Glossary terms: alignment, instruction-tuning, human-feedback-learning, rlhf, preference-optimization, policy, guardrail, evaluation, fine-tuning, system-prompt

choice-draft-alignment-q3-correct [CORRECT]: It corrects the misconception: Aligned means morally good or fully trustworthy.

Feedback: Good distinction. Alignment is clearer when that misconception is separated from the mechanism.

choice-draft-alignment-q3-rag-1 [DISTRACTOR]: It means the model permanently learned the retrieved document.

Feedback: This choice points toward RAG. RAG matters nearby, but this question is asking about Alignment.

Misconception: Not training, fine-tuning, permanent memory, or a guarantee that the answer is true. | Represented term: rag | Source: explicitly confusable glossary metadata

Rationale: Tempting because RAG is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Alignment mechanism or distinction.

choice-draft-alignment-q3-pretraining-2 [DISTRACTOR]: It happens again from scratch during every user prompt.

Feedback: This choice points toward Pretraining. Pretraining matters nearby, but this question is asking about Alignment.

Misconception: Often confused with fine-tuning or perfect source recall. | Represented term: pretraining | Source: explicitly confusable glossary metadata

Rationale: Tempting because Pretraining is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Alignment mechanism or distinction.

choice-draft-alignment-q3-adapter-3 [DISTRACTOR]: It is mainly about Adapter, not the mechanism named in Alignment.

Feedback: This choice points toward Adapter. Adapter matters nearby, but this question is asking about Alignment.

Misconception: Not a prompt and not retrieved context. | Represented term: adapter | Source: explicitly confusable glossary metadata

Rationale: Tempting because Adapter is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Alignment mechanism or distinction.

Question 4: How does Alignment connect to the prompt's Journey?

ID: draft-alignment-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-alignment-q4-correct

Misconception tags: None listed

Glossary terms: alignment, instruction-tuning, human-feedback-learning, rlhf, preference-optimization, policy, guardrail, evaluation, fine-tuning, system-prompt

choice-draft-alignment-q4-correct [CORRECT]: Fine-tuning can shape model behavior.

Feedback: Good connection. Fine-tuning can shape model behavior.

choice-draft-alignment-q4-pretraining-1 [DISTRACTOR]: It happens again from scratch during every user prompt.

Feedback: This choice points toward Pretraining. Pretraining matters nearby, but this question is asking about Alignment.

Misconception: Often confused with fine-tuning or perfect source recall. | Represented term: pretraining | Source: explicitly confusable glossary metadata

Rationale: Tempting because Pretraining is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Alignment mechanism or distinction.

choice-draft-alignment-q4-adapter-2 [DISTRACTOR]: It is mainly about Adapter, not the mechanism named in Alignment.

Feedback: This choice points toward Adapter. Adapter matters nearby, but this question is asking about Alignment.

Misconception: Not a prompt and not retrieved context. | Represented term: adapter | Source: explicitly confusable glossary metadata

Rationale: Tempting because Adapter is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Alignment mechanism or distinction.

choice-draft-alignment-q4-training-3 [DISTRACTOR]: It permanently updates model weights during this ordinary response.

Feedback: This choice points toward Training. Training matters nearby, but this question is asking about Alignment.

Misconception: Ordinary chat inference is not training unless a separate training process updates weights. | Represented term: training | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Alignment mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

9. Inference

Stage: 2. Morning Commute | **Card ID:** inference | **Card in stage:** 1

Primary learning objective: Explain Inference as part of the day in the life of a prompt.

Secondary objectives: Inference is normal model use: fixed learned weights process the current context to produce scores for the next response token., Every time a deployed model answers a prompt., Inference creates temporary activations, hidden states, and next-token scores for this run., Prompt tokens, retrieved context, conversation history, and response-so-far tokens can all be current context; inference generates the next response token., Batch 1 explained how weights are shaped before use.

Key distinctions: Inference creates temporary activations, hidden states, and next-token scores for this run., Prompt tokens, retrieved context, conversation history, and response-so-far tokens can all be current context; inference generates the next response token., Normal chat automatically trains the model.

Misconceptions to correct: Normal chat automatically trains the model.

Related glossary terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling

Nearby glossary distractor candidates: vector, embedding-table, feature, embedding, distributed-representation

Why this matters: This is the line between using a trained model and training a model.

Existing Live Checkpoint

Question count: 1

Question: Which part is temporary during ordinary inference?

Correct: Temporary activations

- Temporary activations (correct)
- The model weights permanently Not quite. Ordinary inference normally does not rewrite weights or training data.
- The training dataset Not quite. Ordinary inference normally does not rewrite weights or training data.
- Every document the model has seen Not quite. Ordinary inference normally does not rewrite weights or training data.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: Which part is temporary during ordinary inference?

ID: draft-inference-q1 | **Type:** definition | **Correct choice ID:** choice-draft-inference-q1-correct

Misconception tags: None listed

Glossary terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling

choice-draft-inference-q1-correct [CORRECT]: Temporary activations

Feedback: Insight strengthened. Inference uses fixed weights but creates temporary internal states for the current run.

choice-draft-inference-q1-the-model-weights-permanently [DISTRACTOR]: The model weights permanently

Feedback: Not quite. Ordinary inference normally does not rewrite weights or training data.

Misconception: The model weights permanently | **Represented term:** training | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-inference-q1-the-training-dataset [DISTRACTOR]: The training dataset

Feedback: Not quite. Ordinary inference normally does not rewrite weights or training data.

Misconception: The training dataset | **Represented term:** training | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-inference-q1-every-document-the-model-has-seen [DISTRACTOR]: Every document the model has seen

Feedback: Not quite. Ordinary inference normally does not rewrite weights or training data.

Misconception: Every document the model has seen | **Represented term:** rag | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Inference most directly teach?

ID: draft-inference-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-inference-q2-correct

Misconception tags: None listed

Glossary terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling

choice-draft-inference-q2-correct [CORRECT]: Inference creates temporary activations, hidden states, and next-token scores for this run.

Feedback: Good distinction. Inference creates temporary activations, hidden states, and next-token scores for this run.

choice-draft-inference-q2-fine-tuning-1 [DISTRACTOR]: It is the same as adding outside text into the current prompt.

Feedback: This choice points toward Fine-tuning. Fine-tuning matters nearby, but this question is asking about Inference.

Misconception: Different from prompting; prompting changes temporary context, not durable weights. | Represented term: fine-tuning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Fine-tuning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Inference mechanism or distinction.

choice-draft-inference-q2-context-window-2 [DISTRACTOR]: It is permanent model memory.

Feedback: This choice points toward Context window. Context window matters nearby, but this question is asking about Inference.

Misconception: Not permanent memory; it is temporary working context. | Represented term: context window | Source: explicitly confusable glossary metadata

Rationale: Tempting because Context window is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Inference mechanism or distinction.

choice-draft-inference-q2-prompt-3 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about Inference.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Inference mechanism or distinction.

Question 3: Which common confusion does Inference help learners avoid?

ID: draft-inference-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-inference-q3-correct

Misconception tags: None listed

Glossary terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling

choice-draft-inference-q3-correct [CORRECT]: It corrects the misconception: Normal chat automatically trains the model.

Feedback: Good distinction. Inference is clearer when that misconception is separated from the mechanism.

choice-draft-inference-q3-context-window-1 [DISTRACTOR]: It is permanent model memory.

Feedback: This choice points toward Context window. Context window matters nearby, but this question is asking about Inference.

Misconception: Not permanent memory; it is temporary working context. | Represented term: context window | Source: explicitly confusable glossary metadata

Rationale: Tempting because Context window is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Inference mechanism or distinction.

choice-draft-inference-q3-prompt-2 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about Inference.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Inference mechanism or distinction.

choice-draft-inference-q3-memory-3 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Inference.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Inference mechanism or distinction.

Question 4: How does Inference connect to the prompt's Journey?

ID: draft-inference-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-inference-q4-correct

Misconception tags: None listed

Glossary terms: inference, forward-pass, input-context, weight, activation, hidden state, logits, sampling

choice-draft-inference-q4-correct [CORRECT]: Batch 1 explained how weights are shaped before use.

Feedback: Good connection. Batch 1 explained how weights are shaped before use.

choice-draft-inference-q4-prompt-1 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about Inference.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Inference mechanism or distinction.

choice-draft-inference-q4-memory-2 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Inference.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Inference mechanism or distinction.

choice-draft-inference-q4-system-prompt-3 [DISTRACTOR]: It is mainly about System prompt, not the mechanism named in Inference.

Feedback: This choice points toward System prompt. System prompt matters nearby, but this question is asking about Inference.

Misconception: Not fine-tuning, training, permanent memory, or conscience. | Represented term: system-prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because System prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Inference mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

10. Prompt vs Response

Stage: 2. Morning Commute | Card ID: prompt-response | Card in stage: 2

Primary learning objective: Explain Prompt vs Response as part of the day in the life of a prompt.

Secondary objectives: Prompt tokens are given to the model; response tokens are generated by the model one at a time and appended to the context., During inference and response generation., Prompt and response-so-far tokens are temporary context., This is the distinction itself: the user prompt is given context; "... " is a generated response token appended to the response-so-far., After we distinguish prompt from response, we can see how both are represented as tokens.

Key distinctions: Prompt and response-so-far tokens are temporary context., This is the distinction itself: the user prompt is given context; "... " is a generated response token appended to the response-so-far., The model generates the whole response at once.

Misconceptions to correct: The model generates the whole response at once.

Related glossary terms: prompt, response, prompt-tokens, response-tokens, generated-token, input-context, completion

Nearby glossary distractor candidates: response-so-far, decoding-step, model-output, token, tokenizer, tokenization, forward-pass, inference

Why this matters: It prevents the myth that the model writes a full answer all at once.

Existing Live Checkpoint

Question count: 1

Question: What happens after one response token is generated?

Correct: It is appended to the context

- It is appended to the context (correct)
- The whole answer appears at once Not quite. A generated token becomes part of the next context, but it is not a durable weight update.
- The model permanently learns it Not quite. A generated token becomes part of the next context, but it is not a durable weight update.
- Softmax disappears Not quite. A generated token becomes part of the next context, but it is not a durable weight update.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: What happens after one response token is generated in Prompt vs Response?

ID: draft-prompt-response-q1 | Type: definition | Correct choice ID: choice-draft-prompt-response-q1-correct

Misconception tags: None listed

Glossary terms: prompt, response, prompt-tokens, response-tokens, generated-token, input-context, completion

choice-draft-prompt-response-q1-correct [CORRECT]: It is appended to the context

Feedback: The response grows by next token, append, repeat.

choice-draft-prompt-response-q1-the-whole-answer-appears-at-once [DISTRACTOR]: The whole answer appears at once

Feedback: Not quite. A generated token becomes part of the next context, but it is not a durable weight update.

Misconception: The whole answer appears at once | Represented term: system-prompt | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-prompt-response-q1-the-model-permanently-learns-it [DISTRACTOR]: The model permanently learns it

Feedback: Not quite. A generated token becomes part of the next context, but it is not a durable weight update.

Misconception: The model permanently learns it | Represented term: context window | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-prompt-response-q1-softmax-disappears [DISTRACTOR]: Softmax disappears

Feedback: Not quite. A generated token becomes part of the next context, but it is not a durable weight update.

Misconception: Softmax disappears | Represented term: softmax | Source: learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Prompt vs Response most directly teach?

ID: draft-prompt-response-q2 | Type: mechanism | Correct choice ID: choice-draft-prompt-response-q2-correct

Misconception tags: None listed

Glossary terms: prompt, response, prompt-tokens, response-tokens, generated-token, input-context, completion

choice-draft-prompt-response-q2-correct [CORRECT]: Prompt and response-so-far tokens are temporary context.

Feedback: Good distinction. Prompt and response-so-far tokens are temporary context.

choice-draft-prompt-response-q2-autoregression-1 [DISTRACTOR]: It denoises a whole image from noise.

Feedback: This choice points toward Autoregression. Autoregression matters nearby, but this question is asking about Prompt vs Response.

Misconception: Autoregression is being treated as the same idea as Prompt vs Response. | Represented term: autoregression | Source: explicitly confusable glossary metadata

Rationale: Tempting because Autoregression is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Prompt vs Response mechanism or distinction.

choice-draft-prompt-response-q2-sampling-2 [DISTRACTOR]: It writes the whole answer at once.

Feedback: This choice points toward Sampling. Sampling matters nearby, but this question is asking about Prompt vs Response.

Misconception: Not writing the whole response, permanently teaching the model, or proving truth. | Represented term: sampling | Source: explicitly confusable glossary metadata

Rationale: Tempting because Sampling is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Prompt vs Response mechanism or distinction.

choice-draft-prompt-response-q2-memory-3 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Prompt vs Response.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Prompt vs Response mechanism or distinction.

Question 3: Which common confusion does Prompt vs Response help learners avoid?

ID: draft-prompt-response-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-prompt-response-q3-correct

Misconception tags: None listed

Glossary terms: prompt, response, prompt-tokens, response-tokens, generated-token, input-context, completion

choice-draft-prompt-response-q3-correct [CORRECT]: It corrects the misconception: The model generates the whole response at once.

Feedback: Good distinction. Prompt vs Response is clearer when that misconception is separated from the mechanism.

choice-draft-prompt-response-q3-sampling-1 [DISTRACTOR]: It writes the whole answer at once.

Feedback: This choice points toward Sampling. Sampling matters nearby, but this question is asking about Prompt vs Response.

Misconception: Not writing the whole response, permanently teaching the model, or proving truth. | Represented term: sampling | Source: explicitly confusable glossary metadata

Rationale: Tempting because Sampling is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Prompt vs Response mechanism or distinction.

choice-draft-prompt-response-q3-memory-2 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Prompt vs Response.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Prompt vs Response mechanism or distinction.

choice-draft-prompt-response-q3-response-so-far-3 [DISTRACTOR]: It is mainly about Response-so-far, not the mechanism named in Prompt vs Response.

Feedback: This choice points toward Response-so-far. Response-so-far matters nearby, but this question is asking about Prompt vs Response.

Misconception: Not the whole final answer, permanent memory, or training data. | Represented term: response-so-far | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response-so-far is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Prompt vs Response mechanism or distinction.

Question 4: How does Prompt vs Response connect to the prompt's Journey?

ID: draft-prompt-response-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-prompt-response-q4-correct

Misconception tags: None listed

Glossary terms: prompt, response, prompt-tokens, response-tokens, generated-token, input-context, completion

choice-draft-prompt-response-q4-correct [CORRECT]: After we distinguish prompt from response, we can see how both are represented as tokens.

Feedback: Good connection. After we distinguish prompt from response, we can see how both are represented as tokens.

choice-draft-prompt-response-q4-memory-1 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Prompt vs Response.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Prompt vs Response mechanism or distinction.

choice-draft-prompt-response-q4-response-so-far-2 [DISTRACTOR]: It is mainly about Response-so-far, not the mechanism named in Prompt vs Response.

Feedback: This choice points toward Response-so-far. Response-so-far matters nearby, but this question is asking about Prompt vs Response.

Misconception: Not the whole final answer, permanent memory, or training data. | Represented term: response-so-far | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response-so-far is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Prompt vs Response mechanism or distinction.

choice-draft-prompt-response-q4-retrieval-3 [DISTRACTOR]: It is the same as fine-tuning the model on new data.

Feedback: This choice points toward Retrieval. Retrieval matters nearby, but this question is asking about Prompt vs Response.

Misconception: Retrieval is being treated as the same idea as Prompt vs Response. | Represented term: retrieval | Source: explicitly confusable glossary metadata

Rationale: Tempting because Retrieval is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Prompt vs Response mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

11. Tokenization

Stage: 2. Morning Commute | **Card ID:** tokens | **Card in stage:** 3

Primary learning objective: Explain Tokenization as part of the day in the life of a prompt.

Secondary objectives: Tokenization splits text into chunks that can be represented as token IDs., Before embedding lookup and before transformer layers process the current context., Tokenization itself does not change model weights., Both prompt text and generated response text are represented as tokens., Tokens become token IDs, and token IDs retrieve embeddings.

Key distinctions: Tokenization itself does not change model weights., Both prompt text and generated response text are represented as tokens., Tokens are always whole words or human concepts.

Misconceptions to correct: Tokens are always whole words or human concepts.

Related glossary terms: tokenization, tokenizer, token, prompt-tokens, response-tokens, token-id, embedding

Nearby glossary distractor candidates: model-output, decoding-step, input-context, completion, generated-token, response-so-far, response, prompt

Why this matters: Without tokenization, the model has no numerical pieces to process.

Existing Live Checkpoint

Question count: 1

Question: What can a token be?

Correct: A word, word piece, punctuation mark, or other text chunk

- A word, word piece, punctuation mark, or other text chunk (correct)
- Only a complete word Not quite. Tokens are chunks chosen by a tokenizer, and they do not have to match human word boundaries.
- A permanent memory Not quite. Tokens are chunks chosen by a tokenizer, and they do not have to match human word boundaries.
- A whole finished answer Not quite. Tokens are chunks chosen by a tokenizer, and they do not have to match human word boundaries.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: What can a token be in Tokenization?

ID: draft-tokens-q1 | **Type:** definition | **Correct choice ID:** choice-draft-tokens-q1-correct

Misconception tags: None listed

Glossary terms: tokenization, tokenizer, token, prompt-tokens, response-tokens, token-id, embedding

choice-draft-tokens-q1-correct [CORRECT]: A word, word piece, punctuation mark, or other text chunk

Feedback: Insight strengthened. Tokens are model-readable chunks, not always human words.

choice-draft-tokens-q1-only-a-complete-word [DISTRACTOR]: Only a complete word

Feedback: Not quite. Tokens are chunks chosen by a tokenizer, and they do not have to match human word boundaries.

Misconception: Only a complete word | **Represented term:** token | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-tokens-q1-a-permanent-memory [DISTRACTOR]: A permanent memory

Feedback: Not quite. Tokens are chunks chosen by a tokenizer, and they do not have to match human word boundaries.

Misconception: A permanent memory | **Represented term:** memory | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-tokens-q1-a-whole-finished-answer [DISTRACTOR]: A whole finished answer

Feedback: Not quite. Tokens are chunks chosen by a tokenizer, and they do not have to match human word boundaries.

Misconception: A whole finished answer | **Represented term:** tokenization | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Tokenization most directly teach?

ID: draft-tokens-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-tokens-q2-correct

Misconception tags: None listed

Glossary terms: tokenization, tokenizer, token, prompt-tokens, response-tokens, token-id, embedding

choice-draft-tokens-q2-correct [CORRECT]: Tokenization itself does not change model weights.

Feedback: Good distinction. Tokenization itself does not change model weights.

choice-draft-tokens-q2-response-1 [DISTRACTOR]: It is the complete input given to the model before generation.

Feedback: This choice points toward Response. Response matters nearby, but this question is asking about Tokenization.

Misconception: Not all at once; response tokens are appended step by step. | Represented term: response | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Tokenization mechanism or distinction.

choice-draft-tokens-q2-generated-token-2 [DISTRACTOR]: It is mainly about Generated token, not the mechanism named in Tokenization.

Feedback: This choice points toward Generated token. Generated token matters nearby, but this question is asking about Tokenization.

Misconception: Not a permanent memory or training example by itself. | Represented term: generated-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Generated token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Tokenization mechanism or distinction.

choice-draft-tokens-q2-autoregression-3 [DISTRACTOR]: It denoises a whole image from noise.

Feedback: This choice points toward Autoregression. Autoregression matters nearby, but this question is asking about Tokenization.

Misconception: Autoregression is being treated as the same idea as Tokenization. | Represented term: autoregression | Source: explicitly confusable glossary metadata

Rationale: Tempting because Autoregression is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Tokenization mechanism or distinction.

Question 3: Which common confusion does Tokenization help learners avoid?

ID: draft-tokens-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-tokens-q3-correct

Misconception tags: None listed

Glossary terms: tokenization, tokenizer, token, prompt-tokens, response-tokens, token-id, embedding

choice-draft-tokens-q3-correct [CORRECT]: It corrects the misconception: Tokens are always whole words or human concepts.

Feedback: Good distinction. Tokenization is clearer when that misconception is separated from the mechanism.

choice-draft-tokens-q3-generated-token-1 [DISTRACTOR]: It is mainly about Generated token, not the mechanism named in Tokenization.

Feedback: This choice points toward Generated token. Generated token matters nearby, but this question is asking about Tokenization.

Misconception: Not a permanent memory or training example by itself. | Represented term: generated-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Generated token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Tokenization mechanism or distinction.

choice-draft-tokens-q3-autoregression-2 [DISTRACTOR]: It denoises a whole image from noise.

Feedback: This choice points toward Autoregression. Autoregression matters nearby, but this question is asking about Tokenization.

Misconception: Autoregression is being treated as the same idea as Tokenization. | Represented term: autoregression | Source: explicitly confusable glossary metadata

Rationale: Tempting because Autoregression is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Tokenization mechanism or distinction.

choice-draft-tokens-q3-embedding-table-3 [DISTRACTOR]: It is mainly about Embedding table, not the mechanism named in Tokenization.

Feedback: This choice points toward Embedding table. Embedding table matters nearby, but this question is asking about Tokenization.

Misconception: Not the tokenizer and not temporary context. | Represented term: embedding-table | Source: explicitly confusable glossary metadata

Rationale: Tempting because Embedding table is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Tokenization mechanism or distinction.

Questions to Review Checklist

☐ Are the learning objectives accurate?

☐ Does each question restate the topic clearly?

☐ Is one answer clearly best?

☐ Are distractors plausible but not unfair?

☐ Are the distractors close enough to teach a boundary?

☐ Does feedback explain the misconception?

☐ Are any questions redundant?

☐ Are any important questions missing?

☐ Is the question count right for topic density?

☐ Is the reading level right?

12. Token IDs

Stage: 2. Morning Commute | **Card ID:** token-ids | **Card in stage:** 4

Primary learning objective: Explain Token IDs as part of the day in the life of a prompt.

Secondary objectives: A token ID is the number assigned to a token so the model can look up its learned starting vector., Between tokenization and embedding lookup., Token IDs are part of representation during inference., Prompt tokens become token IDs before processing., Token IDs point to rows in the embedding table.

Key distinctions: Token IDs are part of representation during inference., Prompt tokens become token IDs before processing., The token ID is the meaning.

Misconceptions to correct: The token ID is the meaning.

Related glossary terms: token-id, token, tokenizer, embedding-table, embedding

Nearby glossary distractor candidates: tokenization

Why this matters: Token IDs are the bridge from text chunks to numerical computation.

Existing Live Checkpoint

Question count: 1

Question: Why does the model need token IDs?

Correct: To look up embeddings

- To look up embeddings (correct)
- To store a permanent memory Not quite. The ID is just a lookup key; the meaning comes from learned numerical patterns, not the number itself.
- To skip tensors Not quite. The ID is just a lookup key; the meaning comes from learned numerical patterns, not the number itself.
- To guarantee truth Not quite. The ID is just a lookup key; the meaning comes from learned numerical patterns, not the number itself.

Proposed Draft Question Bank

Proposed count: 2. **Rationale:** Light or narrow concept card; two checks should confirm the core distinction without overloading the learner.

Question 1: Why does the model need token IDs?

ID: draft-token-ids-q1 | **Type:** definition | **Correct choice ID:** choice-draft-token-ids-q1-correct

Misconception tags: None listed

Glossary terms: token-id, token, tokenizer, embedding-table, embedding

choice-draft-token-ids-q1-correct [CORRECT]: To look up embeddings

Feedback: Token IDs connect text chunks to learned embedding vectors.

choice-draft-token-ids-q1-to-store-a-permanent-memory [DISTRACTOR]: To store a permanent memory

Feedback: Not quite. The ID is just a lookup key; the meaning comes from learned numerical patterns, not the number itself.

Misconception: To store a permanent memory | **Represented term:** memory | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-token-ids-q1-to-skip-tensors [DISTRACTOR]: To skip tensors

Feedback: Not quite. The ID is just a lookup key; the meaning comes from learned numerical patterns, not the number itself.

Misconception: To skip tensors | **Represented term:** embedding-table | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-token-ids-q1-to-guarantee-truth [DISTRACTOR]: To guarantee truth

Feedback: Not quite. The ID is just a lookup key; the meaning comes from learned numerical patterns, not the number itself.

Misconception: To guarantee truth | **Represented term:** probability | **Source:** learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Token IDs most directly teach?

ID: draft-token-ids-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-token-ids-q2-correct

Misconception tags: None listed

Glossary terms: token-id, token, tokenizer, embedding-table, embedding

choice-draft-token-ids-q2-correct [CORRECT]: Token IDs are part of representation during inference.

Feedback: Good distinction. Token IDs are part of representation during inference.

choice-draft-token-ids-q2-hidden-state-1 [DISTRACTOR]: It is the token's fixed starting vector before context matters.

Feedback: This choice points toward Hidden state. Hidden state matters nearby, but this question is asking about Token IDs.

Misconception: Often confused with embeddings, weights, saved memory, or visible response text. | Represented term: hidden state | Source: explicitly confusable glossary metadata

Rationale: Tempting because Hidden state is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Token IDs mechanism or distinction.

choice-draft-token-ids-q2-memory-2 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Token IDs.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Token IDs mechanism or distinction.

choice-draft-token-ids-q2-vector-3 [DISTRACTOR]: It is a written sentence rather than a numerical representation.

Feedback: This choice points toward Vector. Vector matters nearby, but this question is asking about Token IDs.

Misconception: Not a paragraph, chat memory, or simple English definition. | Represented term: vector | Source: explicitly confusable glossary metadata

Rationale: Tempting because Vector is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Token IDs mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

13. Embeddings

Stage: 2. Morning Commute | **Card ID:** embeddings | **Card in stage:** 5

Primary learning objective: Explain Embeddings as part of the day in the life of a prompt.

Secondary objectives: An embedding is a learned starting vector retrieved for a token ID., At the start of model processing, before transformer layers., The embedding table is durable learned model data., Prompt tokens and response-so-far tokens are looked up as embeddings when they are inside the current context., Embeddings are vectors. The next lesson explains vectors more generally.

Key distinctions: The embedding table is durable learned model data., Prompt tokens and response-so-far tokens are looked up as embeddings when they are inside the current context., An embedding is a dictionary definition or a human memory.

Misconceptions to correct: An embedding is a dictionary definition or a human memory.

Related glossary terms: embedding, embedding-table, token-id, vector, hidden state, inference, training

Nearby glossary distractor candidates: tokenization, tokenizer, token

Why this matters: Embeddings are the first step from token IDs into the model's learned numerical space.

Existing Live Checkpoint

Question count: 1

Question: What is an embedding in this app?

Correct: A learned starting vector for a token ID

- A learned starting vector for a token ID (correct)
- A finished sentence Not quite. Embeddings are numerical vectors, not definitions, memories, or hidden English text.
- A human memory Not quite. Embeddings are numerical vectors, not definitions, memories, or hidden English text.
- A secret English definition Not quite. Embeddings are numerical vectors, not definitions, memories, or hidden English text.

Proposed Draft Question Bank

Proposed count: 2. **Rationale:** Light or narrow concept card; two checks should confirm the core distinction without overloading the learner.

Question 1: What is an embedding in this app in Embeddings?

ID: draft-embeddings-q1 | **Type:** definition | **Correct choice ID:** choice-draft-embeddings-q1-correct

Misconception tags: None listed

Glossary terms: embedding, embedding-table, token-id, vector, hidden state, inference, training

choice-draft-embeddings-q1-correct [CORRECT]: A learned starting vector for a token ID

Feedback: Insight strengthened. An embedding starts the token in numerical space before context reshapes it.

choice-draft-embeddings-q1-a-finished-sentence [DISTRACTOR]: A finished sentence

Feedback: Not quite. Embeddings are numerical vectors, not definitions, memories, or hidden English text.

Misconception: A finished sentence | **Represented term:** memory | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-embeddings-q1-a-human-memory [DISTRACTOR]: A human memory

Feedback: Not quite. Embeddings are numerical vectors, not definitions, memories, or hidden English text.

Misconception: A human memory | **Represented term:** memory | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-embeddings-q1-a-secret-english-definition [DISTRACTOR]: A secret English definition

Feedback: Not quite. Embeddings are numerical vectors, not definitions, memories, or hidden English text.

Misconception: A secret English definition | **Represented term:** token-id | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Embeddings most directly teach?

ID: draft-embeddings-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-embeddings-q2-correct

Misconception tags: None listed

Glossary terms: embedding, embedding-table, token-id, vector, hidden state, inference, training

choice-draft-embeddings-q2-correct [CORRECT]: The embedding table is durable learned model data.

Feedback: Good distinction. The embedding table is durable learned model data.

choice-draft-embeddings-q2-tokenizer-1 [DISTRACTOR]: It is mainly about Tokenizer, not the mechanism named in Embeddings.

Feedback: This choice points toward Tokenizer. Tokenizer matters nearby, but this question is asking about Embeddings.

Misconception: Not the model weights and not the embedding table. | Represented term: tokenizer | Source: explicitly confusable glossary metadata

Rationale: Tempting because Tokenizer is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Embeddings mechanism or distinction.

choice-draft-embeddings-q2-tensor-2 [DISTRACTOR]: It is a single word rather than an organized block of numbers.

Feedback: This choice points toward Tensor. Tensor matters nearby, but this question is asking about Embeddings.

Misconception: Not one word, one number, or a mysterious object by default. | Represented term: tensor | Source: explicitly confusable glossary metadata

Rationale: Tempting because Tensor is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Embeddings mechanism or distinction.

choice-draft-embeddings-q2-feature-3 [DISTRACTOR]: It is mainly about Feature, not the mechanism named in Embeddings.

Feedback: This choice points toward Feature. Feature matters nearby, but this question is asking about Embeddings.

Misconception: Not necessarily a single obvious trait like dogness or grammar. | Represented term: feature | Source: explicitly confusable glossary metadata

Rationale: Tempting because Feature is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Embeddings mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

14. Vectors

Stage: 2. Morning Commute | **Card ID:** vectors | **Card in stage:** 6

Primary learning objective: Explain Vectors as part of the day in the life of a prompt.

Secondary objectives: A vector is a list of numbers that represents features of something, such as a token., Throughout the model: embeddings, hidden states, attention calculations, MLP transformations, and output scoring all involve vectors., Some vectors are durable learned parameters, such as embedding rows., Prompt tokens and response-so-far tokens become vectors when processed in the current context., Several token vectors organized together become a tensor.

Key distinctions: Some vectors are durable learned parameters, such as embedding rows., Prompt tokens and response-so-far tokens become vectors when processed in the current context., Each vector dimension has a simple human-readable label.

Misconceptions to correct: Each vector dimension has a simple human-readable label.

Related glossary terms: vector, feature, distributed-representation, embedding, hidden state, activation

Nearby glossary distractor candidates: tensor, weight-tensor

Why this matters: Vectors let the model compute with many fuzzy features at once instead of using plain-English labels.

Existing Live Checkpoint

Question count: 1

Question: Which statement is most accurate?

Correct: Embeddings and hidden states are both vectors

- Embeddings and hidden states are both vectors (correct)
- Vectors are whole paragraphs Not quite. Vectors are numerical representations; their features are distributed and not usually human-labeled.
- Vectors are permanent chats Not quite. Vectors are numerical representations; their features are distributed and not usually human-labeled.
- Every vector dimension is a neat English meaning Not quite. Vectors are numerical representations; their features are distributed and not usually human-labeled.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: Which statement best explains Vectors in this learning card?

ID: draft-vectors-q1 | **Type:** definition | **Correct choice ID:** choice-draft-vectors-q1-correct

Misconception tags: None listed

Glossary terms: vector, feature, distributed-representation, embedding, hidden state, activation

choice-draft-vectors-q1-correct [CORRECT]: Embeddings and hidden states are both vectors

Feedback: Insight strengthened. Vectors let the model compute with many fuzzy numerical features at once.

choice-draft-vectors-q1-vectors-are-whole-paragraphs [DISTRACTOR]: Vectors are whole paragraphs

Feedback: Not quite. Vectors are numerical representations; their features are distributed and not usually human-labeled.

Misconception: Vectors are whole paragraphs | Represented term: embedding | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-vectors-q1-vectors-are-permanent-chats [DISTRACTOR]: Vectors are permanent chats

Feedback: Not quite. Vectors are numerical representations; their features are distributed and not usually human-labeled.

Misconception: Vectors are permanent chats | Represented term: feature | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-vectors-q1-every-vector-dimension-is-a-neat-english-meaning [DISTRACTOR]: Every vector dimension is a neat English meaning

Feedback: Not quite. Vectors are numerical representations; their features are distributed and not usually human-labeled.

Misconception: Every vector dimension is a neat English meaning | Represented term: hidden state | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Vectors most directly teach?

ID: draft-vectors-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-vectors-q2-correct

Misconception tags: None listed

Glossary terms: vector, feature, distributed-representation, embedding, hidden state, activation

choice-draft-vectors-q2-correct [CORRECT]: Some vectors are durable learned parameters, such as embedding rows.

Feedback: Good distinction. Some vectors are durable learned parameters, such as embedding rows.

choice-draft-vectors-q2-memory-1 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Vectors.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Vectors mechanism or distinction.

choice-draft-vectors-q2-embedding-table-2 [DISTRACTOR]: It is mainly about Embedding table, not the mechanism named in Vectors.

Feedback: This choice points toward Embedding table. Embedding table matters nearby, but this question is asking about Vectors.

Misconception: Not the tokenizer and not temporary context. | Represented term: embedding-table | Source: explicitly confusable glossary metadata

Rationale: Tempting because Embedding table is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Vectors mechanism or distinction.

choice-draft-vectors-q2-token-id-3 [DISTRACTOR]: It is the meaning of the word itself.

Feedback: This choice points toward Token ID. Token ID matters nearby, but this question is asking about Vectors.

Misconception: The ID is not the meaning, not a memory, and not a truth score. | Represented term: token-id | Source: explicitly confusable glossary metadata

Rationale: Tempting because Token ID is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Vectors mechanism or distinction.

Question 3: Which common confusion does Vectors help learners avoid?

ID: draft-vectors-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-vectors-q3-correct

Misconception tags: None listed

Glossary terms: vector, feature, distributed-representation, embedding, hidden state, activation

choice-draft-vectors-q3-correct [CORRECT]: It corrects the misconception: Each vector dimension has a simple human-readable label.

Feedback: Good distinction. Vectors is clearer when that misconception is separated from the mechanism.

choice-draft-vectors-q3-embedding-table-1 [DISTRACTOR]: It is mainly about Embedding table, not the mechanism named in Vectors.

Feedback: This choice points toward Embedding table. Embedding table matters nearby, but this question is asking about Vectors.

Misconception: Not the tokenizer and not temporary context. | Represented term: embedding-table | Source: explicitly confusable glossary metadata

Rationale: Tempting because Embedding table is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Vectors mechanism or distinction.

choice-draft-vectors-q3-token-id-2 [DISTRACTOR]: It is the meaning of the word itself.

Feedback: This choice points toward Token ID. Token ID matters nearby, but this question is asking about Vectors.

Misconception: The ID is not the meaning, not a memory, and not a truth score. | Represented term: token-id | Source: explicitly confusable glossary metadata

Rationale: Tempting because Token ID is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Vectors mechanism or distinction.

choice-draft-vectors-q3-layer-3 [DISTRACTOR]: It is a permanent memory store for the conversation.

Feedback: This choice points toward Layer. Layer matters nearby, but this question is asking about Vectors.

Misconception: Not a conscious stage of thought. | Represented term: layer | Source: explicitly confusable glossary metadata

Rationale: Tempting because Layer is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Vectors mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?

- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

15. Tensors

Stage: 2. Morning Commute | **Card ID:** tensors | **Card in stage:** 7

Primary learning objective: Explain Tensors as part of the day in the life of a prompt.

Secondary objectives: A tensor is an organized block of numbers, often holding many token vectors at once., Tensors carry data through the transformer layers during a forward pass., Weight tensors are durable learned parameters., The current context tensor may include the complete user prompt, retrieved tokens, prior conversation tokens, and response-so-far tokens such as "A jealous dog chased a startled cat across the kitchen"., Now that current context is in tensor form, transformer layers can apply attention and MLPs.

Key distinctions: Weight tensors are durable learned parameters., The current context tensor may include the complete user prompt, retrieved tokens, prior conversation tokens, and response-so-far tokens such as "A jealous dog chased a startled cat across the kitchen"., A tensor is one word, one number, or a mysterious object.

Misconceptions to correct: A tensor is one word, one number, or a mysterious object.

Related glossary terms: tensor, vector, activation, weight-tensor, input-context, layer

Nearby glossary distractor candidates: distributed-representation, feature

Why this matters: The transformer transforms tensors, not raw words.

Existing Live Checkpoint

Question count: 1

Question: Which is the better definition of tensor here?

Correct: An organized block of numbers

- An organized block of numbers (correct)
- A word in a dictionary Not quite. In this app, tensor means a shaped numerical block, not a word, rule, or memory.
- A rule written by a programmer Not quite. In this app, tensor means a shaped numerical block, not a word, rule, or memory.
- A permanent memory of the prompt Not quite. In this app, tensor means a shaped numerical block, not a word, rule, or memory.

Proposed Draft Question Bank

Proposed count: 2. **Rationale:** Light or narrow concept card; two checks should confirm the core distinction without overloading the learner.

Question 1: Which definition best describes a tensor in the model processing path?

ID: draft-tensors-q1 | **Type:** definition | **Correct choice ID:** choice-draft-tensors-q1-correct

Misconception tags: None listed

Glossary terms: tensor, vector, activation, weight-tensor, input-context, layer

choice-draft-tensors-q1-correct [CORRECT]: An organized block of numbers

Feedback: Insight strengthened. Tensors organize many vectors so model layers can process them.

choice-draft-tensors-q1-a-word-in-a-dictionary [DISTRACTOR]: A word in a dictionary

Feedback: Not quite. In this app, tensor means a shaped numerical block, not a word, rule, or memory.

Misconception: A word in a dictionary | **Represented term:** token | **Source:** same-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-tensors-q1-a-rule-written-by-a-programmer [DISTRACTOR]: A rule written by a programmer

Feedback: Not quite. In this app, tensor means a shaped numerical block, not a word, rule, or memory.

Misconception: A rule written by a programmer | **Represented term:** activation | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-tensors-q1-a-permanent-memory-of-the-prompt [DISTRACTOR]: A permanent memory of the prompt

Feedback: Not quite. In this app, tensor means a shaped numerical block, not a word, rule, or memory.

Misconception: A permanent memory of the prompt | **Represented term:** memory | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Tensors most directly teach?

ID: draft-tensors-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-tensors-q2-correct

Misconception tags: None listed

Glossary terms: tensor, vector, activation, weight-tensor, input-context, layer

choice-draft-tensors-q2-correct [CORRECT]: Weight tensors are durable learned parameters.

Feedback: Good distinction. Weight tensors are durable learned parameters.

choice-draft-tensors-q2-hidden-state-1 [DISTRACTOR]: It is the token's fixed starting vector before context matters.

Feedback: This choice points toward Hidden state. Hidden state matters nearby, but this question is asking about Tensors.

Misconception: Often confused with embeddings, weights, saved memory, or visible response text. | Represented term: hidden state | Source: explicitly confusable glossary metadata

Rationale: Tempting because Hidden state is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Tensors mechanism or distinction.

choice-draft-tensors-q2-distributed-representation-2 [DISTRACTOR]: It is mainly about Distributed representation, not the mechanism named in Tensors.

Feedback: This choice points toward Distributed representation. Distributed representation matters nearby, but this question is asking about Tensors.

Misconception: Not one dimension per English concept. | Represented term: distributed-representation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Distributed representation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Tensors mechanism or distinction.

choice-draft-tensors-q2-inference-3 [DISTRACTOR]: It durably trains the model on the current prompt.

Feedback: This choice points toward Inference. Inference matters nearby, but this question is asking about Tensors.

Misconception: Inference can use context during one run, but it does not normally train the model. | Represented term: inference | Source: explicitly confusable glossary metadata

Rationale: Tempting because Inference is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Tensors mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

16. Attention

Stage: 3. Workday | Card ID: attention | Card in stage: 1

Primary learning objective: Explain Attention as part of the day in the life of a prompt.

Secondary objectives: Attention is a mechanism that assigns relevance weights between token positions in the current context., Inside transformer layers during a forward pass., Attention uses learned weights, but the attention pattern for a prompt is temporary during inference., Prompt tokens, retrieved context, and response-so-far tokens can all attend to one another if they are inside the current context., After tokens become vectors and tensors, attention is one main operation inside transformer layers.

Key distinctions: Attention uses learned weights, but the attention pattern for a prompt is temporary during inference., Prompt tokens, retrieved context, and response-so-far tokens can all attend to one another if they are inside the current context., Attention means consciousness or awareness.

Misconceptions to correct: Attention means consciousness or awareness.

Related glossary terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer

Nearby glossary distractor candidates: distributed-representation, feature, vector, tensor, activation, weight-tensor, embedding-table

Why this matters: Attention lets one token position use information from other visible token positions in the current context.

Existing Live Checkpoint

Question count: 1

Question: What does attention do in a transformer?

Correct: It weights relevance between token positions.

- It weights relevance between token positions. (correct)
- It gives the model consciousness. Not quite. Attention is not awareness. It is temporary relevance weighting inside the forward pass.
- It permanently stores memories. Not quite. Attention patterns are temporary during inference; they do not store permanent memories.
- It chooses the final answer by itself. Not quite. Attention helps shape hidden states; logits, softmax, and sampling come later.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: What does attention do in a transformer?

ID: draft-attention-q1 | Type: definition | Correct choice ID: choice-draft-attention-q1-correct

Misconception tags: None listed

Glossary terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer

choice-draft-attention-q1-correct [CORRECT]: It weights relevance between token positions.

Feedback: Attention weights relevance between token positions in the current context.

choice-draft-attention-q1-it-gives-the-model-consciousness [DISTRACTOR]: It gives the model consciousness.

Feedback: Not quite. Attention is not awareness. It is temporary relevance weighting inside the forward pass.

Misconception: It gives the model consciousness. | Represented term: llm | Source: learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-attention-q1-it-permanently-stores-memories [DISTRACTOR]: It permanently stores memories.

Feedback: Not quite. Attention patterns are temporary during inference; they do not store permanent memories.

Misconception: It permanently stores memories. | Represented term: layer | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-attention-q1-it-chooses-the-final-answer-by-itself [DISTRACTOR]: It chooses the final answer by itself.

Feedback: Not quite. Attention helps shape hidden states; logits, softmax, and sampling come later.

Misconception: It chooses the final answer by itself. | Represented term: sampling | Source: nearby-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Attention most directly teach?

ID: draft-attention-q2 | Type: mechanism | Correct choice ID: choice-draft-attention-q2-correct

Misconception tags: None listed

Glossary terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer

choice-draft-attention-q2-correct [CORRECT]: Attention uses learned weights, but the attention pattern for a prompt is temporary during inference.

Feedback: Good distinction. Attention uses learned weights, but the attention pattern for a prompt is temporary during inference.

choice-draft-attention-q2-prompt-1 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about Attention.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Attention mechanism or distinction.

choice-draft-attention-q2-memory-2 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Attention.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Attention mechanism or distinction.

choice-draft-attention-q2-system-prompt-3 [DISTRACTOR]: It is mainly about System prompt, not the mechanism named in Attention.

Feedback: This choice points toward System prompt. System prompt matters nearby, but this question is asking about Attention.

Misconception: Not fine-tuning, training, permanent memory, or conscience. | Represented term: system-prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because System prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Attention mechanism or distinction.

Question 3: Which common confusion does Attention help learners avoid?

ID: draft-attention-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-attention-q3-correct

Misconception tags: None listed

Glossary terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer

choice-draft-attention-q3-correct [CORRECT]: It corrects the misconception: Attention means consciousness or awareness.

Feedback: Good distinction. Attention is clearer when that misconception is separated from the mechanism.

choice-draft-attention-q3-memory-1 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Attention.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Attention mechanism or distinction.

choice-draft-attention-q3-system-prompt-2 [DISTRACTOR]: It is mainly about System prompt, not the mechanism named in Attention.

Feedback: This choice points toward System prompt. System prompt matters nearby, but this question is asking about Attention.

Misconception: Not fine-tuning, training, permanent memory, or conscience. | Represented term: system-prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because System prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Attention mechanism or distinction.

choice-draft-attention-q3-response-so-far-3 [DISTRACTOR]: It is mainly about Response-so-far, not the mechanism named in Attention.

Feedback: This choice points toward Response-so-far. Response-so-far matters nearby, but this question is asking about Attention.

Misconception: Not the whole final answer, permanent memory, or training data. | Represented term: response-so-far | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response-so-far is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Attention mechanism or distinction.

Question 4: How does Attention connect to the prompt's Journey?

ID: draft-attention-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-attention-q4-correct

Misconception tags: None listed

Glossary terms: attention, relevance-weight, input-context, prompt-tokens, response-tokens, hidden state, layer

choice-draft-attention-q4-correct [CORRECT]: After tokens become vectors and tensors, attention is one main operation inside transformer layers.

Feedback: Good connection. After tokens become vectors and tensors, attention is one main operation inside transformer layers.

choice-draft-attention-q4-system-prompt-1 [DISTRACTOR]: It is mainly about System prompt, not the mechanism named in Attention.

Feedback: This choice points toward System prompt. System prompt matters nearby, but this question is asking about Attention.

Misconception: Not fine-tuning, training, permanent memory, or conscience. | Represented term: system-prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because System prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Attention mechanism or distinction.

choice-draft-attention-q4-response-so-far-2 [DISTRACTOR]: It is mainly about Response-so-far, not the mechanism named in Attention.

Feedback: This choice points toward Response-so-far. Response-so-far matters nearby, but this question is asking about Attention.

Misconception: Not the whole final answer, permanent memory, or training data. | Represented term: response-so-far | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response-so-far is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Attention mechanism or distinction.

choice-draft-attention-q4-retrieval-3 [DISTRACTOR]: It is the same as fine-tuning the model on new data.

Feedback: This choice points toward Retrieval. Retrieval matters nearby, but this question is asking about Attention.

Misconception: Retrieval is being treated as the same idea as Attention. | Represented term: retrieval | Source: explicitly confusable glossary metadata

Rationale: Tempting because Retrieval is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Attention mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

17. MLP

Stage: 3. Workday | Card ID: mlp | Card in stage: 2

Primary learning objective: Explain MLP as part of the day in the life of a prompt.

Secondary objectives: An MLP, or multi-layer perceptron, is a feed-forward network that reshapes each token's feature vector inside a transformer layer., Inside each transformer layer, usually after attention., The MLP has learned weights, but the feature values it produces during inference are temporary activations., The MLP transforms vectors for prompt tokens, retrieved/context tokens, and response-so-far tokens that are currently inside the context., Attention mixes information across token positions.

Key distinctions: The MLP has learned weights, but the feature values it produces during inference are temporary activations., The MLP transforms vectors for prompt tokens, retrieved/context tokens, and response-so-far tokens that are currently inside the context., The MLP mixes information between different token positions.

Misconceptions to correct: The MLP mixes information between different token positions.

Related glossary terms: MLP, multi-layer-perceptron, feed-forward-network, feature, input-context, hidden state, attention

Nearby glossary distractor candidates: relevance-weight, weight-tensor, activation, tensor

Why this matters: The MLP helps turn mixed context into richer token representations before later layers and output scoring.

Existing Live Checkpoint

Question count: 1

Question: What does the MLP mainly do here?

Correct: Reshapes each token position's feature vector

- Reshapes each token position's feature vector (correct)
- Mixes information between different token positions Not quite. Attention mixes across token positions. The MLP reshapes within each token position.
- Reads private files Not quite. The MLP transforms current-context vectors; it does not retrieve private files.
- Permanently stores the prompt Not quite. The MLP has durable weights, but its inference activations are temporary.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: What does the MLP mainly do here?

ID: draft-mlp-q1 | Type: definition | Correct choice ID: choice-draft-mlp-q1-correct

Misconception tags: None listed

Glossary terms: MLP, multi-layer-perceptron, feed-forward-network, feature, input-context, hidden state, attention

choice-draft-mlp-q1-correct [CORRECT]: Reshapes each token position's feature vector

Feedback: The MLP reshapes features for a token position during the forward pass.

choice-draft-mlp-q1-mixes-information-between-different-token-positions [DISTRACTOR]: Mixes information between different token positions

Feedback: Not quite. Attention mixes across token positions. The MLP reshapes within each token position.

Misconception: Mixes information between different token positions | Represented term: token | Source: nearby-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-mlp-q1-reads-private-files [DISTRACTOR]: Reads private files

Feedback: Not quite. The MLP transforms current-context vectors; it does not retrieve private files.

Misconception: Reads private files | Represented term: feature | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-mlp-q1-permanently-stores-the-prompt [DISTRACTOR]: Permanently stores the prompt

Feedback: Not quite. The MLP has durable weights, but its inference activations are temporary.

Misconception: Permanently stores the prompt | Represented term: hidden state | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does MLP most directly teach?

ID: draft-mlp-q2 | Type: mechanism | Correct choice ID: choice-draft-mlp-q2-correct

Misconception tags: None listed

Glossary terms: MLP, multi-layer-perceptron, feed-forward-network, feature, input-context, hidden state, attention

choice-draft-mlp-q2-correct [CORRECT]: The MLP has learned weights, but the feature values it produces during inference are temporary activations.

Feedback: Good distinction. The MLP has learned weights, but the feature values it produces during inference are temporary activations.

choice-draft-mlp-q2-distributed-representation-1 [DISTRACTOR]: It is mainly about Distributed representation, not the mechanism named in MLP.

Feedback: This choice points toward Distributed representation. Distributed representation matters nearby, but this question is asking about MLP.

Misconception: Not one dimension per English concept. | Represented term: distributed-representation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Distributed representation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific MLP mechanism or distinction.

choice-draft-mlp-q2-context-window-2 [DISTRACTOR]: It is permanent model memory.

Feedback: This choice points toward Context window. Context window matters nearby, but this question is asking about MLP.

Misconception: Not permanent memory; it is temporary working context. | Represented term: context window | Source: explicitly confusable glossary metadata

Rationale: Tempting because Context window is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific MLP mechanism or distinction.

choice-draft-mlp-q2-prompt-3 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about MLP.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific MLP mechanism or distinction.

Question 3: Which common confusion does MLP help learners avoid?

ID: draft-mlp-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-mlp-q3-correct

Misconception tags: None listed

Glossary terms: MLP, multi-layer-perceptron, feed-forward-network, feature, input-context, hidden state, attention

choice-draft-mlp-q3-correct [CORRECT]: It corrects the misconception: The MLP mixes information between different token positions.

Feedback: Good distinction. MLP is clearer when that misconception is separated from the mechanism.

choice-draft-mlp-q3-context-window-1 [DISTRACTOR]: It is permanent model memory.

Feedback: This choice points toward Context window. Context window matters nearby, but this question is asking about MLP.

Misconception: Not permanent memory; it is temporary working context. | Represented term: context window | Source: explicitly confusable glossary metadata

Rationale: Tempting because Context window is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific MLP mechanism or distinction.

choice-draft-mlp-q3-prompt-2 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about MLP.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific MLP mechanism or distinction.

choice-draft-mlp-q3-memory-3 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about MLP.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific MLP mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?

- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

18. Layers

Stage: 3. Workday | Card ID: layers | Card in stage: 3

Primary learning objective: Explain Layers as part of the day in the life of a prompt.

Secondary objectives: A layer is a repeated transformer block that refines hidden states through attention, MLPs, residual paths, and normalization., During each forward pass through the transformer stack., Layer weights are durable learned parameters., All current-context tokens move through the stack during a forward pass., Each layer repeats attention and MLP transformations, while residual paths and normalization help useful signal carry forward.

Key distinctions: Layer weights are durable learned parameters., All current-context tokens move through the stack during a forward pass., Each layer is a human-like thought or reasoning step.

Misconceptions to correct: Each layer is a human-like thought or reasoning step.

Related glossary terms: layer, transformer-layer, attention, MLP, hidden state, residual-path, normalization

Nearby glossary distractor candidates: feed-forward-network, multi-layer-perceptron

Why this matters: One layer is not the whole model. The repeated stack is what gradually reshapes token representations.

Existing Live Checkpoint

Question count: 1

Question: What do layers repeatedly refine?

Correct: Temporary hidden states

- Temporary hidden states (correct)
- Permanent user memory Not quite. Layers process temporary activations during a forward pass; they do not save user memory.
- The app interface Not quite. Layers are inside the model, not the app chrome.
- The training dataset Not quite. The training dataset shaped weights earlier; layers process the current context now.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: What do layers repeatedly refine?

ID: draft-layers-q1 | Type: definition | Correct choice ID: choice-draft-layers-q1-correct

Misconception tags: None listed

Glossary terms: layer, transformer-layer, attention, MLP, hidden state, residual-path, normalization

choice-draft-layers-q1-correct [CORRECT]: Temporary hidden states

Feedback: Layers repeat attention and MLP transformations to refine temporary hidden states.

choice-draft-layers-q1-permanent-user-memory [DISTRACTOR]: Permanent user memory

Feedback: Not quite. Layers process temporary activations during a forward pass; they do not save user memory.

Misconception: Permanent user memory | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-layers-q1-the-app-interface [DISTRACTOR]: The app interface

Feedback: Not quite. Layers are inside the model, not the app chrome.

Misconception: The app interface | Represented term: MLP | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-layers-q1-the-training-dataset [DISTRACTOR]: The training dataset

Feedback: Not quite. The training dataset shaped weights earlier; layers process the current context now.

Misconception: The training dataset | Represented term: training | Source: nearby-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Layers most directly teach?

ID: draft-layers-q2 | Type: mechanism | Correct choice ID: choice-draft-layers-q2-correct

Misconception tags: None listed

Glossary terms: layer, transformer-layer, attention, MLP, hidden state, residual-path, normalization

choice-draft-layers-q2-correct [CORRECT]: Layer weights are durable learned parameters.

Feedback: Good distinction. Layer weights are durable learned parameters.

choice-draft-layers-q2-feature-1 [DISTRACTOR]: It is mainly about Feature, not the mechanism named in Layers.

Feedback: This choice points toward Feature. Feature matters nearby, but this question is asking about Layers.

Misconception: Not necessarily a single obvious trait like dogness or grammar. | Represented term: feature | Source: explicitly confusable glossary metadata

Rationale: Tempting because Feature is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Layers mechanism or distinction.

choice-draft-layers-q2-multi-layer-perceptron-2 [DISTRACTOR]: It is mainly about Multi-layer perceptron, not the mechanism named in Layers.

Feedback: This choice points toward Multi-layer perceptron. Multi-layer perceptron matters nearby, but this question is asking about Layers.

Misconception: Not a biological neuron or a separate chat agent. | Represented term: multi-layer-perceptron | Source: explicitly confusable glossary metadata

Rationale: Tempting because Multi-layer perceptron is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Layers mechanism or distinction.

choice-draft-layers-q2-feed-forward-network-3 [DISTRACTOR]: It is mainly about Feed-forward network, not the mechanism named in Layers.

Feedback: This choice points toward Feed-forward network. Feed-forward network matters nearby, but this question is asking about Layers.

Misconception: Not retrieval, memory, or cross-token attention. | Represented term: feed-forward-network | Source: explicitly confusable glossary metadata

Rationale: Tempting because Feed-forward network is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Layers mechanism or distinction.

Question 3: Which common confusion does Layers help learners avoid?

ID: draft-layers-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-layers-q3-correct

Misconception tags: None listed

Glossary terms: layer, transformer-layer, attention, MLP, hidden state, residual-path, normalization

choice-draft-layers-q3-correct [CORRECT]: It corrects the misconception: Each layer is a human-like thought or reasoning step.

Feedback: Good distinction. Layers is clearer when that misconception is separated from the mechanism.

choice-draft-layers-q3-multi-layer-perceptron-1 [DISTRACTOR]: It is mainly about Multi-layer perceptron, not the mechanism named in Layers.

Feedback: This choice points toward Multi-layer perceptron. Multi-layer perceptron matters nearby, but this question is asking about Layers.

Misconception: Not a biological neuron or a separate chat agent. | Represented term: multi-layer-perceptron | Source: explicitly confusable glossary metadata

Rationale: Tempting because Multi-layer perceptron is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Layers mechanism or distinction.

choice-draft-layers-q3-feed-forward-network-2 [DISTRACTOR]: It is mainly about Feed-forward network, not the mechanism named in Layers.

Feedback: This choice points toward Feed-forward network. Feed-forward network matters nearby, but this question is asking about Layers.

Misconception: Not retrieval, memory, or cross-token attention. | Represented term: feed-forward-network | Source: explicitly confusable glossary metadata

Rationale: Tempting because Feed-forward network is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Layers mechanism or distinction.

choice-draft-layers-q3-embedding-3 [DISTRACTOR]: It is the temporary context-shaped hidden state.

Feedback: This choice points toward Embedding. Embedding matters nearby, but this question is asking about Layers.

Misconception: Often confused with a dictionary definition, a human memory, or a hidden state. | Represented term: embedding | Source: explicitly confusable glossary metadata

Rationale: Tempting because Embedding is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Layers mechanism or distinction.

Question 4: How does Layers connect to the prompt's Journey?

ID: draft-layers-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-layers-q4-correct

Misconception tags: None listed

Glossary terms: layer, transformer-layer, attention, MLP, hidden state, residual-path, normalization

choice-draft-layers-q4-correct [CORRECT]: Each layer repeats attention and MLP transformations, while residual paths and normalization help useful signal carry forward.

Feedback: Good connection. Each layer repeats attention and MLP transformations, while residual paths and normalization help useful signal carry forward.

choice-draft-layers-q4-feed-forward-network-1 [DISTRACTOR]: It is mainly about Feed-forward network, not the mechanism named in Layers.

Feedback: This choice points toward Feed-forward network. Feed-forward network matters nearby, but this question is asking about Layers.

Misconception: Not retrieval, memory, or cross-token attention. | Represented term: feed-forward-network | Source: explicitly confusable glossary metadata

Rationale: Tempting because Feed-forward network is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Layers mechanism or distinction.

choice-draft-layers-q4-embedding-2 [DISTRACTOR]: It is the temporary context-shaped hidden state.

Feedback: This choice points toward Embedding. Embedding matters nearby, but this question is asking about Layers.

Misconception: Often confused with a dictionary definition, a human memory, or a hidden state. | Represented term: embedding | Source: explicitly confusable glossary metadata

Rationale: Tempting because Embedding is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Layers mechanism or distinction.

choice-draft-layers-q4-memory-3 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Layers.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Layers mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

19. Hidden States

Stage: 3. Workday | Card ID: hidden-states | Card in stage: 4

Primary learning objective: Explain Hidden States as part of the day in the life of a prompt.

Secondary objectives: A hidden state is a temporary context-shaped vector for a token at a particular layer., After embeddings enter the transformer layer stack during a forward pass., Hidden states are temporary activations., Prompt tokens, retrieved tokens, and response-so-far tokens all have hidden states while the model processes the current context., Embeddings start token vectors.

Key distinctions: Hidden states are temporary activations., Prompt tokens, retrieved tokens, and response-so-far tokens all have hidden states while the model processes the current context., Hidden states are secret memories, encrypted English, or stored thoughts.

Misconceptions to correct: Hidden states are secret memories, encrypted English, or stored thoughts.

Related glossary terms: hidden state, activation, temporary-activation, embedding, weight, input-context, forward-pass, layer, inference

Nearby glossary distractor candidates: embedding-table, token-id, vector, feature, distributed-representation

Why this matters: Hidden states are where the current context changes token representations before next-token scoring.

Existing Live Checkpoint

Question count: 1

Question: Why is it called hidden state?

Correct: It is internal numbers, not visible text

- It is internal numbers, not visible text (correct)
- It is encrypted English Not quite. Hidden states are numerical activations, not a hidden English sentence.
- It is a secret memory file Not quite. Hidden states are temporary during the forward pass, not saved user memory.
- It is the training data Not quite. Training data shaped weights earlier; hidden states are created for the current context.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: Why is it called hidden state in Hidden States?

ID: draft-hidden-states-q1 | Type: definition | Correct choice ID: choice-draft-hidden-states-q1-correct

Misconception tags: None listed

Glossary terms: hidden state, activation, temporary-activation, embedding, weight, input-context, forward-pass, layer, inference

choice-draft-hidden-states-q1-correct [CORRECT]: It is internal numbers, not visible text

Feedback: Insight unlocked. Hidden means inside the model's working representation, not secret memory.

choice-draft-hidden-states-q1-it-is-encrypted-english [DISTRACTOR]: It is encrypted English

Feedback: Not quite. Hidden states are numerical activations, not a hidden English sentence.

Misconception: It is encrypted English | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-hidden-states-q1-it-is-a-secret-memory-file [DISTRACTOR]: It is a secret memory file

Feedback: Not quite. Hidden states are temporary during the forward pass, not saved user memory.

Misconception: It is a secret memory file | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-hidden-states-q1-it-is-the-training-data [DISTRACTOR]: It is the training data

Feedback: Not quite. Training data shaped weights earlier; hidden states are created for the current context.

Misconception: It is the training data | Represented term: training | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Hidden States most directly teach?

ID: draft-hidden-states-q2 | Type: mechanism | Correct choice ID: choice-draft-hidden-states-q2-correct

Misconception tags: None listed

Glossary terms: hidden state, activation, temporary-activation, embedding, weight, input-context, forward-pass, layer, inference

choice-draft-hidden-states-q2-correct [CORRECT]: Hidden states are temporary activations.

Feedback: Good distinction. Hidden states are temporary activations.

choice-draft-hidden-states-q2-tensor-1 [DISTRACTOR]: It is a single word rather than an organized block of numbers.

Feedback: This choice points toward Tensor. Tensor matters nearby, but this question is asking about Hidden States.

Misconception: Not one word, one number, or a mysterious object by default. | Represented term: tensor | Source: explicitly confusable glossary metadata

Rationale: Tempting because Tensor is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hidden States mechanism or distinction.

choice-draft-hidden-states-q2-attention-2 [DISTRACTOR]: It means the model is paying human-like attention.

Feedback: This choice points toward Attention. Attention matters nearby, but this question is asking about Hidden States.

Misconception: Attention is not consciousness, awareness, desire, or human focus. | Represented term: attention | Source: explicitly confusable glossary metadata

Rationale: Tempting because Attention is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hidden States mechanism or distinction.

choice-draft-hidden-states-q2-embedding-table-3 [DISTRACTOR]: It is mainly about Embedding table, not the mechanism named in Hidden States.

Feedback: This choice points toward Embedding table. Embedding table matters nearby, but this question is asking about Hidden States.

Misconception: Not the tokenizer and not temporary context. | Represented term: embedding-table | Source: explicitly confusable glossary metadata

Rationale: Tempting because Embedding table is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hidden States mechanism or distinction.

Question 3: Which common confusion does Hidden States help learners avoid?

ID: draft-hidden-states-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-hidden-states-q3-correct

Misconception tags: None listed

Glossary terms: hidden state, activation, temporary-activation, embedding, weight, input-context, forward-pass, layer, inference

choice-draft-hidden-states-q3-correct [CORRECT]: It corrects the misconception: Hidden states are secret memories, encrypted English, or stored thoughts.

Feedback: Good distinction. Hidden States is clearer when that misconception is separated from the mechanism.

choice-draft-hidden-states-q3-attention-1 [DISTRACTOR]: It means the model is paying human-like attention.

Feedback: This choice points toward Attention. Attention matters nearby, but this question is asking about Hidden States.

Misconception: Attention is not consciousness, awareness, desire, or human focus. | Represented term: attention | Source: explicitly confusable glossary metadata

Rationale: Tempting because Attention is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hidden States mechanism or distinction.

choice-draft-hidden-states-q3-embedding-table-2 [DISTRACTOR]: It is mainly about Embedding table, not the mechanism named in Hidden States.

Feedback: This choice points toward Embedding table. Embedding table matters nearby, but this question is asking about Hidden States.

Misconception: Not the tokenizer and not temporary context. | Represented term: embedding-table | Source: explicitly confusable glossary metadata

Rationale: Tempting because Embedding table is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hidden States mechanism or distinction.

choice-draft-hidden-states-q3-token-id-3 [DISTRACTOR]: It is the meaning of the word itself.

Feedback: This choice points toward Token ID. Token ID matters nearby, but this question is asking about Hidden States.

Misconception: The ID is not the meaning, not a memory, and not a truth score. | Represented term: token-id | Source: explicitly confusable glossary metadata

Rationale: Tempting because Token ID is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hidden States mechanism or distinction.

Question 4: How does Hidden States connect to the prompt's Journey?

ID: draft-hidden-states-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-hidden-states-q4-correct

Misconception tags: None listed

Glossary terms: hidden state, activation, temporary-activation, embedding, weight, input-context, forward-pass, layer, inference

choice-draft-hidden-states-q4-correct [CORRECT]: Embeddings start token vectors.

Feedback: Good connection. Embeddings start token vectors.

choice-draft-hidden-states-q4-embedding-table-1 [DISTRACTOR]: It is mainly about Embedding table, not the mechanism named in Hidden States.

Feedback: This choice points toward Embedding table. Embedding table matters nearby, but this question is asking about Hidden States.

Misconception: Not the tokenizer and not temporary context. | Represented term: embedding-table | Source: explicitly confusable glossary metadata

Rationale: Tempting because Embedding table is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hidden States mechanism or distinction.

choice-draft-hidden-states-q4-token-id-2 [DISTRACTOR]: It is the meaning of the word itself.

Feedback: This choice points toward Token ID. Token ID matters nearby, but this question is asking about Hidden States.

Misconception: The ID is not the meaning, not a memory, and not a truth score. | Represented term: token-id | Source: explicitly confusable glossary metadata

Rationale: Tempting because Token ID is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hidden States mechanism or distinction.

choice-draft-hidden-states-q4-parameter-3 [DISTRACTOR]: It is mainly about Parameter, not the mechanism named in Hidden States.

Feedback: This choice points toward Parameter. Parameter matters nearby, but this question is asking about Hidden States.

Misconception: Often used interchangeably with weights in casual explanations. | Represented term: parameter | Source: explicitly confusable glossary metadata

Rationale: Tempting because Parameter is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hidden States mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

20. Logits

Stage: 4. Decision Room | **Card ID:** logits | **Card in stage:** 1

Primary learning objective: Explain Logits as part of the day in the life of a prompt.

Secondary objectives: Logits are raw scores for possible next tokens before they are turned into probabilities., Near the end of a forward pass, after hidden states are processed and before softmax., Logits are temporary scores for this forward pass., Logits score possible next response tokens based on the current context, including the prompt and response-so-far., Hidden states produce logits.

Key distinctions: Logits are temporary scores for this forward pass., Logits score possible next response tokens based on the current context, including the prompt and response-so-far., Logits are probabilities or truth scores.

Misconceptions to correct: Logits are probabilities or truth scores.

Related glossary terms: logits, hidden state, next-token, generated-token, softmax, probability

Nearby glossary distractor candidates: relevance-weight, attention, weight-tensor, MLP, activation, multi-layer-perceptron, tensor, feed-forward-network

Why this matters: Logits are the model's raw candidate scores before probability conversion and sampling.

Existing Live Checkpoint

Question count: 1

Question: What are logits?

Correct: Raw scores before probabilities.

- Raw scores before probabilities. (correct)
- Probabilities that sum to one. Not quite. Softmax comes next and turns logits into probabilities.
- Permanent model memories. Not quite. Logits are temporary scores during this forward pass, not stored memories.
- The whole generated answer. Not quite. Logits score candidate next tokens; they do not contain the full response.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: What are logits?

ID: draft-logits-q1 | **Type:** definition | **Correct choice ID:** choice-draft-logits-q1-correct

Misconception tags: None listed

Glossary terms: logits, hidden state, next-token, generated-token, softmax, probability

choice-draft-logits-q1-correct [CORRECT]: Raw scores before probabilities.

Feedback: Insight strengthened. Logits are temporary raw scores for possible next tokens.

choice-draft-logits-q1-probabilities-that-sum-to-one [DISTRACTOR]: Probabilities that sum to one.

Feedback: Not quite. Softmax comes next and turns logits into probabilities.

Misconception: Probabilities that sum to one. | Represented term: softmax | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-logits-q1-permanent-model-memories [DISTRACTOR]: Permanent model memories.

Feedback: Not quite. Logits are temporary scores during this forward pass, not stored memories.

Misconception: Permanent model memories. | Represented term: sampling | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-logits-q1-the-whole-generated-answer [DISTRACTOR]: The whole generated answer.

Feedback: Not quite. Logits score candidate next tokens; they do not contain the full response.

Misconception: The whole generated answer. | Represented term: next-token | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Logits most directly teach?

ID: draft-logits-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-logits-q2-correct

Misconception tags: None listed

Glossary terms: logits, hidden state, next-token, generated-token, softmax, probability

choice-draft-logits-q2-correct [CORRECT]: Logits are temporary scores for this forward pass.

Feedback: Good distinction. Logits are temporary scores for this forward pass.

choice-draft-logits-q2-memory-1 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Logits.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Logits mechanism or distinction.

choice-draft-logits-q2-activation-2 [DISTRACTOR]: It is mainly about Activation, not the mechanism named in Logits.

Feedback: This choice points toward Activation. Activation matters nearby, but this question is asking about Logits.

Misconception: Not a learned weight, though weights help produce activations. | Represented term: activation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Activation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Logits mechanism or distinction.

choice-draft-logits-q2-vector-3 [DISTRACTOR]: It is a written sentence rather than a numerical representation.

Feedback: This choice points toward Vector. Vector matters nearby, but this question is asking about Logits.

Misconception: Not a paragraph, chat memory, or simple English definition. | Represented term: vector | Source: explicitly confusable glossary metadata

Rationale: Tempting because Vector is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Logits mechanism or distinction.

Question 3: Which common confusion does Logits help learners avoid?

ID: draft-logits-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-logits-q3-correct

Misconception tags: None listed

Glossary terms: logits, hidden state, next-token, generated-token, softmax, probability

choice-draft-logits-q3-correct [CORRECT]: It corrects the misconception: Logits are probabilities or truth scores.

Feedback: Good distinction. Logits is clearer when that misconception is separated from the mechanism.

choice-draft-logits-q3-activation-1 [DISTRACTOR]: It is mainly about Activation, not the mechanism named in Logits.

Feedback: This choice points toward Activation. Activation matters nearby, but this question is asking about Logits.

Misconception: Not a learned weight, though weights help produce activations. | Represented term: activation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Activation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Logits mechanism or distinction.

choice-draft-logits-q3-vector-2 [DISTRACTOR]: It is a written sentence rather than a numerical representation.

Feedback: This choice points toward Vector. Vector matters nearby, but this question is asking about Logits.

Misconception: Not a paragraph, chat memory, or simple English definition. | Represented term: vector | Source: explicitly confusable glossary metadata

Rationale: Tempting because Vector is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Logits mechanism or distinction.

choice-draft-logits-q3-tensor-3 [DISTRACTOR]: It is a single word rather than an organized block of numbers.

Feedback: This choice points toward Tensor. Tensor matters nearby, but this question is asking about Logits.

Misconception: Not one word, one number, or a mysterious object by default. | Represented term: tensor | Source: explicitly confusable glossary metadata

Rationale: Tempting because Tensor is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Logits mechanism or distinction.

Questions to Review Checklist

☐ Are the learning objectives accurate?

☐ Does each question restate the topic clearly?

☐ Is one answer clearly best?

☐ Are distractors plausible but not unfair?

☐ Are the distractors close enough to teach a boundary?

☐ Does feedback explain the misconception?

☐ Are any questions redundant?

☐ Are any important questions missing?

☐ Is the question count right for topic density?

☐ Is the reading level right?

21. Softmax

Stage: 4. Decision Room | **Card ID:** softmax | **Card in stage:** 2

Primary learning objective: Explain Softmax as part of the day in the life of a prompt.

Secondary objectives: Softmax is a function that turns raw next-token scores into probabilities that sum to one., During decoding, after logits and before sampling., Softmax creates temporary probabilities for the current next-token decision., Softmax helps choose the next response token from candidates shaped by the current context., Logits go in. Probabilities come out.

Key distinctions: Softmax creates temporary probabilities for the current next-token decision., Softmax helps choose the next response token from candidates shaped by the current context., Softmax guarantees truth or certainty.

Misconceptions to correct: Softmax guarantees truth or certainty.

Related glossary terms: softmax, logits, decoding-step, sampling, probability

Nearby glossary distractor candidates: hidden state, temporary-activation, normalization, residual-path, transformer-layer, layer

Why this matters: Softmax turns a scoreboard into a choice-ready probability cloud.

Existing Live Checkpoint

Question count: 1

Question: What does softmax do?

Correct: Turns raw scores into probabilities.

- Turns raw scores into probabilities. (correct)
- Checks whether an answer is true. Not quite. Softmax normalizes scores into probabilities, but it does not verify truth.
- Permanently updates weights. Not quite. Softmax is temporary inference math, not training.
- Chooses the whole response at once. Not quite. Softmax prepares probabilities for one next-token choice.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: What does softmax do?

ID: draft-softmax-q1 | **Type:** definition | **Correct choice ID:** choice-draft-softmax-q1-correct

Misconception tags: None listed

Glossary terms: softmax, logits, decoding-step, sampling, probability

choice-draft-softmax-q1-correct [CORRECT]: Turns raw scores into probabilities.

Feedback: Insight strengthened. Softmax turns raw scores into probabilities that can be sampled.

choice-draft-softmax-q1-checks-whether-an-answer-is-true [DISTRACTOR]: Checks whether an answer is true.

Feedback: Not quite. Softmax normalizes scores into probabilities, but it does not verify truth.

Misconception: Checks whether an answer is true. | Represented term: probability | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-softmax-q1-permanently-updates-weights [DISTRACTOR]: Permanently updates weights.

Feedback: Not quite. Softmax is temporary inference math, not training.

Misconception: Permanently updates weights. | Represented term: training | Source: nearby-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-softmax-q1-chooses-the-whole-response-at-once [DISTRACTOR]: Chooses the whole response at once.

Feedback: Not quite. Softmax prepares probabilities for one next-token choice.

Misconception: Chooses the whole response at once. | Represented term: sampling | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Softmax most directly teach?

ID: draft-softmax-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-softmax-q2-correct

Misconception tags: None listed

Glossary terms: softmax, logits, decoding-step, sampling, probability

choice-draft-softmax-q2-correct [CORRECT]: Softmax creates temporary probabilities for the current next-token decision.

Feedback: Good distinction. Softmax creates temporary probabilities for the current next-token decision.

choice-draft-softmax-q2-temperature-1 [DISTRACTOR]: It is mainly about Temperature, not the mechanism named in Softmax.

Feedback: This choice points toward Temperature. Temperature matters nearby, but this question is asking about Softmax.

Misconception: Not creativity, truth, or intelligence by itself. | Represented term: temperature | Source: explicitly confusable glossary metadata

Rationale: Tempting because Temperature is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Softmax mechanism or distinction.

choice-draft-softmax-q2-generated-token-2 [DISTRACTOR]: It is mainly about Generated token, not the mechanism named in Softmax.

Feedback: This choice points toward Generated token. Generated token matters nearby, but this question is asking about Softmax.

Misconception: Not a permanent memory or training example by itself. | Represented term: generated-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Generated token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Softmax mechanism or distinction.

choice-draft-softmax-q2-autoregression-3 [DISTRACTOR]: It denoises a whole image from noise.

Feedback: This choice points toward Autoregression. Autoregression matters nearby, but this question is asking about Softmax.

Misconception: Autoregression is being treated as the same idea as Softmax. | Represented term: autoregression | Source: explicitly confusable glossary metadata

Rationale: Tempting because Autoregression is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Softmax mechanism or distinction.

Question 3: Which common confusion does Softmax help learners avoid?

ID: draft-softmax-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-softmax-q3-correct

Misconception tags: None listed

Glossary terms: softmax, logits, decoding-step, sampling, probability

choice-draft-softmax-q3-correct [CORRECT]: It corrects the misconception: Softmax guarantees truth or certainty.

Feedback: Good distinction. Softmax is clearer when that misconception is separated from the mechanism.

choice-draft-softmax-q3-generated-token-1 [DISTRACTOR]: It is mainly about Generated token, not the mechanism named in Softmax.

Feedback: This choice points toward Generated token. Generated token matters nearby, but this question is asking about Softmax.

Misconception: Not a permanent memory or training example by itself. | Represented term: generated-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Generated token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Softmax mechanism or distinction.

choice-draft-softmax-q3-autoregression-2 [DISTRACTOR]: It denoises a whole image from noise.

Feedback: This choice points toward Autoregression. Autoregression matters nearby, but this question is asking about Softmax.

Misconception: Autoregression is being treated as the same idea as Softmax. | Represented term: autoregression | Source: explicitly confusable glossary metadata

Rationale: Tempting because Autoregression is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Softmax mechanism or distinction.

choice-draft-softmax-q3-hallucination-3 [DISTRACTOR]: It means the model intentionally lied.

Feedback: This choice points toward Hallucination. Hallucination matters nearby, but this question is asking about Softmax.

Misconception: Not lying; lying implies intent. | Represented term: hallucination | Source: explicitly confusable glossary metadata

Rationale: Tempting because Hallucination is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Softmax mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?

☐ Is the reading level right?

22. Sampling

Stage: 4. Decision Room | **Card ID:** sampling | **Card in stage:** 3

Primary learning objective: Explain Sampling as part of the day in the life of a prompt.

Secondary objectives: Sampling is the decoding step that chooses one next token from the probability distribution., During decoding, after softmax., Sampling is a temporary choice during inference., Sampling chooses the next response token, not the whole answer at once., Softmax provides probabilities.

Key distinctions: Sampling is a temporary choice during inference., Sampling chooses the next response token, not the whole answer at once., Sampling writes the whole response or permanently teaches the model.

Misconceptions to correct: Sampling writes the whole response or permanently teaches the model.

Related glossary terms: sampling, probability, generated-token, next-token, decoding-step, response-tokens, temperature, top-k, top-p

Nearby glossary distractor candidates: relevance-weight, attention, MLP, multi-layer-perceptron, feed-forward-network, weight-tensor, activation, tensor

Why this matters: Sampling explains why the same prompt can produce different but plausible responses.

Existing Live Checkpoint

Question count: 1

Question: When the model chooses one token, what happens next?

Correct: The token is appended to the response-so-far.

- The token is appended to the response-so-far. (correct)
- The model permanently learns it. Not quite. Sampling is temporary inference behavior, not a durable weight update.
- The whole response appears at once. Not quite. Sampling chooses one token; autoregression repeats that process.
- Softmax proves it is true. Not quite. Probability is not truth. A likely token can still be unsupported.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: When the model chooses one token, what happens next in Sampling?

ID: draft-sampling-q1 | **Type:** definition | **Correct choice ID:** choice-draft-sampling-q1-correct

Misconception tags: None listed

Glossary terms: sampling, probability, generated-token, next-token, decoding-step, response-tokens, temperature, top-k, top-p

choice-draft-sampling-q1-correct [CORRECT]: The token is appended to the response-so-far.

Feedback: Insight strengthened. The selected response token becomes part of the context for the next forward pass.

choice-draft-sampling-q1-the-model-permanently-learns-it [DISTRACTOR]: The model permanently learns it.

Feedback: Not quite. Sampling is temporary inference behavior, not a durable weight update.

Misconception: The model permanently learns it. | Represented term: temperature | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-sampling-q1-the-whole-response-appears-at-once [DISTRACTOR]: The whole response appears at once.

Feedback: Not quite. Sampling chooses one token; autoregression repeats that process.

Misconception: The whole response appears at once. | Represented term: probability | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-sampling-q1-softmax-proves-it-is-true [DISTRACTOR]: Softmax proves it is true.

Feedback: Not quite. Probability is not truth. A likely token can still be unsupported.

Misconception: Softmax proves it is true. | Represented term: softmax | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Sampling most directly teach?

ID: draft-sampling-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-sampling-q2-correct

Misconception tags: None listed

Glossary terms: sampling, probability, generated-token, next-token, decoding-step, response-tokens, temperature, top-k, top-p

choice-draft-sampling-q2-correct [CORRECT]: Sampling is a temporary choice during inference.

Feedback: Good distinction. Sampling is a temporary choice during inference.

choice-draft-sampling-q2-hallucination-1 [DISTRACTOR]: It means the model intentionally lied.

Feedback: This choice points toward Hallucination. Hallucination matters nearby, but this question is asking about Sampling.

Misconception: Not lying; lying implies intent. | Represented term: hallucination | Source: explicitly confusable glossary metadata

Rationale: Tempting because Hallucination is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Sampling mechanism or distinction.

choice-draft-sampling-q2-logits-2 [DISTRACTOR]: It is already a probability distribution that sums to one.

Feedback: This choice points toward Logits. Logits matters nearby, but this question is asking about Sampling.

Misconception: Not probabilities, truth scores, permanent memories, or the sampled token. | Represented term: logits | Source: explicitly confusable glossary metadata

Rationale: Tempting because Logits is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Sampling mechanism or distinction.

choice-draft-sampling-q2-response-3 [DISTRACTOR]: It is the complete input given to the model before generation.

Feedback: This choice points toward Response. Response matters nearby, but this question is asking about Sampling.

Misconception: Not all at once; response tokens are appended step by step. | Represented term: response | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Sampling mechanism or distinction.

Question 3: Which common confusion does Sampling help learners avoid?

ID: draft-sampling-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-sampling-q3-correct

Misconception tags: None listed

Glossary terms: sampling, probability, generated-token, next-token, decoding-step, response-tokens, temperature, top-k, top-p

choice-draft-sampling-q3-correct [CORRECT]: It corrects the misconception: Sampling writes the whole response or permanently teaches the model.

Feedback: Good distinction. Sampling is clearer when that misconception is separated from the mechanism.

choice-draft-sampling-q3-logits-1 [DISTRACTOR]: It is already a probability distribution that sums to one.

Feedback: This choice points toward Logits. Logits matters nearby, but this question is asking about Sampling.

Misconception: Not probabilities, truth scores, permanent memories, or the sampled token. | Represented term: logits | Source: explicitly confusable glossary metadata

Rationale: Tempting because Logits is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Sampling mechanism or distinction.

choice-draft-sampling-q3-response-2 [DISTRACTOR]: It is the complete input given to the model before generation.

Feedback: This choice points toward Response. Response matters nearby, but this question is asking about Sampling.

Misconception: Not all at once; response tokens are appended step by step. | Represented term: response | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Sampling mechanism or distinction.

choice-draft-sampling-q3-hidden-state-3 [DISTRACTOR]: It is the token's fixed starting vector before context matters.

Feedback: This choice points toward Hidden state. Hidden state matters nearby, but this question is asking about Sampling.

Misconception: Often confused with embeddings, weights, saved memory, or visible response text. | Represented term: hidden state | Source: same-stage glossary term

Rationale: Tempting because Hidden state is same-stage for this part of the Journey. It is not correct because the question asks for the specific Sampling mechanism or distinction.

Question 4: How does Sampling connect to the prompt's Journey?

ID: draft-sampling-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-sampling-q4-correct

Misconception tags: None listed

Glossary terms: sampling, probability, generated-token, next-token, decoding-step, response-tokens, temperature, top-k, top-p

choice-draft-sampling-q4-correct [CORRECT]: Softmax provides probabilities.

Feedback: Good connection. Softmax provides probabilities.

choice-draft-sampling-q4-response-1 [DISTRACTOR]: It is the complete input given to the model before generation.

Feedback: This choice points toward Response. Response matters nearby, but this question is asking about Sampling.

Misconception: Not all at once; response tokens are appended step by step. | Represented term: response | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Sampling mechanism or distinction.

choice-draft-sampling-q4-hidden-state-2 [DISTRACTOR]: It is the token's fixed starting vector before context matters.

Feedback: This choice points toward Hidden state. Hidden state matters nearby, but this question is asking about Sampling.

Misconception: Often confused with embeddings, weights, saved memory, or visible response text. | Represented term: hidden state | Source: same-stage glossary term

Rationale: Tempting because Hidden state is same-stage for this part of the Journey. It is not correct because the question asks for the specific Sampling mechanism or distinction.

choice-draft-sampling-q4-relevance-weight-3 [DISTRACTOR]: It is awareness rather than a numerical relevance weight.

Feedback: This choice points toward Relevance weight. Relevance weight matters nearby, but this question is asking about Sampling.

Misconception: Not truth, confidence, or human attention. | Represented term: relevance-weight | Source: nearby-stage glossary term

Rationale: Tempting because Relevance weight is nearby-stage for this part of the Journey. It is not correct because the question asks for the specific Sampling mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

23. Autoregression

Stage: 5. The Day Repeats | **Card ID:** autoregression | **Card in stage:** 1

Primary learning objective: Explain Autoregression as part of the day in the life of a prompt.

Secondary objectives: Autoregression is response generation by repeatedly choosing one next token, appending it to the context, and running the model again., During response generation after each next-token choice., Autoregression updates the temporary context, not the model weights., The generated response grows token by token., Sampling chooses one token.

Key distinctions: Autoregression updates the temporary context, not the model weights., The generated response grows token by token., The model writes the whole response at once.

Misconceptions to correct: The model writes the whole response at once.

Related glossary terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference

Nearby glossary distractor candidates: embedding-table, embedding, vector, feature, distributed-representation, token-id

Why this matters: It prevents the misconception that the model writes the whole answer at once.

Existing Live Checkpoint

Question count: 1

Question: What happens after a token is generated?

Correct: It is appended to the context and the model can run again.

- It is appended to the context and the model can run again. (correct)
- The model permanently learns it. Not quite. Autoregression updates temporary context, not durable weights.
- The whole answer appears at once. Not quite. The response grows one token at a time.
- The token becomes a training example. Not quite. Ordinary generation is inference, not training.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: What happens after a token is generated in Autoregression?

ID: draft-autoregression-q1 | **Type:** definition | **Correct choice ID:** choice-draft-autoregression-q1-correct

Misconception tags: None listed

Glossary terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference

choice-draft-autoregression-q1-correct [CORRECT]: It is appended to the context and the model can run again.

Feedback: Insight strengthened. Autoregression means the response grows by choose, append, and repeat.

choice-draft-autoregression-q1-the-model-permanently-learns-it [DISTRACTOR]: The model permanently learns it.

Feedback: Not quite. Autoregression updates temporary context, not durable weights.

Misconception: The model permanently learns it. | **Represented term:** diffusion | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-autoregression-q1-the-whole-answer-appears-at-once [DISTRACTOR]: The whole answer appears at once.

Feedback: Not quite. The response grows one token at a time.

Misconception: The whole answer appears at once. | **Represented term:** generated-token | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-autoregression-q1-the-token-becomes-a-training-example [DISTRACTOR]: The token becomes a training example.

Feedback: Not quite. Ordinary generation is inference, not training.

Misconception: The token becomes a training example. | **Represented term:** training | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Autoregression most directly teach?

ID: draft-autoregression-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-autoregression-q2-correct

Misconception tags: None listed

Glossary terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference

choice-draft-autoregression-q2-correct [CORRECT]: Autoregression updates the temporary context, not the model weights.

Feedback: Good distinction. Autoregression updates the temporary context, not the model weights.

choice-draft-autoregression-q2-context-window-1 [DISTRACTOR]: It is permanent model memory.

Feedback: This choice points toward Context window. Context window matters nearby, but this question is asking about Autoregression.

Misconception: Not permanent memory; it is temporary working context. | Represented term: context window | Source: explicitly confusable glossary metadata

Rationale: Tempting because Context window is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Autoregression mechanism or distinction.

choice-draft-autoregression-q2-prompt-2 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about Autoregression.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Autoregression mechanism or distinction.

choice-draft-autoregression-q2-memory-3 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Autoregression.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Autoregression mechanism or distinction.

Question 3: Which common confusion does Autoregression help learners avoid?

ID: draft-autoregression-q3 | Type: boundary distinction | Correct choice ID: choice-draft-autoregression-q3-correct

Misconception tags: None listed

Glossary terms: autoregression, completion, generated-token, response-tokens, response-so-far, input-context, sampling, inference

choice-draft-autoregression-q3-correct [CORRECT]: It corrects the misconception: The model writes the whole response at once.

Feedback: Good distinction. Autoregression is clearer when that misconception is separated from the mechanism.

choice-draft-autoregression-q3-prompt-1 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about Autoregression.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Autoregression mechanism or distinction.

choice-draft-autoregression-q3-memory-2 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Autoregression.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Autoregression mechanism or distinction.

choice-draft-autoregression-q3-system-prompt-3 [DISTRACTOR]: It is mainly about System prompt, not the mechanism named in Autoregression.

Feedback: This choice points toward System prompt. System prompt matters nearby, but this question is asking about Autoregression.

Misconception: Not fine-tuning, training, permanent memory, or conscience. | Represented term: system-prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because System prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Autoregression mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?

☐ Is the question count right for topic density?

☐ Is the reading level right?

24. Context Window

Stage: 5. The Day Repeats | **Card ID:** context-window | **Card in stage:** 2

Primary learning objective: Explain Context Window as part of the day in the life of a prompt.

Secondary objectives: The context window is the limited amount of tokens or media the model can consider at one time., During inference, whenever the model processes the current context., The context window is temporary., The context window can contain prompt tokens, retrieved context, prior conversation, and response-so-far tokens., Autoregression makes response-so-far grow.

Key distinctions: The context window is temporary., The context window can contain prompt tokens, retrieved context, prior conversation, and response-so-far tokens., The context window is permanent memory.

Misconceptions to correct: The context window is permanent memory.

Related glossary terms: context window, input-context, response-so-far, retrieval, prompt-tokens, response-tokens, memory, rag, autoregression, inference

Nearby glossary distractor candidates: relevance-weight, attention, MLP, multi-layer-perceptron, feed-forward-network

Why this matters: The model cannot use what has fallen out of context.

Existing Live Checkpoint

Question count: 1

Question: What happens to information outside the context window?

Correct: The model cannot directly use it in the current run.

- The model cannot directly use it in the current run. (correct)
- It becomes permanent memory. Not quite. Context can feel like memory, but it is temporary input unless a separate system saves it.
- It updates the weights. Not quite. Ordinary context does not train or rewrite weights.
- It is always retrieved automatically. Not quite. Retrieval is a separate system step, not a guarantee.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: What happens to information outside the context window?

ID: draft-context-window-q1 | **Type:** definition | **Correct choice ID:** choice-draft-context-window-q1-correct

Misconception tags: None listed

Glossary terms: context window, input-context, response-so-far, retrieval, prompt-tokens, response-tokens, memory, rag, autoregression, inference

choice-draft-context-window-q1-correct [CORRECT]: The model cannot directly use it in the current run.

Feedback: Insight strengthened. The model can only use what remains visible in the current context.

choice-draft-context-window-q1-it-becomes-permanent-memory [DISTRACTOR]: It becomes permanent memory.

Feedback: Not quite. Context can feel like memory, but it is temporary input unless a separate system saves it.

Misconception: It becomes permanent memory. | Represented term: memory | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-context-window-q1-it-updates-the-weights [DISTRACTOR]: It updates the weights.

Feedback: Not quite. Ordinary context does not train or rewrite weights.

Misconception: It updates the weights. | Represented term: training | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-context-window-q1-it-is-always-retrieved-automatically [DISTRACTOR]: It is always retrieved automatically.

Feedback: Not quite. Retrieval is a separate system step, not a guarantee.

Misconception: It is always retrieved automatically. | Represented term: rag | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Context Window most directly teach?

ID: draft-context-window-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-context-window-q2-correct

Misconception tags: None listed

Glossary terms: context window, input-context, response-so-far, retrieval, prompt-tokens, response-tokens, memory, rag, autoregression, inference

choice-draft-context-window-q2-correct [CORRECT]: The context window is temporary.

Feedback: Good distinction. The context window is temporary.

choice-draft-context-window-q2-system-prompt-1 [DISTRACTOR]: It is mainly about System prompt, not the mechanism named in Context Window.

Feedback: This choice points toward System prompt. System prompt matters nearby, but this question is asking about Context Window.

Misconception: Not fine-tuning, training, permanent memory, or conscience. | Represented term: system-prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because System prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Context Window mechanism or distinction.

choice-draft-context-window-q2-forward-pass-2 [DISTRACTOR]: It is mainly about Forward pass, not the mechanism named in Context Window.

Feedback: This choice points toward Forward pass. Forward pass matters nearby, but this question is asking about Context Window.

Misconception: Not backpropagation or a training update. | Represented term: forward-pass | Source: explicitly confusable glossary metadata

Rationale: Tempting because Forward pass is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Context Window mechanism or distinction.

choice-draft-context-window-q2-response-3 [DISTRACTOR]: It is the complete input given to the model before generation.

Feedback: This choice points toward Response. Response matters nearby, but this question is asking about Context Window.

Misconception: Not all at once; response tokens are appended step by step. | Represented term: response | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Context Window mechanism or distinction.

Question 3: Which common confusion does Context Window help learners avoid?

ID: draft-context-window-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-context-window-q3-correct

Misconception tags: None listed

Glossary terms: context window, input-context, response-so-far, retrieval, prompt-tokens, response-tokens, memory, rag, autoregression, inference

choice-draft-context-window-q3-correct [CORRECT]: It corrects the misconception: The context window is permanent memory.

Feedback: Good distinction. Context Window is clearer when that misconception is separated from the mechanism.

choice-draft-context-window-q3-forward-pass-1 [DISTRACTOR]: It is mainly about Forward pass, not the mechanism named in Context Window.

Feedback: This choice points toward Forward pass. Forward pass matters nearby, but this question is asking about Context Window.

Misconception: Not backpropagation or a training update. | Represented term: forward-pass | Source: explicitly confusable glossary metadata

Rationale: Tempting because Forward pass is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Context Window mechanism or distinction.

choice-draft-context-window-q3-response-2 [DISTRACTOR]: It is the complete input given to the model before generation.

Feedback: This choice points toward Response. Response matters nearby, but this question is asking about Context Window.

Misconception: Not all at once; response tokens are appended step by step. | Represented term: response | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Context Window mechanism or distinction.

choice-draft-context-window-q3-generated-token-3 [DISTRACTOR]: It is mainly about Generated token, not the mechanism named in Context Window.

Feedback: This choice points toward Generated token. Generated token matters nearby, but this question is asking about Context Window.

Misconception: Not a permanent memory or training example by itself. | Represented term: generated-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Generated token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Context Window mechanism or distinction.

Question 4: How does Context Window connect to the prompt's Journey?

ID: draft-context-window-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-context-window-q4-correct

Misconception tags: None listed

Glossary terms: context window, input-context, response-so-far, retrieval, prompt-tokens, response-tokens, memory, rag, autoregression, inference

choice-draft-context-window-q4-correct [CORRECT]: Autoregression makes response-so-far grow.

Feedback: Good connection. Autoregression makes response-so-far grow.

choice-draft-context-window-q4-response-1 [DISTRACTOR]: It is the complete input given to the model before generation.

Feedback: This choice points toward Response. Response matters nearby, but this question is asking about Context Window.

Misconception: Not all at once; response tokens are appended step by step. | Represented term: response | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Context Window mechanism or distinction.

choice-draft-context-window-q4-generated-token-2 [DISTRACTOR]: It is mainly about Generated token, not the mechanism named in Context Window.

Feedback: This choice points toward Generated token. Generated token matters nearby, but this question is asking about Context Window.

Misconception: Not a permanent memory or training example by itself. | Represented term: generated-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Generated token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Context Window mechanism or distinction.

choice-draft-context-window-q4-training-3 [DISTRACTOR]: It permanently updates model weights during this ordinary response.

Feedback: This choice points toward Training. Training matters nearby, but this question is asking about Context Window.

Misconception: Ordinary chat inference is not training unless a separate training process updates weights. | Represented term: training | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Context Window mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

25. RAG and Retrieval

Stage: 5. The Day Repeats | **Card ID:** rag-retrieval | **Card in stage:** 3

Primary learning objective: Explain RAG and Retrieval as part of the day in the life of a prompt.

Secondary objectives: Retrieval-augmented generation, or RAG, retrieves outside information and places it into the model's current context before the model generates a response., During inference, before or during response generation., RAG usually does not change model weights., Retrieved material becomes prompt/context tokens., RAG depends on the context window.

Key distinctions: RAG usually does not change model weights., Retrieved material becomes prompt/context tokens., RAG is training, permanent memory, guaranteed truth, or consciousness.

Misconceptions to correct: RAG is training, permanent memory, guaranteed truth, or consciousness.

Related glossary terms: rag, retrieval, grounding, context window, input-context, prompt-tokens, response-tokens, inference, fine-tuning, training, hallucination

Nearby glossary distractor candidates: relevance-weight, attention, MLP, multi-layer-perceptron, feed-forward-network, weight-tensor, activation, tensor

Why this matters: RAG explains many "chat with your documents" systems. The model is not magically knowing private files; the system is retrieving relevant material and showing it to the model.

Existing Live Checkpoint

Question count: 1

Question: What does RAG usually do?

Correct: Retrieves outside information and places it into the current context

- Retrieves outside information and places it into the current context (correct)
- Permanently updates model weights Not quite. RAG is retrieval plus context. It is not training, permanent memory, consciousness, or a truth guarantee.
- Guarantees every answer is true Not quite. RAG can improve grounding, but retrieved context can still be incomplete, stale, or misused.
- Makes the model aware of documents Not quite. Retrieved documents become temporary context tokens, not consciousness.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: What does RAG usually do in RAG and Retrieval?

ID: draft-rag-retrieval-q1 | **Type:** definition | **Correct choice ID:** choice-draft-rag-retrieval-q1-correct

Misconception tags: None listed

Glossary terms: rag, retrieval, grounding, context window, input-context, prompt-tokens, response-tokens, inference, fine-tuning, training, hallucination

choice-draft-rag-retrieval-q1-correct [CORRECT]: Retrieves outside information and places it into the current context

Feedback: Insight strengthened. RAG adds temporary evidence to context; it does not rewrite model weights.

choice-draft-rag-retrieval-q1-permanently-updates-model-weights [DISTRACTOR]: Permanently updates model weights

Feedback: Not quite. RAG is retrieval plus context. It is not training, permanent memory, consciousness, or a truth guarantee.

Misconception: Permanently updates model weights | **Represented term:** training | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-rag-retrieval-q1-guarantees-every-answer-is-true [DISTRACTOR]: Guarantees every answer is true

Feedback: Not quite. RAG can improve grounding, but retrieved context can still be incomplete, stale, or misused.

Misconception: Guarantees every answer is true | **Represented term:** probability | **Source:** nearby-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-rag-retrieval-q1-makes-the-model-aware-of-documents [DISTRACTOR]: Makes the model aware of documents

Feedback: Not quite. Retrieved documents become temporary context tokens, not consciousness.

Misconception: Makes the model aware of documents | **Represented term:** rag | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does RAG and Retrieval most directly teach?

ID: draft-rag-retrieval-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-rag-retrieval-q2-correct

Misconception tags: None listed

Glossary terms: rag, retrieval, grounding, context window, input-context, prompt-tokens, response-tokens, inference, fine-tuning, training, hallucination

choice-draft-rag-retrieval-q2-correct [CORRECT]: RAG usually does not change model weights.

Feedback: Good distinction. RAG usually does not change model weights.

choice-draft-rag-retrieval-q2-citation-1 [DISTRACTOR]: It automatically proves the cited claim.

Feedback: This choice points toward Citation. Citation matters nearby, but this question is asking about RAG and Retrieval.

Misconception: Not proof by itself and not the same as training data. | Represented term: citation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Citation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific RAG and Retrieval mechanism or distinction.

choice-draft-rag-retrieval-q2-training-data-2 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in RAG and Retrieval.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about RAG and Retrieval.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific RAG and Retrieval mechanism or distinction.

choice-draft-rag-retrieval-q2-system-prompt-3 [DISTRACTOR]: It is mainly about System prompt, not the mechanism named in RAG and Retrieval.

Feedback: This choice points toward System prompt. System prompt matters nearby, but this question is asking about RAG and Retrieval.

Misconception: Not fine-tuning, training, permanent memory, or conscience. | Represented term: system-prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because System prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific RAG and Retrieval mechanism or distinction.

Question 3: Which common confusion does RAG and Retrieval help learners avoid?

ID: draft-rag-retrieval-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-rag-retrieval-q3-correct

Misconception tags: None listed

Glossary terms: rag, retrieval, grounding, context window, input-context, prompt-tokens, response-tokens, inference, fine-tuning, training, hallucination

choice-draft-rag-retrieval-q3-correct [CORRECT]: It corrects the misconception: RAG is training, permanent memory, guaranteed truth, or consciousness.

Feedback: Good distinction. RAG and Retrieval is clearer when that misconception is separated from the mechanism.

choice-draft-rag-retrieval-q3-training-data-1 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in RAG and Retrieval.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about RAG and Retrieval.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific RAG and Retrieval mechanism or distinction.

choice-draft-rag-retrieval-q3-system-prompt-2 [DISTRACTOR]: It is mainly about System prompt, not the mechanism named in RAG and Retrieval.

Feedback: This choice points toward System prompt. System prompt matters nearby, but this question is asking about RAG and Retrieval.

Misconception: Not fine-tuning, training, permanent memory, or conscience. | Represented term: system-prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because System prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific RAG and Retrieval mechanism or distinction.

choice-draft-rag-retrieval-q3-response-so-far-3 [DISTRACTOR]: It is mainly about Response-so-far, not the mechanism named in RAG and Retrieval.

Feedback: This choice points toward Response-so-far. Response-so-far matters nearby, but this question is asking about RAG and Retrieval.

Misconception: Not the whole final answer, permanent memory, or training data. | Represented term: response-so-far | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response-so-far is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific RAG and Retrieval mechanism or distinction.

Question 4: How does RAG and Retrieval connect to the prompt's Journey?

ID: draft-rag-retrieval-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-rag-retrieval-q4-correct

Misconception tags: None listed

Glossary terms: rag, retrieval, grounding, context window, input-context, prompt-tokens, response-tokens, inference, fine-tuning, training, hallucination

choice-draft-rag-retrieval-q4-correct [CORRECT]: RAG depends on the context window.

Feedback: Good connection. RAG depends on the context window.

choice-draft-rag-retrieval-q4-system-prompt-1 [DISTRACTOR]: It is mainly about System prompt, not the mechanism named in RAG and Retrieval.

Feedback: This choice points toward System prompt. System prompt matters nearby, but this question is asking about RAG and Retrieval.

Misconception: Not fine-tuning, training, permanent memory, or conscience. | Represented term: system-prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because System prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific RAG and Retrieval mechanism or distinction.

choice-draft-rag-retrieval-q4-response-so-far-2 [DISTRACTOR]: It is mainly about Response-so-far, not the mechanism named in RAG and Retrieval.

Feedback: This choice points toward Response-so-far. Response-so-far matters nearby, but this question is asking about RAG and Retrieval.

Misconception: Not the whole final answer, permanent memory, or training data. | Represented term: response-so-far | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response-so-far is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific RAG and Retrieval mechanism or distinction.

choice-draft-rag-retrieval-q4-forward-pass-3 [DISTRACTOR]: It is mainly about Forward pass, not the mechanism named in RAG and Retrieval.

Feedback: This choice points toward Forward pass. Forward pass matters nearby, but this question is asking about RAG and Retrieval.

Misconception: Not backpropagation or a training update. | Represented term: forward-pass | Source: explicitly confusable glossary metadata

Rationale: Tempting because Forward pass is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific RAG and Retrieval mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

26. Grounding

Stage: 5. The Day Repeats | **Card ID:** grounding | **Card in stage:** 4

Primary learning objective: Explain Grounding as part of the day in the life of a prompt.

Secondary objectives: Grounding means connecting a model's response to available evidence, such as retrieved documents, citations, data, or tool results., During inference and response generation, usually through retrieved context, tools, data, or system design., Grounding usually supplies temporary evidence in the current context., Evidence becomes prompt/context material., RAG can place evidence into context; grounding asks whether the generated answer actually stays connected to that evidence.

Key distinctions: Grounding usually supplies temporary evidence in the current context., Evidence becomes prompt/context material., A citation-looking answer is automatically grounded.

Misconceptions to correct: A citation-looking answer is automatically grounded.

Related glossary terms: grounding, rag, retrieval, context window, input-context, citation, hallucination, inference

Nearby glossary distractor candidates: sampling, probability, temperature, softmax, top-k, top-p

Why this matters: Grounding helps learners ask whether a fluent answer is tied to evidence rather than merely sounding right.

Existing Live Checkpoint

Question count: 1

Question: What does grounding try to do?

Correct: Connect an answer to available evidence

- Connect an answer to available evidence (correct)
- Make every answer true Not quite. Grounding can reduce unsupported answers, but it is not a guarantee.
- Permanently train the model Not quite. Grounding usually supplies temporary evidence in context.
- Make the model conscious Not quite. Grounding is a system design pattern, not awareness.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: What does grounding try to do?

ID: draft-grounding-q1 | **Type:** definition | **Correct choice ID:** choice-draft-grounding-q1-correct

Misconception tags: None listed

Glossary terms: grounding, rag, retrieval, context window, input-context, citation, hallucination, inference

choice-draft-grounding-q1-correct [CORRECT]: Connect an answer to available evidence

Feedback: Insight strengthened. Grounding ties claims to evidence, but the evidence still needs review.

choice-draft-grounding-q1-make-every-answer-true [DISTRACTOR]: Make every answer true

Feedback: Not quite. Grounding can reduce unsupported answers, but it is not a guarantee.

Misconception: Make every answer true | **Represented term:** probability | **Source:** nearby-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-grounding-q1-permanently-train-the-model [DISTRACTOR]: Permanently train the model

Feedback: Not quite. Grounding usually supplies temporary evidence in context.

Misconception: Permanently train the model | **Represented term:** training | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-grounding-q1-make-the-model-conscious [DISTRACTOR]: Make the model conscious

Feedback: Not quite. Grounding is a system design pattern, not awareness.

Misconception: Make the model conscious | **Represented term:** llm | **Source:** learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Grounding most directly teach?

ID: draft-grounding-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-grounding-q2-correct

Misconception tags: None listed

Glossary terms: grounding, rag, retrieval, context window, input-context, citation, hallucination, inference

choice-draft-grounding-q2-correct [CORRECT]: Grounding usually supplies temporary evidence in the current context.

Feedback: Good distinction. Grounding usually supplies temporary evidence in the current context.

choice-draft-grounding-q2-prompt-1 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about Grounding.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Grounding mechanism or distinction.

choice-draft-grounding-q2-memory-2 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Grounding.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Grounding mechanism or distinction.

choice-draft-grounding-q2-training-data-3 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Grounding.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Grounding.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Grounding mechanism or distinction.

Question 3: Which common confusion does Grounding help learners avoid?

ID: draft-grounding-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-grounding-q3-correct

Misconception tags: None listed

Glossary terms: grounding, rag, retrieval, context window, input-context, citation, hallucination, inference

choice-draft-grounding-q3-correct [CORRECT]: It corrects the misconception: A citation-looking answer is automatically grounded.

Feedback: Good distinction. Grounding is clearer when that misconception is separated from the mechanism.

choice-draft-grounding-q3-memory-1 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Grounding.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Grounding mechanism or distinction.

choice-draft-grounding-q3-training-data-2 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Grounding.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Grounding.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Grounding mechanism or distinction.

choice-draft-grounding-q3-prompt-tokens-3 [DISTRACTOR]: It is a response token the model generated during this answer.

Feedback: This choice points toward Prompt token. Prompt token matters nearby, but this question is asking about Grounding.

Misconception: Not generated by the model during this response step. | Represented term: prompt-tokens | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Grounding mechanism or distinction.

Question 4: How does Grounding connect to the prompt's Journey?

ID: draft-grounding-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-grounding-q4-correct

Misconception tags: None listed

Glossary terms: grounding, rag, retrieval, context window, input-context, citation, hallucination, inference

choice-draft-grounding-q4-correct [CORRECT]: RAG can place evidence into context; grounding asks whether the generated answer actually stays connected to that evidence.

Feedback: Good connection. RAG can place evidence into context; grounding asks whether the generated answer actually stays connected to that evidence.

choice-draft-grounding-q4-training-data-1 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Grounding.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Grounding.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Grounding mechanism or distinction.

choice-draft-grounding-q4-prompt-tokens-2 [DISTRACTOR]: It is a response token the model generated during this answer.

Feedback: This choice points toward Prompt token. Prompt token matters nearby, but this question is asking about Grounding.

Misconception: Not generated by the model during this response step. | Represented term: prompt-tokens | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Grounding mechanism or distinction.

choice-draft-grounding-q4-response-tokens-3 [DISTRACTOR]: It is input text that existed before generation began.

Feedback: This choice points toward Response token. Response token matters nearby, but this question is asking about Grounding.

Misconception: Not a durable weight update or saved memory. | Represented term: response-tokens | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Grounding mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

27. Hallucinations

Stage: 5. The Day Repeats | **Card ID:** hallucinations | **Card in stage:** 5

Primary learning objective: Explain Hallucinations as part of the day in the life of a prompt.

Secondary objectives: A hallucination is a fluent model output that is unsupported, false, fabricated, or not grounded in the available evidence., During response generation., A hallucination is an output event during inference., The hallucination appears in generated response tokens., Grounding tries to reduce unsupported answers.

Key distinctions: A hallucination is an output event during inference., The hallucination appears in generated response tokens., A hallucination means the model is lying or malicious.

Misconceptions to correct: A hallucination means the model is lying or malicious.

Related glossary terms: hallucination, grounding, rag, logits, softmax, sampling, inference, uncertainty, human-review

Nearby glossary distractor candidates: retrieval, citation, memory, context window

Why this matters: Hallucinations are one of the most practical reasons model literacy matters: fluency is not the same as truth.

Existing Live Checkpoint

Question count: 1

Question: Why can hallucinations happen?

Correct: The model can generate plausible text without enough grounding

- The model can generate plausible text without enough grounding (correct)
- The model is always trying to deceive Not quite. Lying implies intent; the model is generating likely text.
- Softmax guarantees truth Not quite. Softmax turns scores into probabilities, not truth.
- RAG permanently teaches the model Not quite. RAG usually changes temporary context, not weights.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: Why can hallucinations happen?

ID: draft-hallucinations-q1 | **Type:** definition | **Correct choice ID:** choice-draft-hallucinations-q1-correct

Misconception tags: None listed

Glossary terms: hallucination, grounding, rag, logits, softmax, sampling, inference, uncertainty, human-review

choice-draft-hallucinations-q1-correct [CORRECT]: The model can generate plausible text without enough grounding

Feedback: Insight strengthened. A fluent claim can still need grounding or review.

choice-draft-hallucinations-q1-the-model-is-always-trying-to-deceive [DISTRACTOR]: The model is always trying to deceive

Feedback: Not quite. Lying implies intent; the model is generating likely text.

Misconception: The model is always trying to deceive | Represented term: grounding | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-hallucinations-q1-softmax-guarantees-truth [DISTRACTOR]: Softmax guarantees truth

Feedback: Not quite. Softmax turns scores into probabilities, not truth.

Misconception: Softmax guarantees truth | Represented term: softmax | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-hallucinations-q1-rag-permanently-teaches-the-model [DISTRACTOR]: RAG permanently teaches the model

Feedback: Not quite. RAG usually changes temporary context, not weights.

Misconception: RAG permanently teaches the model | Represented term: human-review | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Hallucinations most directly teach?

ID: draft-hallucinations-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-hallucinations-q2-correct

Misconception tags: None listed

Glossary terms: hallucination, grounding, rag, logits, softmax, sampling, inference, uncertainty, human-review

choice-draft-hallucinations-q2-correct [CORRECT]: A hallucination is an output event during inference.

Feedback: Good distinction. A hallucination is an output event during inference.

choice-draft-hallucinations-q2-fine-tuning-1 [DISTRACTOR]: It is the same as adding outside text into the current prompt.

Feedback: This choice points toward Fine-tuning. Fine-tuning matters nearby, but this question is asking about Hallucinations.

Misconception: Different from prompting; prompting changes temporary context, not durable weights. | Represented term: fine-tuning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Fine-tuning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hallucinations mechanism or distinction.

choice-draft-hallucinations-q2-training-2 [DISTRACTOR]: It permanently updates model weights during this ordinary response.

Feedback: This choice points toward Training. Training matters nearby, but this question is asking about Hallucinations.

Misconception: Ordinary chat inference is not training unless a separate training process updates weights. | Represented term: training | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hallucinations mechanism or distinction.

choice-draft-hallucinations-q2-input-context-3 [DISTRACTOR]: It is mainly about Input context, not the mechanism named in Hallucinations.

Feedback: This choice points toward Input context. Input context matters nearby, but this question is asking about Hallucinations.

Misconception: Not the training dataset or durable model weights. | Represented term: input-context | Source: explicitly confusable glossary metadata

Rationale: Tempting because Input context is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hallucinations mechanism or distinction.

Question 3: Which common confusion does Hallucinations help learners avoid?

ID: draft-hallucinations-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-hallucinations-q3-correct

Misconception tags: None listed

Glossary terms: hallucination, grounding, rag, logits, softmax, sampling, inference, uncertainty, human-review

choice-draft-hallucinations-q3-correct [CORRECT]: It corrects the misconception: A hallucination means the model is lying or malicious.

Feedback: Good distinction. Hallucinations is clearer when that misconception is separated from the mechanism.

choice-draft-hallucinations-q3-training-1 [DISTRACTOR]: It permanently updates model weights during this ordinary response.

Feedback: This choice points toward Training. Training matters nearby, but this question is asking about Hallucinations.

Misconception: Ordinary chat inference is not training unless a separate training process updates weights. | Represented term: training | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hallucinations mechanism or distinction.

choice-draft-hallucinations-q3-input-context-2 [DISTRACTOR]: It is mainly about Input context, not the mechanism named in Hallucinations.

Feedback: This choice points toward Input context. Input context matters nearby, but this question is asking about Hallucinations.

Misconception: Not the training dataset or durable model weights. | Represented term: input-context | Source: explicitly confusable glossary metadata

Rationale: Tempting because Input context is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hallucinations mechanism or distinction.

choice-draft-hallucinations-q3-context-window-3 [DISTRACTOR]: It is permanent model memory.

Feedback: This choice points toward Context window. Context window matters nearby, but this question is asking about Hallucinations.

Misconception: Not permanent memory; it is temporary working context. | Represented term: context window | Source: explicitly confusable glossary metadata

Rationale: Tempting because Context window is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hallucinations mechanism or distinction.

Question 4: How does Hallucinations connect to the prompt's Journey?

ID: draft-hallucinations-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-hallucinations-q4-correct

Misconception tags: None listed

Glossary terms: hallucination, grounding, rag, logits, softmax, sampling, inference, uncertainty, human-review

choice-draft-hallucinations-q4-correct [CORRECT]: Grounding tries to reduce unsupported answers.

Feedback: Good connection. Grounding tries to reduce unsupported answers.

choice-draft-hallucinations-q4-input-context-1 [DISTRACTOR]: It is mainly about Input context, not the mechanism named in Hallucinations.

Feedback: This choice points toward Input context. Input context matters nearby, but this question is asking about Hallucinations.

Misconception: Not the training dataset or durable model weights. | Represented term: input-context | Source: explicitly confusable glossary metadata

Rationale: Tempting because Input context is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hallucinations mechanism or distinction.

choice-draft-hallucinations-q4-context-window-2 [DISTRACTOR]: It is permanent model memory.

Feedback: This choice points toward Context window. Context window matters nearby, but this question is asking about Hallucinations.

Misconception: Not permanent memory; it is temporary working context. | Represented term: context window | Source: explicitly confusable glossary metadata

Rationale: Tempting because Context window is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hallucinations mechanism or distinction.

choice-draft-hallucinations-q4-prompt-3 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about Hallucinations.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Hallucinations mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

28. How AI Learns

Stage: 6. Twilight: The Wider Landscape | **Card ID:** how-ai-learns | **Card in stage:** 1

Primary learning objective: Explain How AI Learns as part of the day in the life of a prompt.

Secondary objectives: AI systems can change or behave differently through durable training, human feedback, retrieval, prompting, or temporary in-context steering., Across training, fine-tuning, evaluation, retrieval systems, and ordinary inference., Training, fine-tuning, and adapter training are durable., Prompts and examples can steer the response during inference, but they usually do not rewrite the model., Earlier cards taught training, fine-tuning, inference, RAG, and context.

Key distinctions: Training, fine-tuning, and adapter training are durable., Prompts and examples can steer the response during inference, but they usually do not rewrite the model., Any useful response means the model permanently learned something.

Misconceptions to correct: Any useful response means the model permanently learned something.

Related glossary terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference

Nearby glossary distractor candidates: embedding, embedding-table, token-id

Why this matters: People often say a model "learned" something when it may only be using temporary context.

Existing Live Checkpoint

Question count: 1

Question: Which action usually changes model weights?

Correct: Fine-tuning on examples.

- Fine-tuning on examples. (correct)
- Adding a prompt instruction. Not quite. Prompting steers the current context; it does not normally update weights.
- Retrieving a document into context. Not quite. Retrieval adds temporary context, not durable model memory.
- Choosing a sampled token. Not quite. Sampling chooses a response token during inference.

Proposed Draft Question Bank

Proposed count: 5. **Rationale:** Synthesis or badge-critical card; five checks cover multiple mechanisms and responsible-use distinctions.

Question 1: Which AI system action usually changes model weights?

ID: draft-how-ai-learns-q1 | **Type:** definition | **Correct choice ID:** choice-draft-how-ai-learns-q1-correct

Misconception tags: None listed

Glossary terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference

choice-draft-how-ai-learns-q1-correct [CORRECT]: Fine-tuning on examples.

Feedback: Insight strengthened. Fine-tuning is a training process that can durably adjust weights or adapter weights.

choice-draft-how-ai-learns-q1-adding-a-prompt-instruction [DISTRACTOR]: Adding a prompt instruction.

Feedback: Not quite. Prompting steers the current context; it does not normally update weights.

Misconception: Adding a prompt instruction. | Represented term: fine-tuning | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-how-ai-learns-q1-retrieving-a-document-into-context [DISTRACTOR]: Retrieving a document into context.

Feedback: Not quite. Retrieval adds temporary context, not durable model memory.

Misconception: Retrieving a document into context. | Represented term: rag | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-how-ai-learns-q1-choosing-a-sampled-token [DISTRACTOR]: Choosing a sampled token.

Feedback: Not quite. Sampling chooses a response token during inference.

Misconception: Choosing a sampled token. | Represented term: token | Source: learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does How AI Learns most directly teach?

ID: draft-how-ai-learns-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-how-ai-learns-q2-correct

Misconception tags: None listed

Glossary terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference

choice-draft-how-ai-learns-q2-correct [CORRECT]: Training, fine-tuning, and adapter training are durable.

Feedback: Good distinction. Training, fine-tuning, and adapter training are durable.

choice-draft-how-ai-learns-q2-training-data-1 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in How AI Learns.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about How AI Learns.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

choice-draft-how-ai-learns-q2-next-token-2 [DISTRACTOR]: It is mainly about Next token, not the mechanism named in How AI Learns.

Feedback: This choice points toward Next token. Next token matters nearby, but this question is asking about How AI Learns.

Misconception: Not the whole answer, a permanent memory, or a guarantee of truth. | Represented term: next-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Next token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

choice-draft-how-ai-learns-q2-prompt-3 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about How AI Learns.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

Question 3: Which common confusion does How AI Learns help learners avoid?

ID: draft-how-ai-learns-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-how-ai-learns-q3-correct

Misconception tags: None listed

Glossary terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference

choice-draft-how-ai-learns-q3-correct [CORRECT]: It corrects the misconception: Any useful response means the model permanently learned something.

Feedback: Good distinction. How AI Learns is clearer when that misconception is separated from the mechanism.

choice-draft-how-ai-learns-q3-next-token-1 [DISTRACTOR]: It is mainly about Next token, not the mechanism named in How AI Learns.

Feedback: This choice points toward Next token. Next token matters nearby, but this question is asking about How AI Learns.

Misconception: Not the whole answer, a permanent memory, or a guarantee of truth. | Represented term: next-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Next token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

choice-draft-how-ai-learns-q3-prompt-2 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about How AI Learns.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

choice-draft-how-ai-learns-q3-alignment-3 [DISTRACTOR]: It is mainly about Alignment, not the mechanism named in How AI Learns.

Feedback: This choice points toward Alignment. Alignment matters nearby, but this question is asking about How AI Learns.

Misconception: Aligned behavior does not guarantee truth or mean the model understands morality. | Represented term: alignment | Source: explicitly confusable glossary metadata

Rationale: Tempting because Alignment is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

Question 4: How does How AI Learns connect to the prompt's Journey?

ID: draft-how-ai-learns-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-how-ai-learns-q4-correct

Misconception tags: None listed

Glossary terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference

choice-draft-how-ai-learns-q4-correct [CORRECT]: Earlier cards taught training, fine-tuning, inference, RAG, and context.

Feedback: Good connection. Earlier cards taught training, fine-tuning, inference, RAG, and context.

choice-draft-how-ai-learns-q4-prompt-1 [DISTRACTOR]: It is text generated by the model after the answer begins.

Feedback: This choice points toward Prompt. Prompt matters nearby, but this question is asking about How AI Learns.

Misconception: Not the same as the generated response or durable model memory. | Represented term: prompt | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

choice-draft-how-ai-learns-q4-alignment-2 [DISTRACTOR]: It is mainly about Alignment, not the mechanism named in How AI Learns.

Feedback: This choice points toward Alignment. Alignment matters nearby, but this question is asking about How AI Learns.

Misconception: Aligned behavior does not guarantee truth or mean the model understands morality. | Represented term: alignment | Source: explicitly confusable glossary metadata

Rationale: Tempting because Alignment is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

choice-draft-how-ai-learns-q4-instruction-tuning-3 [DISTRACTOR]: It is mainly about Instruction tuning, not the mechanism named in How AI Learns.

Feedback: This choice points toward Instruction tuning. Instruction tuning matters nearby, but this question is asking about How AI Learns.

Misconception: Not the same as one instruction inside a prompt. | Represented term: instruction-tuning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Instruction tuning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

Question 5: A learner uses How AI Learns to explain an AI answer. What should they say?

ID: draft-how-ai-learns-q5 | Type: applied scenario | Correct choice ID: choice-draft-how-ai-learns-q5-correct

Misconception tags: None listed

Glossary terms: training, pretraining, fine-tuning, adapter, human-feedback-learning, in-context learning, rag, retrieval, evaluation, inference

choice-draft-how-ai-learns-q5-correct [CORRECT]: People often say a model "learned" something when it may only be using temporary context.

Feedback: Good model-literacy move. People often say a model "learned" something when it may only be using temporary context.

choice-draft-how-ai-learns-q5-alignment-1 [DISTRACTOR]: It is mainly about Alignment, not the mechanism named in How AI Learns.

Feedback: This choice points toward Alignment. Alignment matters nearby, but this question is asking about How AI Learns.

Misconception: Aligned behavior does not guarantee truth or mean the model understands morality. | Represented term: alignment | Source: explicitly confusable glossary metadata

Rationale: Tempting because Alignment is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

choice-draft-how-ai-learns-q5-instruction-tuning-2 [DISTRACTOR]: It is mainly about Instruction tuning, not the mechanism named in How AI Learns.

Feedback: This choice points toward Instruction tuning. Instruction tuning matters nearby, but this question is asking about How AI Learns.

Misconception: Not the same as one instruction inside a prompt. | Represented term: instruction-tuning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Instruction tuning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

choice-draft-how-ai-learns-q5-parameter-3 [DISTRACTOR]: It is mainly about Parameter, not the mechanism named in How AI Learns.

Feedback: This choice points toward Parameter. Parameter matters nearby, but this question is asking about How AI Learns.

Misconception: Often used interchangeably with weights in casual explanations. | Represented term: parameter | Source: explicitly confusable glossary metadata

Rationale: Tempting because Parameter is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific How AI Learns mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

29. Diffusion vs Autoregression

Stage: 6. Twilight: The Wider Landscape | **Card ID:** diffusion | **Card in stage:** 2

Primary learning objective: Explain Diffusion vs Autoregression as part of the day in the life of a prompt.

Secondary objectives: Autoregressive models generate one token at a time, while diffusion models often generate by gradually denoising a noisy pattern., During generation, after the system has been trained., Both are generation-time processes., An LLM response grows token by token., Decision Room and Day Repeats explained autoregressive token generation.

Key distinctions: Both are generation-time processes., An LLM response grows token by token., All generative AI works like ChatGPT.

Misconceptions to correct: All generative AI works like ChatGPT.

Related glossary terms: diffusion, autoregression, generative-ai, sampling, denoising

Nearby glossary distractor candidates: attention, relevance-weight, weight-tensor, activation, tensor

Why this matters: Generative AI is not one mechanism. LLMs and diffusion systems are neighboring families with different generation patterns.

Existing Live Checkpoint

Question count: 1

Question: Which statement is most accurate?

Correct: LLMs often generate text one token at a time; diffusion models often refine from noise.

- LLMs often generate text one token at a time; diffusion models often refine from noise. (correct)
- All generative AI uses token-by-token prediction. Not quite. Diffusion-style generation often refines a noisy representation instead.
- Diffusion models are just larger LLMs. Not quite. They are neighboring generative AI families with different generation patterns.
- Softmax proves image outputs are true. Not quite. Decoding mechanics do not guarantee truth.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: Which statement best explains Diffusion vs Autoregression in this learning card?

ID: draft-diffusion-q1 | **Type:** definition | **Correct choice ID:** choice-draft-diffusion-q1-correct

Misconception tags: None listed

Glossary terms: diffusion, autoregression, generative-ai, sampling, denoising

choice-draft-diffusion-q1-correct [CORRECT]: LLMs often generate text one token at a time; diffusion models often refine from noise.

Feedback: Insight strengthened. Autoregression adds tokens; diffusion refines from noise.

choice-draft-diffusion-q1-all-generative-ai-uses-token-by-token-prediction [DISTRACTOR]: All generative AI uses token-by-token prediction.

Feedback: Not quite. Diffusion-style generation often refines a noisy representation instead.

Misconception: All generative AI uses token-by-token prediction. | Represented term: token | Source: learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-diffusion-q1-diffusion-models-are-just-larger-llms [DISTRACTOR]: Diffusion models are just larger LLMs.

Feedback: Not quite. They are neighboring generative AI families with different generation patterns.

Misconception: Diffusion models are just larger LLMs. | Represented term: diffusion | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-diffusion-q1-softmax-proves-image-outputs-are-true [DISTRACTOR]: Softmax proves image outputs are true.

Feedback: Not quite. Decoding mechanics do not guarantee truth.

Misconception: Softmax proves image outputs are true. | Represented term: softmax | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Diffusion vs Autoregression most directly teach?

ID: draft-diffusion-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-diffusion-q2-correct

Misconception tags: None listed

Glossary terms: diffusion, autoregression, generative-ai, sampling, denoising

choice-draft-diffusion-q2-correct [CORRECT]: Both are generation-time processes.

Feedback: Good distinction. Both are generation-time processes.

choice-draft-diffusion-q2-completion-1 [DISTRACTOR]: It is mainly about Completion, not the mechanism named in Diffusion vs Autoregression.

Feedback: This choice points toward Completion. Completion matters nearby, but this question is asking about Diffusion vs Autoregression.

Misconception: Not a training update. | Represented term: completion | Source: explicitly confusable glossary metadata

Rationale: Tempting because Completion is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Diffusion vs Autoregression mechanism or distinction.

choice-draft-diffusion-q2-generated-token-2 [DISTRACTOR]: It is mainly about Generated token, not the mechanism named in Diffusion vs Autoregression.

Feedback: This choice points toward Generated token. Generated token matters nearby, but this question is asking about Diffusion vs Autoregression.

Misconception: Not a permanent memory or training example by itself. | Represented term: generated-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Generated token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Diffusion vs Autoregression mechanism or distinction.

choice-draft-diffusion-q2-response-tokens-3 [DISTRACTOR]: It is input text that existed before generation began.

Feedback: This choice points toward Response token. Response token matters nearby, but this question is asking about Diffusion vs Autoregression.

Misconception: Not a durable weight update or saved memory. | Represented term: response-tokens | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Diffusion vs Autoregression mechanism or distinction.

Question 3: Which common confusion does Diffusion vs Autoregression help learners avoid?

ID: draft-diffusion-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-diffusion-q3-correct

Misconception tags: None listed

Glossary terms: diffusion, autoregression, generative-ai, sampling, denoising

choice-draft-diffusion-q3-correct [CORRECT]: It corrects the misconception: All generative AI works like ChatGPT.

Feedback: Good distinction. Diffusion vs Autoregression is clearer when that misconception is separated from the mechanism.

choice-draft-diffusion-q3-generated-token-1 [DISTRACTOR]: It is mainly about Generated token, not the mechanism named in Diffusion vs Autoregression.

Feedback: This choice points toward Generated token. Generated token matters nearby, but this question is asking about Diffusion vs Autoregression.

Misconception: Not a permanent memory or training example by itself. | Represented term: generated-token | Source: explicitly confusable glossary metadata

Rationale: Tempting because Generated token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Diffusion vs Autoregression mechanism or distinction.

choice-draft-diffusion-q3-response-tokens-2 [DISTRACTOR]: It is input text that existed before generation began.

Feedback: This choice points toward Response token. Response token matters nearby, but this question is asking about Diffusion vs Autoregression.

Misconception: Not a durable weight update or saved memory. | Represented term: response-tokens | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Diffusion vs Autoregression mechanism or distinction.

choice-draft-diffusion-q3-softmax-3 [DISTRACTOR]: It proves the answer is true.

Feedback: This choice points toward Softmax. Softmax matters nearby, but this question is asking about Diffusion vs Autoregression.

Misconception: Not truth-checking, certainty, training, or choosing the whole response. | Represented term: softmax | Source: explicitly confusable glossary metadata

Rationale: Tempting because Softmax is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Diffusion vs Autoregression mechanism or distinction.

Questions to Review Checklist

☐ Are the learning objectives accurate?

☐ Does each question restate the topic clearly?

☐ Is one answer clearly best?

☐ Are distractors plausible but not unfair?

☐ Are the distractors close enough to teach a boundary?

- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

30. Multimodal AI

Stage: 6. Twilight: The Wider Landscape | **Card ID:** multimodal | **Card in stage:** 3

Primary learning objective: Explain Multimodal AI as part of the day in the life of a prompt.

Secondary objectives: Multimodal AI works with more than one kind of input or output, such as text, images, audio, video, or code., In systems designed to encode, connect, or generate multiple media types., The ability comes from training and architecture., The prompt may include text, images, audio, or other media., Earlier cards focused on LLM text generation.

Key distinctions: The ability comes from training and architecture., The prompt may include text, images, audio, or other media., Multimodal AI has human-like perception or feelings.

Misconceptions to correct: Multimodal AI has human-like perception or feelings.

Related glossary terms: multimodal, representation, embedding, vector, generative-ai, foundation-model

Nearby glossary distractor candidates: model-output, decoding-step, input-context, completion, token, generated-token, tokenizer, response-so-far, tokenization, response-tokens, prompt-tokens, response

Why this matters: Not all AI is text-only. Modern systems increasingly connect language with other media.

Existing Live Checkpoint

Question count: 1

Question: What does multimodal mean?

Correct: The system can work with more than one media type.

- The system can work with more than one media type. (correct)
- The model has feelings. Not quite. Multimodal systems process media representations; they do not gain feelings.
- The context window is larger. Not quite. Multimodal means media types, not window size.
- The model permanently remembers images. Not quite. A specific image prompt is usually temporary input context.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: What does multimodal mean in Multimodal AI?

ID: draft-multimodal-q1 | **Type:** definition | **Correct choice ID:** choice-draft-multimodal-q1-correct

Misconception tags: None listed

Glossary terms: multimodal, representation, embedding, vector, generative-ai, foundation-model

choice-draft-multimodal-q1-correct [CORRECT]: The system can work with more than one media type.

Feedback: Insight strengthened. Multimodal systems connect more than one media type through learned representations or linked components.

choice-draft-multimodal-q1-the-model-has-feelings [DISTRACTOR]: The model has feelings.

Feedback: Not quite. Multimodal systems process media representations; they do not gain feelings.

Misconception: The model has feelings. | Represented term: diffusion | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-multimodal-q1-the-context-window-is-larger [DISTRACTOR]: The context window is larger.

Feedback: Not quite. Multimodal means media types, not window size.

Misconception: The context window is larger. | Represented term: context window | Source: nearby-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-multimodal-q1-the-model-permanently-remembers-images [DISTRACTOR]: The model permanently remembers images.

Feedback: Not quite. A specific image prompt is usually temporary input context.

Misconception: The model permanently remembers images. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Multimodal AI most directly teach?

ID: draft-multimodal-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-multimodal-q2-correct

Misconception tags: None listed

Glossary terms: multimodal, representation, embedding, vector, generative-ai, foundation-model

choice-draft-multimodal-q2-correct [CORRECT]: The ability comes from training and architecture.

Feedback: Good distinction. The ability comes from training and architecture.

choice-draft-multimodal-q2-hidden-state-1 [DISTRACTOR]: It is the token's fixed starting vector before context matters.

Feedback: This choice points toward Hidden state. Hidden state matters nearby, but this question is asking about Multimodal AI.

Misconception: Often confused with embeddings, weights, saved memory, or visible response text. | Represented term: hidden state | Source: explicitly confusable glossary metadata

Rationale: Tempting because Hidden state is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Multimodal AI mechanism or distinction.

choice-draft-multimodal-q2-memory-2 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Multimodal AI.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Multimodal AI mechanism or distinction.

choice-draft-multimodal-q2-embedding-table-3 [DISTRACTOR]: It is mainly about Embedding table, not the mechanism named in Multimodal AI.

Feedback: This choice points toward Embedding table. Embedding table matters nearby, but this question is asking about Multimodal AI.

Misconception: Not the tokenizer and not temporary context. | Represented term: embedding-table | Source: explicitly confusable glossary metadata

Rationale: Tempting because Embedding table is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Multimodal AI mechanism or distinction.

Question 3: Which common confusion does Multimodal AI help learners avoid?

ID: draft-multimodal-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-multimodal-q3-correct

Misconception tags: None listed

Glossary terms: multimodal, representation, embedding, vector, generative-ai, foundation-model

choice-draft-multimodal-q3-correct [CORRECT]: It corrects the misconception: Multimodal AI has human-like perception or feelings.

Feedback: Good distinction. Multimodal AI is clearer when that misconception is separated from the mechanism.

choice-draft-multimodal-q3-memory-1 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Multimodal AI.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Multimodal AI mechanism or distinction.

choice-draft-multimodal-q3-embedding-table-2 [DISTRACTOR]: It is mainly about Embedding table, not the mechanism named in Multimodal AI.

Feedback: This choice points toward Embedding table. Embedding table matters nearby, but this question is asking about Multimodal AI.

Misconception: Not the tokenizer and not temporary context. | Represented term: embedding-table | Source: explicitly confusable glossary metadata

Rationale: Tempting because Embedding table is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Multimodal AI mechanism or distinction.

choice-draft-multimodal-q3-token-id-3 [DISTRACTOR]: It is the meaning of the word itself.

Feedback: This choice points toward Token ID. Token ID matters nearby, but this question is asking about Multimodal AI.

Misconception: The ID is not the meaning, not a memory, and not a truth score. | Represented term: token-id | Source: explicitly confusable glossary metadata

Rationale: Tempting because Token ID is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Multimodal AI mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?

- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

31. The Perfect Storm

Stage: 6. Twilight: The Wider Landscape | **Card ID:** perfect-storm | **Card in stage:** 4

Primary learning objective: Explain The Perfect Storm as part of the day in the life of a prompt.

Secondary objectives: Modern LLMs became possible when human-created digital data, powerful compute and storage, deep-learning methods, human labor, and economic incentives converged., This is model history and system context, after learners know the mechanics of training, inference, context, and RAG., The storm made durable training possible., A prompt reaches a system built from that larger storm; the response is still generated one token at a time., Now that the mechanics are visible, the Journey widens to ask whose data, labor, infrastructure, and choices made those mechanics useful.

Key distinctions: The storm made durable training possible., A prompt reaches a system built from that larger storm; the response is still generated one token at a time., LLMs appeared because of one magic breakthrough.

Misconceptions to correct: LLMs appeared because of one magic breakthrough.

Related glossary terms: perfect-storm-term, training-data, compute, storage, deep-learning, human-feedback-labor, economic-incentives, data-center, training

Nearby glossary distractor candidates: sampling, temperature, top-k, top-p, probability, softmax

Why this matters: It lowers mystery and hype at once. LLMs are not sudden magic objects; they are technical and social systems.

Existing Live Checkpoint

Question count: 1

Question: Which answer best explains why modern LLMs arrived now?

Correct: Data, compute, methods, labor, and incentives converged

- Data, compute, methods, labor, and incentives converged (correct)
- One magic breakthrough happened Not quite. The transformer mattered, but the larger storm included data, compute, methods, labor, and incentives.
- Softmax became conscious Not quite. LLMs are trained systems, not conscious softmax machines.
- Prompts started training themselves Not quite. Prompting shapes current context; it does not explain why LLMs became possible.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: Which answer best explains why modern LLM capability arrived now?

ID: draft-perfect-storm-q1 | **Type:** definition | **Correct choice ID:** choice-draft-perfect-storm-q1-correct

Misconception tags: None listed

Glossary terms: perfect-storm-term, training-data, compute, storage, deep-learning, human-feedback-labor, economic-incentives, data-center, training

choice-draft-perfect-storm-q1-correct [CORRECT]: Data, compute, methods, labor, and incentives converged

Feedback: Insight unlocked: LLMs came from a convergence, not a single magic moment.

choice-draft-perfect-storm-q1-one-magic-breakthrough-happened [DISTRACTOR]: One magic breakthrough happened

Feedback: Not quite. The transformer mattered, but the larger storm included data, compute, methods, labor, and incentives.

Misconception: One magic breakthrough happened | **Represented term:** compute | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-perfect-storm-q1-softmax-became-conscious [DISTRACTOR]: Softmax became conscious

Feedback: Not quite. LLMs are trained systems, not conscious softmax machines.

Misconception: Softmax became conscious | **Represented term:** llm | **Source:** learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-perfect-storm-q1-prompts-started-training-themselves [DISTRACTOR]: Prompts started training themselves

Feedback: Not quite. Prompting shapes current context; it does not explain why LLMs became possible.

Misconception: Prompts started training themselves | **Represented term:** training | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does The Perfect Storm most directly teach?

ID: draft-perfect-storm-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-perfect-storm-q2-correct

Misconception tags: None listed

Glossary terms: perfect-storm-term, training-data, compute, storage, deep-learning, human-feedback-labor, economic-incentives, data-center, training

choice-draft-perfect-storm-q2-correct [CORRECT]: The storm made durable training possible.

Feedback: Good distinction. The storm made durable training possible.

choice-draft-perfect-storm-q2-validation-data-1 [DISTRACTOR]: It is mainly about Validation data, not the mechanism named in The Perfect Storm.

Feedback: This choice points toward Validation data. Validation data matters nearby, but this question is asking about The Perfect Storm.

Misconception: Not training data if it is kept out of weight updates. | Represented term: validation-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Validation data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific The Perfect Storm mechanism or distinction.

choice-draft-perfect-storm-q2-energy-use-2 [DISTRACTOR]: It is mainly about Energy use, not the mechanism named in The Perfect Storm.

Feedback: This choice points toward Energy use. Energy use matters nearby, but this question is asking about The Perfect Storm.

Misconception: Not identical for every model or every response. | Represented term: energy-use | Source: explicitly confusable glossary metadata

Rationale: Tempting because Energy use is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific The Perfect Storm mechanism or distinction.

choice-draft-perfect-storm-q2-efficient-inference-3 [DISTRACTOR]: It is mainly about Efficient inference, not the mechanism named in The Perfect Storm.

Feedback: This choice points toward Efficient inference. Efficient inference matters nearby, but this question is asking about The Perfect Storm.

Misconception: Not automatically worse or automatically safe. | Represented term: efficient-inference | Source: explicitly confusable glossary metadata

Rationale: Tempting because Efficient inference is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific The Perfect Storm mechanism or distinction.

Question 3: Which common confusion does The Perfect Storm help learners avoid?

ID: draft-perfect-storm-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-perfect-storm-q3-correct

Misconception tags: None listed

Glossary terms: perfect-storm-term, training-data, compute, storage, deep-learning, human-feedback-labor, economic-incentives, data-center, training

choice-draft-perfect-storm-q3-correct [CORRECT]: It corrects the misconception: LLMs appeared because of one magic breakthrough.

Feedback: Good distinction. The Perfect Storm is clearer when that misconception is separated from the mechanism.

choice-draft-perfect-storm-q3-energy-use-1 [DISTRACTOR]: It is mainly about Energy use, not the mechanism named in The Perfect Storm.

Feedback: This choice points toward Energy use. Energy use matters nearby, but this question is asking about The Perfect Storm.

Misconception: Not identical for every model or every response. | Represented term: energy-use | Source: explicitly confusable glossary metadata

Rationale: Tempting because Energy use is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific The Perfect Storm mechanism or distinction.

choice-draft-perfect-storm-q3-efficient-inference-2 [DISTRACTOR]: It is mainly about Efficient inference, not the mechanism named in The Perfect Storm.

Feedback: This choice points toward Efficient inference. Efficient inference matters nearby, but this question is asking about The Perfect Storm.

Misconception: Not automatically worse or automatically safe. | Represented term: efficient-inference | Source: explicitly confusable glossary metadata

Rationale: Tempting because Efficient inference is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific The Perfect Storm mechanism or distinction.

choice-draft-perfect-storm-q3-model-checkpoint-3 [DISTRACTOR]: It is mainly about Model Checkpoint, not the mechanism named in The Perfect Storm.

Feedback: This choice points toward Model Checkpoint. Model Checkpoint matters nearby, but this question is asking about The Perfect Storm.

Misconception: Different from a checkpoint quiz in this app. | Represented term: model-checkpoint | Source: explicitly confusable glossary metadata

Rationale: Tempting because Model Checkpoint is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific The Perfect Storm mechanism or distinction.

Question 4: How does The Perfect Storm connect to the prompt's Journey?

ID: draft-perfect-storm-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-perfect-storm-q4-correct

Misconception tags: None listed

Glossary terms: perfect-storm-term, training-data, compute, storage, deep-learning, human-feedback-labor, economic-incentives, data-center, training

choice-draft-perfect-storm-q4-correct [CORRECT]: Now that the mechanics are visible, the Journey widens to ask whose data, labor, infrastructure, and choices made those mechanics useful.

Feedback: Good connection. Now that the mechanics are visible, the Journey widens to ask whose data, labor, infrastructure, and choices made those mechanics useful.

choice-draft-perfect-storm-q4-efficient-inference-1 [DISTRACTOR]: It is mainly about Efficient inference, not the mechanism named in The Perfect Storm.

Feedback: This choice points toward Efficient inference. Efficient inference matters nearby, but this question is asking about The Perfect Storm.

Misconception: Not automatically worse or automatically safe. | Represented term: efficient-inference | Source: explicitly confusable glossary metadata

Rationale: Tempting because Efficient inference is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific The Perfect Storm mechanism or distinction.

choice-draft-perfect-storm-q4-model-checkpoint-2 [DISTRACTOR]: It is mainly about Model Checkpoint, not the mechanism named in The Perfect Storm.

Feedback: This choice points toward Model Checkpoint. Model Checkpoint matters nearby, but this question is asking about The Perfect Storm.

Misconception: Different from a checkpoint quiz in this app. | Represented term: model-checkpoint | Source: explicitly confusable glossary metadata

Rationale: Tempting because Model Checkpoint is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific The Perfect Storm mechanism or distinction.

choice-draft-perfect-storm-q4-machine-learning-3 [DISTRACTOR]: It means all AI and machine learning are exactly the same thing.

Feedback: This choice points toward Machine learning. Machine learning matters nearby, but this question is asking about The Perfect Storm.

Misconception: Not exactly the same thing as all AI. | Represented term: machine-learning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Machine learning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific The Perfect Storm mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

32. Collective Intelligence, Extracted

Stage: 7. Midnight Ledger | **Card ID:** collective-intelligence | **Card in stage:** 1

Primary learning objective: Explain Collective Intelligence, Extracted as part of the day in the life of a prompt.

Secondary objectives: Generative AI depends on patterns learned from human-created language, art, code, documents, research, and culture., During data collection and training, before ordinary use., Training on data can durably shape weights., The response may feel original, but its fluency depends partly on patterns learned from human-created prompt-like and document-like data., The perfect storm explains how LLMs became possible; this card asks whose culture, work, and expression powered the data side of that storm.

Key distinctions: Training on data can durably shape weights., The response may feel original, but its fluency depends partly on patterns learned from human-created prompt-like and document-like data., The model created its abilities alone.

Misconceptions to correct: The model created its abilities alone.

Related glossary terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability

Nearby glossary distractor candidates: denoising, in-context learning, representation

Why this matters: It makes provenance, consent, compensation, attribution, and accountability part of model literacy.

Existing Live Checkpoint

Question count: 1

Question: What powers much of a model's usefulness?

Correct: Patterns from human-created data.

- Patterns from human-created data. (correct)
- The model's consciousness.
- A secret live internet memory. Not quite. Training can durably shape weights, but ordinary use is not a live memory of every source.
- The model inventing language from nothing. Not quite. Model fluency depends on patterns learned from human-created language and culture.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: What powers much of a model's usefulness in Collective Intelligence, Extracted?

ID: draft-collective-intelligence-q1 | **Type:** definition | **Correct choice ID:** choice-draft-collective-intelligence-q1-correct

Misconception tags: None listed

Glossary terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability

choice-draft-collective-intelligence-q1-correct [CORRECT]: Patterns from human-created data.

Feedback: Human-created data can shape model patterns without giving the model a mind.

choice-draft-collective-intelligence-q1-the-model-s-consciousness [DISTRACTOR]: The model's consciousness.

Feedback: This choice is tempting, but Collective Intelligence, Extracted asks for Generative AI depends on patterns learned from human-created language, art, code, documents, research, and culture.

Misconception: The model's consciousness. | Represented term: llm | Source: learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-collective-intelligence-q1-a-secret-live-internet-memory [DISTRACTOR]: A secret live internet memory.

Feedback: Not quite. Training can durably shape weights, but ordinary use is not a live memory of every source.

Misconception: A secret live internet memory. | Represented term: memory | Source: learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-collective-intelligence-q1-the-model-inventing-language-from-nothing [DISTRACTOR]: The model inventing language from nothing.

Feedback: Not quite. Model fluency depends on patterns learned from human-created language and culture.

Misconception: The model inventing language from nothing. | Represented term: compensation | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Collective Intelligence, Extracted most directly teach?

ID: draft-collective-intelligence-q2 | Type: mechanism | Correct choice ID: choice-draft-collective-intelligence-q2-correct

Misconception tags: None listed

Glossary terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability

choice-draft-collective-intelligence-q2-correct [CORRECT]: Training on data can durably shape weights.

Feedback: Good distinction. Training on data can durably shape weights.

choice-draft-collective-intelligence-q2-validation-data-1 [DISTRACTOR]: It is mainly about Validation data, not the mechanism named in Collective Intelligence, Extracted.

Feedback: This choice points toward Validation data. Validation data matters nearby, but this question is asking about Collective Intelligence, Extracted.

Misconception: Not training data if it is kept out of weight updates. | Represented term: validation-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Validation data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Collective Intelligence, Extracted mechanism or distinction.

choice-draft-collective-intelligence-q2-governance-2 [DISTRACTOR]: It is only a technical parameter inside the model.

Feedback: This choice points toward Governance. Governance matters nearby, but this question is asking about Collective Intelligence, Extracted.

Misconception: Not the same as one prompt or one safety filter. | Represented term: governance | Source: explicitly confusable glossary metadata

Rationale: Tempting because Governance is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Collective Intelligence, Extracted mechanism or distinction.

choice-draft-collective-intelligence-q2-human-feedback-learning-3 [DISTRACTOR]: It is mainly about Human feedback learning, not the mechanism named in Collective Intelligence, Extracted.

Feedback: This choice points toward Human feedback learning. Human feedback learning matters nearby, but this question is asking about Collective Intelligence, Extracted.

Misconception: Not moral understanding or guaranteed correctness. | Represented term: human-feedback-learning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Human feedback learning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Collective Intelligence, Extracted mechanism or distinction.

Question 3: Which common confusion does Collective Intelligence, Extracted help learners avoid?

ID: draft-collective-intelligence-q3 | Type: boundary distinction | Correct choice ID: choice-draft-collective-intelligence-q3-correct

Misconception tags: None listed

Glossary terms: collective-intelligence-term, training-data, data-provenance, consent, copyright, compensation, human-feedback-labor, accountability

choice-draft-collective-intelligence-q3-correct [CORRECT]: It corrects the misconception: The model created its abilities alone.

Feedback: Good distinction. Collective Intelligence, Extracted is clearer when that misconception is separated from the mechanism.

choice-draft-collective-intelligence-q3-governance-1 [DISTRACTOR]: It is only a technical parameter inside the model.

Feedback: This choice points toward Governance. Governance matters nearby, but this question is asking about Collective Intelligence, Extracted.

Misconception: Not the same as one prompt or one safety filter. | Represented term: governance | Source: explicitly confusable glossary metadata

Rationale: Tempting because Governance is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Collective Intelligence, Extracted mechanism or distinction.

choice-draft-collective-intelligence-q3-human-feedback-learning-2 [DISTRACTOR]: It is mainly about Human feedback learning, not the mechanism named in Collective Intelligence, Extracted.

Feedback: This choice points toward Human feedback learning. Human feedback learning matters nearby, but this question is asking about Collective Intelligence, Extracted.

Misconception: Not moral understanding or guaranteed correctness. | Represented term: human-feedback-learning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Human feedback learning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Collective Intelligence, Extracted mechanism or distinction.

choice-draft-collective-intelligence-q3-alignment-3 [DISTRACTOR]: It is mainly about Alignment, not the mechanism named in Collective Intelligence, Extracted.

Feedback: This choice points toward Alignment. Alignment matters nearby, but this question is asking about Collective Intelligence, Extracted.

Misconception: Aligned behavior does not guarantee truth or mean the model understands morality. | Represented term: alignment | Source: explicitly confusable glossary metadata

Rationale: Tempting because Alignment is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Collective Intelligence, Extracted mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

33. Costs We Must Count

Stage: 7. Midnight Ledger | **Card ID:** costs-we-must-count | **Card in stage:** 2

Primary learning objective: Explain Costs We Must Count as part of the day in the life of a prompt.

Secondary objectives: Generative AI has physical, social, cultural, and ethical costs that must be counted honestly., Across training, inference, infrastructure, procurement, workplace use, education, and media ecosystems., Some costs come from durable training infrastructure; others come from repeated inference workloads, deployment choices, and institutional use., A short response may hide a long supply chain of compute, data, labor, and policy choices., Benefits can be real; this paired card asks what must be protected, redesigned, or governed while pursuing those benefits.

Key distinctions: Some costs come from durable training infrastructure; others come from repeated inference workloads, deployment choices, and institutional use., A short response may hide a long supply chain of compute, data, labor, and policy choices., Digital means weightless or cost-free.

Misconceptions to correct: Digital means weightless or cost-free.

Related glossary terms: environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste, copyright, labor-disruption, deskilling, privacy, governance

Nearby glossary distractor candidates: information-pollution, bias, risk-literacy, prompt-injection, tool-use, overreliance, vendor-lock-in, concentration-of-power

Why this matters: Model literacy includes stewardship: institutions should ask what they are buying, who is affected, and what tradeoffs are hidden.

Existing Live Checkpoint

Question count: 1

Question: Which statement is most responsible?

Correct: AI costs vary and should be counted before decisions

- AI costs vary and should be counted before decisions (correct)
- Digital systems have no physical costs Not quite. Digital services still run on material infrastructure.
- All AI costs are identical everywhere Not quite. Costs vary by model, workload, hardware, cooling, region, and electricity source.
- One prompt always has the same footprint Not quite. The footprint depends on model, workload, hardware, data center, electricity source, and deployment.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: Which statement is most responsible in Costs We Must Count?

ID: draft-costs-we-must-count-q1 | **Type:** definition | **Correct choice ID:** choice-draft-costs-we-must-count-q1-correct

Misconception tags: None listed

Glossary terms: environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste, copyright, labor-disruption, deskilling, privacy, governance

choice-draft-costs-we-must-count-q1-correct [CORRECT]: AI costs vary and should be counted before decisions

Feedback: Insight unlocked: counting costs supports wiser choices.

choice-draft-costs-we-must-count-q1-digital-systems-have-no-physical-costs [DISTRACTOR]: Digital systems have no physical costs

Feedback: Not quite. Digital services still run on material infrastructure.

Misconception: Digital systems have no physical costs | Represented term: energy-use | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-costs-we-must-count-q1-all-ai-costs-are-identical-everywhere [DISTRACTOR]: All AI costs are identical everywhere

Feedback: Not quite. Costs vary by model, workload, hardware, cooling, region, and electricity source.

Misconception: All AI costs are identical everywhere | Represented term: ai | Source: learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-costs-we-must-count-q1-one-prompt-always-has-the-same-footprint [DISTRACTOR]: One prompt always has the same footprint

Feedback: Not quite. The footprint depends on model, workload, hardware, data center, electricity source, and deployment.

Misconception: One prompt always has the same footprint | Represented term: carbon-emissions | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Costs We Must Count most directly teach?

ID: draft-costs-we-must-count-q2 | Type: mechanism | Correct choice ID: choice-draft-costs-we-must-count-q2-correct

Misconception tags: None listed

Glossary terms: environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste, copyright, labor-disruption, deskilling, privacy, governance

choice-draft-costs-we-must-count-q2-correct [CORRECT]: Some costs come from durable training infrastructure; others come from repeated inference workloads, deployment choices, and institutional use.

Feedback: Good distinction. Some costs come from durable training infrastructure; others come from repeated inference workloads, deployment choices, and institutional use.

choice-draft-costs-we-must-count-q2-data-provenance-1 [DISTRACTOR]: It is mainly about Data provenance, not the mechanism named in Costs We Must Count.

Feedback: This choice points toward Data provenance. Data provenance matters nearby, but this question is asking about Costs We Must Count.

Misconception: Not the same as model accuracy or output citation. | Represented term: data-provenance | Source: explicitly confusable glossary metadata

Rationale: Tempting because Data provenance is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Costs We Must Count mechanism or distinction.

choice-draft-costs-we-must-count-q2-consent-2 [DISTRACTOR]: It is mainly about Consent, not the mechanism named in Costs We Must Count.

Feedback: This choice points toward Consent. Consent matters nearby, but this question is asking about Costs We Must Count.

Misconception: Not the same as public availability. | Represented term: consent | Source: explicitly confusable glossary metadata

Rationale: Tempting because Consent is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Costs We Must Count mechanism or distinction.

choice-draft-costs-we-must-count-q2-compensation-3 [DISTRACTOR]: It is mainly about Compensation, not the mechanism named in Costs We Must Count.

Feedback: This choice points toward Compensation. Compensation matters nearby, but this question is asking about Costs We Must Count.

Misconception: Not guaranteed by model usefulness or by public access to a work. | Represented term: compensation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Compensation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Costs We Must Count mechanism or distinction.

Question 3: Which common confusion does Costs We Must Count help learners avoid?

ID: draft-costs-we-must-count-q3 | Type: boundary distinction | Correct choice ID: choice-draft-costs-we-must-count-q3-correct

Misconception tags: None listed

Glossary terms: environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste, copyright, labor-disruption, deskilling, privacy, governance

choice-draft-costs-we-must-count-q3-correct [CORRECT]: It corrects the misconception: Digital means weightless or cost-free.

Feedback: Good distinction. Costs We Must Count is clearer when that misconception is separated from the mechanism.

choice-draft-costs-we-must-count-q3-consent-1 [DISTRACTOR]: It is mainly about Consent, not the mechanism named in Costs We Must Count.

Feedback: This choice points toward Consent. Consent matters nearby, but this question is asking about Costs We Must Count.

Misconception: Not the same as public availability. | Represented term: consent | Source: explicitly confusable glossary metadata

Rationale: Tempting because Consent is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Costs We Must Count mechanism or distinction.

choice-draft-costs-we-must-count-q3-compensation-2 [DISTRACTOR]: It is mainly about Compensation, not the mechanism named in Costs We Must Count.

Feedback: This choice points toward Compensation. Compensation matters nearby, but this question is asking about Costs We Must Count.

Misconception: Not guaranteed by model usefulness or by public access to a work. | Represented term: compensation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Compensation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Costs We Must Count mechanism or distinction.

choice-draft-costs-we-must-count-q3-human-centered-ai-term-3 [DISTRACTOR]: It is mainly about Human-centered AI, not the mechanism named in Costs We Must Count.

Feedback: This choice points toward Human-centered AI. Human-centered AI matters nearby, but this question is asking about Costs We Must Count.

Misconception: Not AI deciding human values for us. | Represented term: human-centered-ai-term | Source: explicitly confusable glossary metadata

Rationale: Tempting because Human-centered AI is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Costs We Must Count mechanism or distinction.

Question 4: How does Costs We Must Count connect to the prompt's Journey?

ID: draft-costs-we-must-count-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-costs-we-must-count-q4-correct

Misconception tags: None listed

Glossary terms: environmental-footprint, energy-use, water-use, carbon-emissions, data-center, e-waste, copyright, labor-disruption, deskilling, privacy, governance

choice-draft-costs-we-must-count-q4-correct [CORRECT]: Benefits can be real; this paired card asks what must be protected, redesigned, or governed while pursuing those benefits.

Feedback: Good connection. Benefits can be real; this paired card asks what must be protected, redesigned, or governed while pursuing those benefits.

choice-draft-costs-we-must-count-q4-compensation-1 [DISTRACTOR]: It is mainly about Compensation, not the mechanism named in Costs We Must Count.

Feedback: This choice points toward Compensation. Compensation matters nearby, but this question is asking about Costs We Must Count.

Misconception: Not guaranteed by model usefulness or by public access to a work. | **Represented term:** compensation | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Compensation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Costs We Must Count mechanism or distinction.

choice-draft-costs-we-must-count-q4-human-centered-ai-term-2 [DISTRACTOR]: It is mainly about Human-centered AI, not the mechanism named in Costs We Must Count.

Feedback: This choice points toward Human-centered AI. Human-centered AI matters nearby, but this question is asking about Costs We Must Count.

Misconception: Not AI deciding human values for us. | **Represented term:** human-centered-ai-term | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Human-centered AI is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Costs We Must Count mechanism or distinction.

choice-draft-costs-we-must-count-q4-human-review-3 [DISTRACTOR]: It is mainly about Human review, not the mechanism named in Costs We Must Count.

Feedback: This choice points toward Human review. Human review matters nearby, but this question is asking about Costs We Must Count.

Misconception: Not rubber-stamping fluent output. | **Represented term:** human-review | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Human review is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Costs We Must Count mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

34. Risk vs Myth

Stage: 7. Midnight Ledger | **Card ID:** risk-myth | **Card in stage:** 3

Primary learning objective: Explain Risk vs Myth as part of the day in the life of a prompt.

Secondary objectives: Risk literacy means separating practical, mechanism-based AI risks from myths or magical stories about what models are., In AI design, procurement, deployment, use, review, and governance., Risk literacy is a human practice., Prompt data, retrieved context, tool outputs, and generated responses can all create risk depending on what they contain and how people use them., Costs showed what must be counted.

Key distinctions: Risk literacy is a human practice., Prompt data, retrieved context, tool outputs, and generated responses can all create risk depending on what they contain and how people use them., Risk only means sci-fi doom, or every fluent model has a mind.

Misconceptions to correct: Risk only means sci-fi doom, or every fluent model has a mind.

Related glossary terms: risk-literacy, privacy, hallucination, prompt-injection, tool-use, overreliance, vendor-lock-in, concentration-of-power, governance, accountability

Nearby glossary distractor candidates: information-pollution, bias, deskilling, labor-disruption, e-waste, carbon-emissions, water-use, energy-use, environmental-footprint

Why this matters: Institutions make better decisions when they focus on real mechanisms rather than fantasy fears.

Existing Live Checkpoint

Question count: 1

Question: Which is a real institutional risk?

Correct: Uploading private data.

- Uploading private data. (correct)
- Softmax stealing files. Not quite. Softmax converts scores to probabilities; file exposure comes from data handling, tools, permissions, and integrations.
- The model becoming conscious in the chat. Not quite. Fluent output does not imply consciousness.
- The model secretly training itself during every ordinary chat. Not quite. Ordinary inference does not durably update weights unless a separate training process happens.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: Which is a real institutional risk in Risk vs Myth?

ID: draft-risk-myth-q1 | **Type:** definition | **Correct choice ID:** choice-draft-risk-myth-q1-correct

Misconception tags: None listed

Glossary terms: risk-literacy, privacy, hallucination, prompt-injection, tool-use, overreliance, vendor-lock-in, concentration-of-power, governance, accountability

choice-draft-risk-myth-q1-correct [CORRECT]: Uploading private data.

Feedback: Privacy exposure is a practical risk caused by data, context, tools, and institutions.

choice-draft-risk-myth-q1-softmax-stealing-files [DISTRACTOR]: Softmax stealing files.

Feedback: Not quite. Softmax converts scores to probabilities; file exposure comes from data handling, tools, permissions, and integrations.

Misconception: Softmax stealing files. | **Represented term:** softmax | **Source:** nearby-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-risk-myth-q1-the-model-becoming-conscious-in-the-chat [DISTRACTOR]: The model becoming conscious in the chat.

Feedback: Not quite. Fluent output does not imply consciousness.

Misconception: The model becoming conscious in the chat. | **Represented term:** llm | **Source:** learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-risk-myth-q1-the-model-secretly-training-itself-during-every-ordinary [DISTRACTOR]: The model secretly training itself during every ordinary chat.

Feedback: Not quite. Ordinary inference does not durably update weights unless a separate training process happens.

Misconception: The model secretly training itself during every ordinary chat. | **Represented term:** training | **Source:** nearby-stage glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Risk vs Myth most directly teach?

ID: draft-risk-myth-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-risk-myth-q2-correct

Misconception tags: None listed

Glossary terms: risk-literacy, privacy, hallucination, prompt-injection, tool-use, overreliance, vendor-lock-in, concentration-of-power, governance, accountability

choice-draft-risk-myth-q2-correct [CORRECT]: Risk literacy is a human practice.

Feedback: Good distinction. Risk literacy is a human practice.

choice-draft-risk-myth-q2-sampling-1 [DISTRACTOR]: It writes the whole answer at once.

Feedback: This choice points toward Sampling. Sampling matters nearby, but this question is asking about Risk vs Myth.

Misconception: Not writing the whole response, permanently teaching the model, or proving truth. | **Represented term:** sampling | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Sampling is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Risk vs Myth mechanism or distinction.

choice-draft-risk-myth-q2-human-review-2 [DISTRACTOR]: It is mainly about Human review, not the mechanism named in Risk vs Myth.

Feedback: This choice points toward Human review. Human review matters nearby, but this question is asking about Risk vs Myth.

Misconception: Not rubber-stamping fluent output. | **Represented term:** human-review | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Human review is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Risk vs Myth mechanism or distinction.

choice-draft-risk-myth-q2-responsible-ai-3 [DISTRACTOR]: It is mainly about Responsible AI, not the mechanism named in Risk vs Myth.

Feedback: This choice points toward Responsible AI. Responsible AI matters nearby, but this question is asking about Risk vs Myth.

Misconception: Not a guarantee that every output is true or safe. | **Represented term:** responsible-ai | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Responsible AI is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Risk vs Myth mechanism or distinction.

Question 3: Which common confusion does Risk vs Myth help learners avoid?

ID: draft-risk-myth-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-risk-myth-q3-correct

Misconception tags: None listed

Glossary terms: risk-literacy, privacy, hallucination, prompt-injection, tool-use, overreliance, vendor-lock-in, concentration-of-power, governance, accountability

choice-draft-risk-myth-q3-correct [CORRECT]: It corrects the misconception: Risk only means sci-fi doom, or every fluent model has a mind.

Feedback: Good distinction. Risk vs Myth is clearer when that misconception is separated from the mechanism.

choice-draft-risk-myth-q3-human-review-1 [DISTRACTOR]: It is mainly about Human review, not the mechanism named in Risk vs Myth.

Feedback: This choice points toward Human review. Human review matters nearby, but this question is asking about Risk vs Myth.

Misconception: Not rubber-stamping fluent output. | **Represented term:** human-review | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Human review is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Risk vs Myth mechanism or distinction.

choice-draft-risk-myth-q3-responsible-ai-2 [DISTRACTOR]: It is mainly about Responsible AI, not the mechanism named in Risk vs Myth.

Feedback: This choice points toward Responsible AI. Responsible AI matters nearby, but this question is asking about Risk vs Myth.

Misconception: Not a guarantee that every output is true or safe. | **Represented term:** responsible-ai | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Responsible AI is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Risk vs Myth mechanism or distinction.

choice-draft-risk-myth-q3-data-center-3 [DISTRACTOR]: It is mainly about Data center, not the mechanism named in Risk vs Myth.

Feedback: This choice points toward Data center. Data center matters nearby, but this question is asking about Risk vs Myth.

Misconception: Not a weightless cloud or a single laptop. | **Represented term:** data-center | **Source:** explicitly confusable glossary metadata

Rationale: Tempting because Data center is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Risk vs Myth mechanism or distinction.

Question 4: How does Risk vs Myth connect to the prompt's Journey?

ID: draft-risk-myth-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-risk-myth-q4-correct

Misconception tags: None listed

Glossary terms: risk-literacy, privacy, hallucination, prompt-injection, tool-use, overreliance, vendor-lock-in, concentration-of-power, governance, accountability

choice-draft-risk-myth-q4-correct [CORRECT]: Costs showed what must be counted.

Feedback: Good connection. Costs showed what must be counted.

choice-draft-risk-myth-q4-responsible-ai-1 [DISTRACTOR]: It is mainly about Responsible AI, not the mechanism named in Risk vs Myth.

Feedback: This choice points toward Responsible AI. Responsible AI matters nearby, but this question is asking about Risk vs Myth.

Misconception: Not a guarantee that every output is true or safe. | Represented term: responsible-ai | Source: explicitly confusable glossary metadata

Rationale: Tempting because Responsible AI is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Risk vs Myth mechanism or distinction.

choice-draft-risk-myth-q4-data-center-2 [DISTRACTOR]: It is mainly about Data center, not the mechanism named in Risk vs Myth.

Feedback: This choice points toward Data center. Data center matters nearby, but this question is asking about Risk vs Myth.

Misconception: Not a weightless cloud or a single laptop. | Represented term: data-center | Source: explicitly confusable glossary metadata

Rationale: Tempting because Data center is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Risk vs Myth mechanism or distinction.

choice-draft-risk-myth-q4-policy-3 [DISTRACTOR]: It is mainly about Policy, not the mechanism named in Risk vs Myth.

Feedback: This choice points toward Policy. Policy matters nearby, but this question is asking about Risk vs Myth.

Misconception: Not the same as moral understanding. | Represented term: policy | Source: explicitly confusable glossary metadata

Rationale: Tempting because Policy is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Risk vs Myth mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

35. Benefits Worth Taking Seriously

Stage: 8. New Dawn | **Card ID:** benefits-worth-taking-seriously | **Card in stage:** 1

Primary learning objective: Explain Benefits Worth Taking Seriously as part of the day in the life of a prompt.

Secondary objectives: AI benefits are strongest when models amplify human judgment rather than replacing accountability., During deployment and use, when people decide what jobs to give the system and how humans review the output., Benefits during use usually come from trained weights plus temporary context, tools, retrieval, and human review., Good prompts can shape useful context and output format, but the response still needs human judgment., After seeing how human expression powers capability, this card asks what that capability can responsibly help people do.

Key distinctions: Benefits during use usually come from trained weights plus temporary context, tools, retrieval, and human review., Good prompts can shape useful context and output format, but the response still needs human judgment., AI benefits require believing in imminent utopia.

Misconceptions to correct: AI benefits require believing in imminent utopia.

Related glossary terms: benefits, accessibility, translation, summarization, rag, human-review, uncertainty, effective-prompting

Nearby glossary distractor candidates: concentration-of-power, vendor-lock-in, overreliance, tool-use, privacy, prompt-injection, risk-literacy, human-centered-ai-term, dignity, accountability, common-good

Why this matters: A serious view can count real benefits without sliding into hype.

Existing Live Checkpoint

Question count: 1

Question: Which benefit claim is the safest kind to make without hype?

Correct: AI can assist drafts, search, translation, and access under human review.

- AI can assist drafts, search, translation, and access under human review. (correct)
- AI will automatically solve education Not quite. Broad utopian claims need strong evidence and careful limits.
- AI has human judgment now Not quite. A model can assist, but humans remain responsible.
- AI benefits mean no costs matter Not quite. Benefits and costs both need honest accounting.

Proposed Draft Question Bank

Proposed count: 3. **Rationale:** Standard concept card; three checks cover definition, mechanism, and a nearby misconception.

Question 1: Which benefit claim is the safest kind to make without hype in Benefits Worth Taking Seriously?

ID: draft-benefits-worth-taking-seriously-q1 | **Type:** definition | **Correct choice ID:** choice-draft-benefits-worth-taking-seriously-q1-correct

Misconception tags: None listed

Glossary terms: benefits, accessibility, translation, summarization, rag, human-review, uncertainty, effective-prompting

choice-draft-benefits-worth-taking-seriously-q1-correct [CORRECT]: AI can assist drafts, search, translation, and access under human review.

Feedback: Insight unlocked: benefits can be real and bounded.

choice-draft-benefits-worth-taking-seriously-q1-ai-will-automatically-solve-education [DISTRACTOR]: AI will automatically solve education

Feedback: Not quite. Broad utopian claims need strong evidence and careful limits.

Misconception: AI will automatically solve education | **Represented term:** translation | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-benefits-worth-taking-seriously-q1-ai-has-human-judgment-now [DISTRACTOR]: AI has human judgment now

Feedback: Not quite. A model can assist, but humans remain responsible.

Misconception: AI has human judgment now | **Represented term:** summarization | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-benefits-worth-taking-seriously-q1-ai-benefits-mean-no-costs-matter [DISTRACTOR]: AI benefits mean no costs matter

Feedback: Not quite. Benefits and costs both need honest accounting.

Misconception: AI benefits mean no costs matter | **Represented term:** human-review | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Benefits Worth Taking Seriously most directly teach?

ID: draft-benefits-worth-taking-seriously-q2 | Type: mechanism | Correct choice ID: choice-draft-benefits-worth-taking-seriously-q2-correct

Misconception tags: None listed

Glossary terms: benefits, accessibility, translation, summarization, rag, human-review, uncertainty, effective-prompting

choice-draft-benefits-worth-taking-seriously-q2-correct [CORRECT]: Benefits during use usually come from trained weights plus temporary context, tools, retrieval, and human review.

Feedback: Good distinction. Benefits during use usually come from trained weights plus temporary context, tools, retrieval, and human review.

choice-draft-benefits-worth-taking-seriously-q2-retrieval-1 [DISTRACTOR]: It is the same as fine-tuning the model on new data.

Feedback: This choice points toward Retrieval. Retrieval matters nearby, but this question is asking about Benefits Worth Taking Seriously.

Misconception: Retrieval is being treated as the same idea as Benefits Worth Taking Seriously. | Represented term: retrieval | Source: explicitly confusable glossary metadata

Rationale: Tempting because Retrieval is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Benefits Worth Taking Seriously mechanism or distinction.

choice-draft-benefits-worth-taking-seriously-q2-fine-tuning-2 [DISTRACTOR]: It is the same as adding outside text into the current prompt.

Feedback: This choice points toward Fine-tuning. Fine-tuning matters nearby, but this question is asking about Benefits Worth Taking Seriously.

Misconception: Different from prompting; prompting changes temporary context, not durable weights. | Represented term: fine-tuning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Fine-tuning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Benefits Worth Taking Seriously mechanism or distinction.

choice-draft-benefits-worth-taking-seriously-q2-training-3 [DISTRACTOR]: It permanently updates model weights during this ordinary response.

Feedback: This choice points toward Training. Training matters nearby, but this question is asking about Benefits Worth Taking Seriously.

Misconception: Ordinary chat inference is not training unless a separate training process updates weights. | Represented term: training | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Benefits Worth Taking Seriously mechanism or distinction.

Question 3: Which common confusion does Benefits Worth Taking Seriously help learners avoid?

ID: draft-benefits-worth-taking-seriously-q3 | Type: boundary distinction | Correct choice ID: choice-draft-benefits-worth-taking-seriously-q3-correct

Misconception tags: None listed

Glossary terms: benefits, accessibility, translation, summarization, rag, human-review, uncertainty, effective-prompting

choice-draft-benefits-worth-taking-seriously-q3-correct [CORRECT]: It corrects the misconception: AI benefits require believing in imminent utopia.

Feedback: Good distinction. Benefits Worth Taking Seriously is clearer when that misconception is separated from the mechanism.

choice-draft-benefits-worth-taking-seriously-q3-fine-tuning-1 [DISTRACTOR]: It is the same as adding outside text into the current prompt.

Feedback: This choice points toward Fine-tuning. Fine-tuning matters nearby, but this question is asking about Benefits Worth Taking Seriously.

Misconception: Different from prompting; prompting changes temporary context, not durable weights. | Represented term: fine-tuning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Fine-tuning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Benefits Worth Taking Seriously mechanism or distinction.

choice-draft-benefits-worth-taking-seriously-q3-training-2 [DISTRACTOR]: It permanently updates model weights during this ordinary response.

Feedback: This choice points toward Training. Training matters nearby, but this question is asking about Benefits Worth Taking Seriously.

Misconception: Ordinary chat inference is not training unless a separate training process updates weights. | Represented term: training | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Benefits Worth Taking Seriously mechanism or distinction.

choice-draft-benefits-worth-taking-seriously-q3-input-context-3 [DISTRACTOR]: It is mainly about Input context, not the mechanism named in Benefits Worth Taking Seriously.

Feedback: This choice points toward Input context. Input context matters nearby, but this question is asking about Benefits Worth Taking Seriously.

Misconception: Not the training dataset or durable model weights. | Represented term: input-context | Source: explicitly confusable glossary metadata

Rationale: Tempting because Input context is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Benefits Worth Taking Seriously mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

36. Human-Centered AI

Stage: 8. New Dawn | **Card ID:** human-centered-ai | **Card in stage:** 2

Primary learning objective: Explain Human-Centered AI as part of the day in the life of a prompt.

Secondary objectives: Human-centered AI asks whether a system supports dignity, learning, responsibility, creativity, relationships, and the common good., During design, procurement, classroom use, research practice, administration, policy, and evaluation., Human-centered choices can shape durable systems, policies, and training; prompts alone cannot give the model moral understanding., A response can sound ethical, but ethical responsibility belongs to humans and institutions., After counting costs, this card names the values that should guide tradeoffs.

Key distinctions: Human-centered choices can shape durable systems, policies, and training; prompts alone cannot give the model moral understanding., A response can sound ethical, but ethical responsibility belongs to humans and institutions., If AI is powerful, it should decide.

Misconceptions to correct: If AI is powerful, it should decide.

Related glossary terms: human-centered-ai-term, dignity, common-good, human-review, accountability, governance, alignment, responsible-ai

Nearby glossary distractor candidates: water-use, energy-use, carbon-emissions, environmental-footprint, e-waste, labor-disruption, deskilling, bias, information-pollution, copyright, risk-literacy, compensation

Why this matters: This gives learners language for judgment, not just awe or fear.

Existing Live Checkpoint

Question count: 1

Question: What should remain central in high-stakes AI use?

Correct: Human judgment and accountability

- Human judgment and accountability (correct)
- The fastest automated answer Not quite. Speed is not the same as dignity, learning, or responsibility.
- The model's moral feelings
- No review once the output is fluent Not quite. Fluency does not replace human accountability.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: What should remain central in high-stakes AI use in Human-Centered AI?

ID: draft-human-centered-ai-q1 | **Type:** definition | **Correct choice ID:** choice-draft-human-centered-ai-q1-correct

Misconception tags: None listed

Glossary terms: human-centered-ai-term, dignity, common-good, human-review, accountability, governance, alignment, responsible-ai

choice-draft-human-centered-ai-q1-correct [CORRECT]: Human judgment and accountability

Feedback: Insight unlocked: human-centered AI keeps responsibility with people.

choice-draft-human-centered-ai-q1-the-fastest-automated-answer [DISTRACTOR]: The fastest automated answer

Feedback: Not quite. Speed is not the same as dignity, learning, or responsibility.

Misconception: The fastest automated answer | **Represented term:** common-good | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-human-centered-ai-q1-the-model-s-moral-feelings [DISTRACTOR]: The model's moral feelings

Feedback: This choice is tempting, but Human-Centered AI asks for Human-centered AI asks whether a system supports dignity, learning, responsibility, creativity, relationships, and the common good.

Misconception: The model's moral feelings | **Represented term:** human-review | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-human-centered-ai-q1-no-review-once-the-output-is-fluent [DISTRACTOR]: No review once the output is fluent

Feedback: Not quite. Fluency does not replace human accountability.

Misconception: No review once the output is fluent | **Represented term:** governance | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Human-Centered AI most directly teach?

ID: draft-human-centered-ai-q2 | **Type:** mechanism | **Correct choice ID:** choice-draft-human-centered-ai-q2-correct

Misconception tags: None listed

Glossary terms: human-centered-ai-term, dignity, common-good, human-review, accountability, governance, alignment, responsible-ai

choice-draft-human-centered-ai-q2-correct [CORRECT]: Human-centered choices can shape durable systems, policies, and training; prompts alone cannot give the model moral understanding.

Feedback: Good distinction. Human-centered choices can shape durable systems, policies, and training; prompts alone cannot give the model moral understanding.

choice-draft-human-centered-ai-q2-policy-1 [DISTRACTOR]: It is mainly about Policy, not the mechanism named in Human-Centered AI.

Feedback: This choice points toward Policy. Policy matters nearby, but this question is asking about Human-Centered AI.

Misconception: Not the same as moral understanding. | Represented term: policy | Source: explicitly confusable glossary metadata

Rationale: Tempting because Policy is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Human-Centered AI mechanism or distinction.

choice-draft-human-centered-ai-q2-instruction-tuning-2 [DISTRACTOR]: It is mainly about Instruction tuning, not the mechanism named in Human-Centered AI.

Feedback: This choice points toward Instruction tuning. Instruction tuning matters nearby, but this question is asking about Human-Centered AI.

Misconception: Not the same as one instruction inside a prompt. | Represented term: instruction-tuning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Instruction tuning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Human-Centered AI mechanism or distinction.

choice-draft-human-centered-ai-q2-human-feedback-learning-3 [DISTRACTOR]: It is mainly about Human feedback learning, not the mechanism named in Human-Centered AI.

Feedback: This choice points toward Human feedback learning. Human feedback learning matters nearby, but this question is asking about Human-Centered AI.

Misconception: Not moral understanding or guaranteed correctness. | Represented term: human-feedback-learning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Human feedback learning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Human-Centered AI mechanism or distinction.

Question 3: Which common confusion does Human-Centered AI help learners avoid?

ID: draft-human-centered-ai-q3 | **Type:** boundary distinction | **Correct choice ID:** choice-draft-human-centered-ai-q3-correct

Misconception tags: None listed

Glossary terms: human-centered-ai-term, dignity, common-good, human-review, accountability, governance, alignment, responsible-ai

choice-draft-human-centered-ai-q3-correct [CORRECT]: It corrects the misconception: If AI is powerful, it should decide.

Feedback: Good distinction. Human-Centered AI is clearer when that misconception is separated from the mechanism.

choice-draft-human-centered-ai-q3-instruction-tuning-1 [DISTRACTOR]: It is mainly about Instruction tuning, not the mechanism named in Human-Centered AI.

Feedback: This choice points toward Instruction tuning. Instruction tuning matters nearby, but this question is asking about Human-Centered AI.

Misconception: Not the same as one instruction inside a prompt. | Represented term: instruction-tuning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Instruction tuning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Human-Centered AI mechanism or distinction.

choice-draft-human-centered-ai-q3-human-feedback-learning-2 [DISTRACTOR]: It is mainly about Human feedback learning, not the mechanism named in Human-Centered AI.

Feedback: This choice points toward Human feedback learning. Human feedback learning matters nearby, but this question is asking about Human-Centered AI.

Misconception: Not moral understanding or guaranteed correctness. | Represented term: human-feedback-learning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Human feedback learning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Human-Centered AI mechanism or distinction.

choice-draft-human-centered-ai-q3-rlhf-3 [DISTRACTOR]: It is mainly about RLHF, not the mechanism named in Human-Centered AI.

Feedback: This choice points toward RLHF. RLHF matters nearby, but this question is asking about Human-Centered AI.

Misconception: Not a guarantee that every answer is true or safe. | Represented term: rlhf | Source: explicitly confusable glossary metadata

Rationale: Tempting because RLHF is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Human-Centered AI mechanism or distinction.

Question 4: How does Human-Centered AI connect to the prompt's Journey?

ID: draft-human-centered-ai-q4 | **Type:** concept relationship | **Correct choice ID:** choice-draft-human-centered-ai-q4-correct

Misconception tags: None listed

Glossary terms: human-centered-ai-term, dignity, common-good, human-review, accountability, governance, alignment, responsible-ai

choice-draft-human-centered-ai-q4-correct [CORRECT]: After counting costs, this card names the values that should guide tradeoffs.

Feedback: Good connection. After counting costs, this card names the values that should guide tradeoffs.

choice-draft-human-centered-ai-q4-human-feedback-learning-1 [DISTRACTOR]: It is mainly about Human feedback learning, not the mechanism named in Human-Centered AI.

Feedback: This choice points toward Human feedback learning. Human feedback learning matters nearby, but this question is asking about Human-Centered AI.

Misconception: Not moral understanding or guaranteed correctness. | Represented term: human-feedback-learning | Source: explicitly confusable glossary metadata

Rationale: Tempting because Human feedback learning is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Human-Centered AI mechanism or distinction.

choice-draft-human-centered-ai-q4-rlhf-2 [DISTRACTOR]: It is mainly about RLHF, not the mechanism named in Human-Centered AI.

Feedback: This choice points toward RLHF. RLHF matters nearby, but this question is asking about Human-Centered AI.

Misconception: Not a guarantee that every answer is true or safe. | Represented term: rlhf | Source: explicitly confusable glossary metadata

Rationale: Tempting because RLHF is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Human-Centered AI mechanism or distinction.

choice-draft-human-centered-ai-q4-guardrail-3 [DISTRACTOR]: It is mainly about Guardrail, not the mechanism named in Human-Centered AI.

Feedback: This choice points toward Guardrail. Guardrail matters nearby, but this question is asking about Human-Centered AI.

Misconception: Not a perfect safety guarantee. | Represented term: guardrail | Source: explicitly confusable glossary metadata

Rationale: Tempting because Guardrail is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Human-Centered AI mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

37. Better AI Is a Choice

Stage: 8. New Dawn | **Card ID:** better-ai-choice | **Card in stage:** 3

Primary learning objective: Explain Better AI Is a Choice as part of the day in the life of a prompt.

Secondary objectives: The costs are shaped by design, deployment, governance, and incentives., Before deployment, during procurement, inside system design, in policy, and through ongoing evaluation., Some choices durably shape models and institutions; others shape a particular workflow or inference run., Prompting helps individuals use a system better, but system-level choices determine what the system is allowed, optimized, and reviewed to do., Human-centered values become design, policy, governance, and incentive choices.

Key distinctions: Some choices durably shape models and institutions; others shape a particular workflow or inference run., Prompting helps individuals use a system better, but system-level choices determine what the system is allowed, optimized, and reviewed to do., The harms are inevitable if the technology is useful.

Misconceptions to correct: The harms are inevitable if the technology is useful.

Related glossary terms: responsible-ai, model-distillation, efficient-inference, data-provenance, consent, compensation, human-review, accountability, governance, rag

Nearby glossary distractor candidates: concentration-of-power, vendor-lock-in, overreliance, tool-use, privacy, prompt-injection, risk-literacy, benefits, accessibility, translation, summarization

Why this matters: Learners should feel agency without pretending every problem is easy.

Existing Live Checkpoint

Question count: 1

Question: Which choice can reduce some AI risks?

Correct: Use retrieval and human review when they fit the task

- Use retrieval and human review when they fit the task (correct)
- Assume bigger is always better Not quite. Smaller task-specific systems may be better for some goals.
- Remove all human accountability Not quite. High-stakes use needs human responsibility.
- Ignore data provenance Not quite. Provenance, consent, and licensing matter.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: Which choice can reduce some AI risks in Better AI Is a Choice?

ID: draft-better-ai-choice-q1 | **Type:** definition | **Correct choice ID:** choice-draft-better-ai-choice-q1-correct

Misconception tags: None listed

Glossary terms: responsible-ai, model-distillation, efficient-inference, data-provenance, consent, compensation, human-review, accountability, governance, rag

choice-draft-better-ai-choice-q1-correct [CORRECT]: Use retrieval and human review when they fit the task

Feedback: Insight unlocked: better AI is shaped by design and governance choices.

choice-draft-better-ai-choice-q1-assume-bigger-is-always-better [DISTRACTOR]: Assume bigger is always better

Feedback: Not quite. Smaller task-specific systems may be better for some goals.

Misconception: Assume bigger is always better | **Represented term:** human-centered-ai-term | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-better-ai-choice-q1-remove-all-human-accountability [DISTRACTOR]: Remove all human accountability

Feedback: Not quite. High-stakes use needs human responsibility.

Misconception: Remove all human accountability | **Represented term:** human-review | **Source:** same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-better-ai-choice-q1-ignore-data-provenance [DISTRACTOR]: Ignore data provenance

Feedback: Not quite. Provenance, consent, and licensing matter.

Misconception: Ignore data provenance | **Represented term:** probability | **Source:** learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Better AI Is a Choice most directly teach?

ID: draft-better-ai-choice-q2 | Type: mechanism | Correct choice ID: choice-draft-better-ai-choice-q2-correct

Misconception tags: None listed

Glossary terms: responsible-ai, model-distillation, efficient-inference, data-provenance, consent, compensation, human-review, accountability, governance, rag

choice-draft-better-ai-choice-q2-correct [CORRECT]: Some choices durably shape models and institutions; others shape a particular workflow or inference run.

Feedback: Good distinction. Some choices durably shape models and institutions; others shape a particular workflow or inference run.

choice-draft-better-ai-choice-q2-compute-1 [DISTRACTOR]: It means compute by itself is intelligence.

Feedback: This choice points toward Compute. Compute matters nearby, but this question is asking about Better AI Is a Choice.

Misconception: Not intelligence by itself. | Represented term: compute | Source: explicitly confusable glossary metadata

Rationale: Tempting because Compute is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Better AI Is a Choice mechanism or distinction.

choice-draft-better-ai-choice-q2-training-data-2 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Better AI Is a Choice.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Better AI Is a Choice.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Better AI Is a Choice mechanism or distinction.

choice-draft-better-ai-choice-q2-copyright-3 [DISTRACTOR]: It is mainly about Copyright, not the mechanism named in Better AI Is a Choice.

Feedback: This choice points toward Copyright. Copyright matters nearby, but this question is asking about Better AI Is a Choice.

Misconception: Not the same as consent, compensation, or public availability. | Represented term: copyright | Source: explicitly confusable glossary metadata

Rationale: Tempting because Copyright is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Better AI Is a Choice mechanism or distinction.

Question 3: Which common confusion does Better AI Is a Choice help learners avoid?

ID: draft-better-ai-choice-q3 | Type: boundary distinction | Correct choice ID: choice-draft-better-ai-choice-q3-correct

Misconception tags: None listed

Glossary terms: responsible-ai, model-distillation, efficient-inference, data-provenance, consent, compensation, human-review, accountability, governance, rag

choice-draft-better-ai-choice-q3-correct [CORRECT]: It corrects the misconception: The harms are inevitable if the technology is useful.

Feedback: Good distinction. Better AI Is a Choice is clearer when that misconception is separated from the mechanism.

choice-draft-better-ai-choice-q3-training-data-1 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Better AI Is a Choice.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Better AI Is a Choice.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Better AI Is a Choice mechanism or distinction.

choice-draft-better-ai-choice-q3-copyright-2 [DISTRACTOR]: It is mainly about Copyright, not the mechanism named in Better AI Is a Choice.

Feedback: This choice points toward Copyright. Copyright matters nearby, but this question is asking about Better AI Is a Choice.

Misconception: Not the same as consent, compensation, or public availability. | Represented term: copyright | Source: explicitly confusable glossary metadata

Rationale: Tempting because Copyright is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Better AI Is a Choice mechanism or distinction.

choice-draft-better-ai-choice-q3-evaluation-3 [DISTRACTOR]: It is mainly about Evaluation, not the mechanism named in Better AI Is a Choice.

Feedback: This choice points toward Evaluation. Evaluation matters nearby, but this question is asking about Better AI Is a Choice.

Misconception: Not the same as training unless results feed a training process. | Represented term: evaluation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Evaluation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Better AI Is a Choice mechanism or distinction.

Question 4: How does Better AI Is a Choice connect to the prompt's Journey?

ID: draft-better-ai-choice-q4 | Type: concept relationship | Correct choice ID: choice-draft-better-ai-choice-q4-correct

Misconception tags: None listed

Glossary terms: responsible-ai, model-distillation, efficient-inference, data-provenance, consent, compensation, human-review, accountability, governance, rag

choice-draft-better-ai-choice-q4-correct [CORRECT]: Human-centered values become design, policy, governance, and incentive choices.

Feedback: Good connection. Human-centered values become design, policy, governance, and incentive choices.

choice-draft-better-ai-choice-q4-copyright-1 [DISTRACTOR]: It is mainly about Copyright, not the mechanism named in Better AI Is a Choice.

Feedback: This choice points toward Copyright. Copyright matters nearby, but this question is asking about Better AI Is a Choice.

Misconception: Not the same as consent, compensation, or public availability. | Represented term: copyright | Source: explicitly confusable glossary metadata

Rationale: Tempting because Copyright is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Better AI Is a Choice mechanism or distinction.

choice-draft-better-ai-choice-q4-evaluation-2 [DISTRACTOR]: It is mainly about Evaluation, not the mechanism named in Better AI Is a Choice.

Feedback: This choice points toward Evaluation. Evaluation matters nearby, but this question is asking about Better AI Is a Choice.

Misconception: Not the same as training unless results feed a training process. | Represented term: evaluation | Source: explicitly confusable glossary metadata

Rationale: Tempting because Evaluation is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Better AI Is a Choice mechanism or distinction.

choice-draft-better-ai-choice-q4-uncertainty-3 [DISTRACTOR]: It is mainly about Uncertainty, not the mechanism named in Better AI Is a Choice.

Feedback: This choice points toward Uncertainty. Uncertainty matters nearby, but this question is asking about Better AI Is a Choice.

Misconception: Not weakness; it is often part of responsible use. | Represented term: uncertainty | Source: explicitly confusable glossary metadata

Rationale: Tempting because Uncertainty is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Better AI Is a Choice mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

38. Effective Prompting from Model Literacy

Stage: 8. New Dawn | **Card ID:** effective-prompting-literacy | **Card in stage:** 4

Primary learning objective: Explain Effective Prompting from Model Literacy as part of the day in the life of a prompt.

Secondary objectives: Good prompts work because they shape the current context: task, constraints, examples, data, audience, evidence needs, uncertainty, and output format., During inference, when a person or system builds the current input context., Prompting usually changes temporary context, not durable weights., Prompt components become input context; the model still generates response tokens one at a time., After system-level choices, this card gives an individual practice while keeping responsibility visible.

Key distinctions: Prompting usually changes temporary context, not durable weights., Prompt components become input context; the model still generates response tokens one at a time., Prompting is magic wording or permanent teaching.

Misconceptions to correct: Prompting is magic wording or permanent teaching.

Related glossary terms: effective-prompting, prompt, context window, input-context, in-context learning, rag, grounding, uncertainty, human-review

Nearby glossary distractor candidates: citation, hallucination

Why this matters: Learners can use AI better when they understand context, RAG, uncertainty, and human review.

Existing Live Checkpoint

Question count: 1

Question: What does a good prompt usually shape?

Correct: The current context for this run

- The current context for this run (correct)
- The model's permanent weights
- The model's consciousness
- A guarantee that every answer is true Not quite. Better context can help, but answers still need review.

Proposed Draft Question Bank

Proposed count: 4. **Rationale:** Dense mechanism or high-confusion card; four checks cover definition, process, boundary, and relationship.

Question 1: What does a good prompt usually shape in Effective Prompting from Model Literacy?

ID: draft-effective-prompting-literacy-q1 | **Type:** definition | **Correct choice ID:** choice-draft-effective-prompting-literacy-q1-correct

Misconception tags: None listed

Glossary terms: effective-prompting, prompt, context window, input-context, in-context learning, rag, grounding, uncertainty, human-review

choice-draft-effective-prompting-literacy-q1-correct [CORRECT]: The current context for this run

Feedback: Insight unlocked: prompting is context design.

choice-draft-effective-prompting-literacy-q1-the-model-s-permanent-weights [DISTRACTOR]: The model's permanent weights

Feedback: This choice is tempting, but Effective Prompting from Model Literacy asks for Good prompts work because they shape the current context: task, constraints, examples, data, audience, evidence needs, uncertainty, and output format.

Misconception: The model's permanent weights | **Represented term:** training | **Source:** explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-effective-prompting-literacy-q1-the-model-s-consciousness [DISTRACTOR]: The model's consciousness

Feedback: This choice is tempting, but Effective Prompting from Model Literacy asks for Good prompts work because they shape the current context: task, constraints, examples, data, audience, evidence needs, uncertainty, and output format.

Misconception: The model's consciousness | **Represented term:** llm | **Source:** learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-effective-prompting-literacy-q1-a-guarantee-that-every-answer-is-true [DISTRACTOR]: A guarantee that every answer is true

Feedback: Not quite. Better context can help, but answers still need review.

Misconception: A guarantee that every answer is true | **Represented term:** probability | **Source:** learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Effective Prompting from Model Literacy most directly teach?

ID: draft-effective-prompting-literacy-q2 | Type: mechanism | Correct choice ID: choice-draft-effective-prompting-literacy-q2-correct

Misconception tags: None listed

Glossary terms: effective-prompting, prompt, context window, input-context, in-context learning, rag, grounding, uncertainty, human-review

choice-draft-effective-prompting-literacy-q2-correct [CORRECT]: Prompting usually changes temporary context, not durable weights.

Feedback: Good distinction. Prompting usually changes temporary context, not durable weights.

choice-draft-effective-prompting-literacy-q2-prompt-tokens-1 [DISTRACTOR]: It is a response token the model generated during this answer.

Feedback: This choice points toward Prompt token. Prompt token matters nearby, but this question is asking about Effective Prompting from Model Literacy.

Misconception: Not generated by the model during this response step. | Represented term: prompt-tokens | Source: explicitly confusable glossary metadata

Rationale: Tempting because Prompt token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Effective Prompting from Model Literacy mechanism or distinction.

choice-draft-effective-prompting-literacy-q2-memory-2 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Effective Prompting from Model Literacy.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Effective Prompting from Model Literacy mechanism or distinction.

choice-draft-effective-prompting-literacy-q2-training-data-3 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Effective Prompting from Model Literacy.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Effective Prompting from Model Literacy.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Effective Prompting from Model Literacy mechanism or distinction.

Question 3: Which common confusion does Effective Prompting from Model Literacy help learners avoid?

ID: draft-effective-prompting-literacy-q3 | Type: boundary distinction | Correct choice ID: choice-draft-effective-prompting-literacy-q3-correct

Misconception tags: None listed

Glossary terms: effective-prompting, prompt, context window, input-context, in-context learning, rag, grounding, uncertainty, human-review

choice-draft-effective-prompting-literacy-q3-correct [CORRECT]: It corrects the misconception: Prompting is magic wording or permanent teaching.

Feedback: Good distinction. Effective Prompting from Model Literacy is clearer when that misconception is separated from the mechanism.

choice-draft-effective-prompting-literacy-q3-memory-1 [DISTRACTOR]: It is only the temporary visible input context.

Feedback: This choice points toward Memory. Memory matters nearby, but this question is asking about Effective Prompting from Model Literacy.

Misconception: Do not confuse context with model weights or saved user memory. | Represented term: memory | Source: explicitly confusable glossary metadata

Rationale: Tempting because Memory is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Effective Prompting from Model Literacy mechanism or distinction.

choice-draft-effective-prompting-literacy-q3-training-data-2 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Effective Prompting from Model Literacy.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Effective Prompting from Model Literacy.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Effective Prompting from Model Literacy mechanism or distinction.

choice-draft-effective-prompting-literacy-q3-response-tokens-3 [DISTRACTOR]: It is input text that existed before generation began.

Feedback: This choice points toward Response token. Response token matters nearby, but this question is asking about Effective Prompting from Model Literacy.

Misconception: Not a durable weight update or saved memory. | Represented term: response-tokens | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Effective Prompting from Model Literacy mechanism or distinction.

Question 4: How does Effective Prompting from Model Literacy connect to the prompt's Journey?

ID: draft-effective-prompting-literacy-q4 | Type: concept relationship | Correct choice ID: choice-draft-effective-prompting-literacy-q4-correct

Misconception tags: None listed

Glossary terms: effective-prompting, prompt, context window, input-context, in-context learning, rag, grounding, uncertainty, human-review

choice-draft-effective-prompting-literacy-q4-correct [CORRECT]: After system-level choices, this card gives an individual practice while keeping responsibility visible.

Feedback: Good connection. After system-level choices, this card gives an individual practice while keeping responsibility visible.

choice-draft-effective-prompting-literacy-q4-training-data-1 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Effective Prompting from Model Literacy.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Effective Prompting from Model Literacy.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Effective Prompting from Model Literacy mechanism or distinction.

choice-draft-effective-prompting-literacy-q4-response-tokens-2 [DISTRACTOR]: It is input text that existed before generation began.

Feedback: This choice points toward Response token. Response token matters nearby, but this question is asking about Effective Prompting from Model Literacy.

Misconception: Not a durable weight update or saved memory. | Represented term: response-tokens | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response token is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Effective Prompting from Model Literacy mechanism or distinction.

choice-draft-effective-prompting-literacy-q4-response-so-far-3 [DISTRACTOR]: It is mainly about Response-so-far, not the mechanism named in Effective Prompting from Model Literacy.

Feedback: This choice points toward Response-so-far. Response-so-far matters nearby, but this question is asking about Effective Prompting from Model Literacy.

Misconception: Not the whole final answer, permanent memory, or training data. | Represented term: response-so-far | Source: explicitly confusable glossary metadata

Rationale: Tempting because Response-so-far is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Effective Prompting from Model Literacy mechanism or distinction.

Questions to Review Checklist

- ☐ Are the learning objectives accurate?
- ☐ Does each question restate the topic clearly?
- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

39. Model Literate Synthesis

Stage: 8. New Dawn | **Card ID:** model-literate-synthesis | **Card in stage:** 5

Primary learning objective: Explain Model Literate Synthesis as part of the day in the life of a prompt.

Secondary objectives: Model literacy means understanding both the mechanics and the human consequences., At the end of the Journey, when the learner connects mechanism, use, risk, and responsibility., The synthesis keeps durable weight changes, temporary context, retrieved evidence, and generated response tokens separate., A prompt sets up context; a response is generated; human judgment decides what to trust and do next., This final card ties the day in the life of a prompt to the wider human choices around AI.

Key distinctions: The synthesis keeps durable weight changes, temporary context, retrieved evidence, and generated response tokens separate., A prompt sets up context; a response is generated; human judgment decides what to trust and do next., Fluency equals understanding or wisdom.

Misconceptions to correct: Fluency equals understanding or wisdom.

Related glossary terms: model-literacy, training, inference, prompt, token, token-id, embedding, hidden state, logits, softmax, sampling, response-tokens, context window, rag, grounding, hallucination, source-review, human-review, accountability, human-centered-ai-term, governance

Nearby glossary distractor candidates: autoregression, top-p, top-k, temperature

Why this matters: The goal is not trivia. It is a durable mental model for wise use, institutional judgment, and better questions.

Existing Live Checkpoint

Question count: 1

Question: Which statement is model-literate?

Correct: Mechanics matter, and humans remain responsible

- Mechanics matter, and humans remain responsible (correct)
- Fluent output proves wisdom Not quite. Fluency is not wisdom.
- Ethics replaces learning the mechanics Not quite. Good judgment needs a clear mechanism.
- Mechanics alone settle every use decision Not quite. Mechanisms inform human choices; they do not replace judgment.

Proposed Draft Question Bank

Proposed count: 5. **Rationale:** Synthesis or badge-critical card; five checks cover multiple mechanisms and responsible-use distinctions.

Question 1: Which statement is model-literate in Model Literate Synthesis?

ID: draft-model-literate-synthesis-q1 | **Type:** definition | **Correct choice ID:** choice-draft-model-literate-synthesis-q1-correct

Misconception tags: None listed

Glossary terms: model-literacy, training, inference, prompt, token, token-id, embedding, hidden state, logits, softmax, sampling, response-tokens, context window, rag, grounding, hallucination, source-review, human-review, accountability, human-centered-ai-term, governance

choice-draft-model-literate-synthesis-q1-correct [CORRECT]: Mechanics matter, and humans remain responsible

Feedback: Insight unlocked: model literacy joins mechanism with judgment.

choice-draft-model-literate-synthesis-q1-fluent-output-proves-wisdom [DISTRACTOR]: Fluent output proves wisdom

Feedback: Not quite. Fluency is not wisdom.

Misconception: Fluent output proves wisdom | Represented term: llm | Source: learning-path-near glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-model-literate-synthesis-q1-ethics-replaces-learning-the-mechanics [DISTRACTOR]: Ethics replaces learning the mechanics

Feedback: Not quite. Good judgment needs a clear mechanism.

Misconception: Ethics replaces learning the mechanics | Represented term: context window | Source: same-card glossary term

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

choice-draft-model-literate-synthesis-q1-mechanics-alone-settle-every-use-decision [DISTRACTOR]: Mechanics alone settle every use decision

Feedback: Not quite. Mechanisms inform human choices; they do not replace judgment.

Misconception: Mechanics alone settle every use decision | Represented term: responsible-ai | Source: explicitly confusable glossary metadata

Rationale: Existing live checkpoint distractor retained for review because it captures a plausible misunderstanding.

Question 2: Which mechanism or distinction does Model Literate Synthesis most directly teach?

ID: draft-model-literate-synthesis-q2 | Type: mechanism | Correct choice ID: choice-draft-model-literate-synthesis-q2-correct

Misconception tags: None listed

Glossary terms: model-literacy, training, inference, prompt, token, token-id, embedding, hidden state, logits, softmax, sampling, response-tokens, context window, rag, grounding, hallucination, source-review, human-review, accountability, human-centered-ai-term, governance

choice-draft-model-literate-synthesis-q2-correct [CORRECT]: The synthesis keeps durable weight changes, temporary context, retrieved evidence, and generated response tokens separate.

Feedback: Good distinction. The synthesis keeps durable weight changes, temporary context, retrieved evidence, and generated response tokens separate.

choice-draft-model-literate-synthesis-q2-loss-1 [DISTRACTOR]: It is mainly about Loss, not the mechanism named in Model Literate Synthesis.

Feedback: This choice points toward Loss. Loss matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Truth, confidence, punishment, or a human sense of failure. | Represented term: loss | Source: explicitly confusable glossary metadata

Rationale: Tempting because Loss is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

choice-draft-model-literate-synthesis-q2-weight-update-2 [DISTRACTOR]: It is mainly about Weight update, not the mechanism named in Model Literate Synthesis.

Feedback: This choice points toward Weight update. Weight update matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Not the same as adding a prompt to context. | Represented term: weight-update | Source: explicitly confusable glossary metadata

Rationale: Tempting because Weight update is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

choice-draft-model-literate-synthesis-q2-training-data-3 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Model Literate Synthesis.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

Question 3: Which common confusion does Model Literate Synthesis help learners avoid?

ID: draft-model-literate-synthesis-q3 | Type: boundary distinction | Correct choice ID: choice-draft-model-literate-synthesis-q3-correct

Misconception tags: None listed

Glossary terms: model-literacy, training, inference, prompt, token, token-id, embedding, hidden state, logits, softmax, sampling, response-tokens, context window, rag, grounding, hallucination, source-review, human-review, accountability, human-centered-ai-term, governance

choice-draft-model-literate-synthesis-q3-correct [CORRECT]: It corrects the misconception: Fluency equals understanding or wisdom.

Feedback: Good distinction. Model Literate Synthesis is clearer when that misconception is separated from the mechanism.

choice-draft-model-literate-synthesis-q3-weight-update-1 [DISTRACTOR]: It is mainly about Weight update, not the mechanism named in Model Literate Synthesis.

Feedback: This choice points toward Weight update. Weight update matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Not the same as adding a prompt to context. | Represented term: weight-update | Source: explicitly confusable glossary metadata

Rationale: Tempting because Weight update is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

choice-draft-model-literate-synthesis-q3-training-data-2 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Model Literate Synthesis.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

choice-draft-model-literate-synthesis-q3-pretraining-3 [DISTRACTOR]: It happens again from scratch during every user prompt.

Feedback: This choice points toward Pretraining. Pretraining matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Often confused with fine-tuning or perfect source recall. | Represented term: pretraining | Source: explicitly confusable glossary metadata

Rationale: Tempting because Pretraining is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

Question 4: How does Model Literate Synthesis connect to the prompt's Journey?

ID: draft-model-literate-synthesis-q4 | Type: concept relationship | Correct choice ID: choice-draft-model-literate-synthesis-q4-correct

Misconception tags: None listed

Glossary terms: model-literacy, training, inference, prompt, token, token-id, embedding, hidden state, logits, softmax, sampling, response-tokens, context window, rag, grounding, hallucination, source-review, human-review, accountability, human-centered-ai-term, governance

choice-draft-model-literate-synthesis-q4-correct [CORRECT]: This final card ties the day in the life of a prompt to the wider human choices around AI.

Feedback: Good connection. This final card ties the day in the life of a prompt to the wider human choices around AI.

choice-draft-model-literate-synthesis-q4-training-data-1 [DISTRACTOR]: It is mainly about Training data, not the mechanism named in Model Literate Synthesis.

Feedback: This choice points toward Training data. Training data matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Not the same as the current prompt or context window. | Represented term: training-data | Source: explicitly confusable glossary metadata

Rationale: Tempting because Training data is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

choice-draft-model-literate-synthesis-q4-pretraining-2 [DISTRACTOR]: It happens again from scratch during every user prompt.

Feedback: This choice points toward Pretraining. Pretraining matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Often confused with fine-tuning or perfect source recall. | Represented term: pretraining | Source: explicitly confusable glossary metadata

Rationale: Tempting because Pretraining is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

choice-draft-model-literate-synthesis-q4-forward-pass-3 [DISTRACTOR]: It is mainly about Forward pass, not the mechanism named in Model Literate Synthesis.

Feedback: This choice points toward Forward pass. Forward pass matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Not backpropagation or a training update. | Represented term: forward-pass | Source: explicitly confusable glossary metadata

Rationale: Tempting because Forward pass is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

Question 5: A learner uses Model Literate Synthesis to explain an AI answer. What should they say?

ID: draft-model-literate-synthesis-q5 | **Type:** applied scenario | **Correct choice ID:** choice-draft-model-literate-synthesis-q5-correct

Misconception tags: None listed

Glossary terms: model-literacy, training, inference, prompt, token, token-id, embedding, hidden state, logits, softmax, sampling, response-tokens, context window, rag, grounding, hallucination, source-review, human-review, accountability, human-centered-ai-term, governance

choice-draft-model-literate-synthesis-q5-correct [CORRECT]: The goal is not trivia. It is a durable mental model for wise use, institutional judgment, and better questions.

Feedback: Good model-literacy move. The goal is not trivia. It is a durable mental model for wise use, institutional judgment, and better questions.

choice-draft-model-literate-synthesis-q5-pretraining-1 [DISTRACTOR]: It happens again from scratch during every user prompt.

Feedback: This choice points toward Pretraining. Pretraining matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Often confused with fine-tuning or perfect source recall. | Represented term: pretraining | Source: explicitly confusable glossary metadata

Rationale: Tempting because Pretraining is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

choice-draft-model-literate-synthesis-q5-forward-pass-2 [DISTRACTOR]: It is mainly about Forward pass, not the mechanism named in Model Literate Synthesis.

Feedback: This choice points toward Forward pass. Forward pass matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Not backpropagation or a training update. | Represented term: forward-pass | Source: explicitly confusable glossary metadata

Rationale: Tempting because Forward pass is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

choice-draft-model-literate-synthesis-q5-input-context-3 [DISTRACTOR]: It is mainly about Input context, not the mechanism named in Model Literate Synthesis.

Feedback: This choice points toward Input context. Input context matters nearby, but this question is asking about Model Literate Synthesis.

Misconception: Not the training dataset or durable model weights. | Represented term: input-context | Source: explicitly confusable glossary metadata

Rationale: Tempting because Input context is explicitly confusable glossary metadata for this part of the Journey. It is not correct because the question asks for the specific Model Literate Synthesis mechanism or distinction.

Questions to Review Checklist

[] Are the learning objectives accurate?

[] Does each question restate the topic clearly?

- ☐ Is one answer clearly best?
- ☐ Are distractors plausible but not unfair?
- ☐ Are the distractors close enough to teach a boundary?
- ☐ Does feedback explain the misconception?
- ☐ Are any questions redundant?
- ☐ Are any important questions missing?
- ☐ Is the question count right for topic density?
- ☐ Is the reading level right?

Deep Research Handoff

Review this Prompt Life checkpoint question bank. For each Journey learning card, evaluate the stated learning objectives, question count, question wording, correct answers, distractors, misconception feedback, and glossary-path-near distractor choices. Recommend edits, removals, additions, and any terms or misconceptions that should be better represented.

Recommended review stance: look for vague wording, unfair distractors, missing durable-vs-temporary distinctions, underrepresented glossary-near confusions, reading-level drift, and redundant questions within the same card.